United States Patent Application Publication	20250276001
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	Fitzgerald; Kevin et al.

SERUM AMYLOID P COMPONENT (APCS) iRNA COMPOSITIONS AND METHODS OF USE THEREOF

Abstract

The invention relates to iRNA, e.g., double stranded ribonucleic acid (dsRNA), compositions targeting the serum amyloid P component (APCS) gene, and methods of using such iRNA, e.g., dsRNA, compositions to inhibit expression of an APCS gene and to treat subjects having an APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.

Inventors: **Fitzgerald; Kevin (Brookline, MA), Hinkle; Gregory (Plymouth, MA)**

Applicant: **Alnylam Pharmaceuticals, Inc.** (Cambridge, MA)

Family ID: **64665390**

Appl. No.: **18/907649**

Filed: **October 07, 2024**

Related U.S. Application Data

parent US continuation 17544984 20211208 ABANDONED child US 18907649
parent US continuation 16877586 20200519 parent-grant-document US 11229663 child US 17544984
parent US continuation PCT/US2018/061906 20181120 PENDING child US 16877586
us-provisional-application US 62588506 20171120

Publication Classification

Int. Cl.: **A61K31/713** (20060101); **A61K47/54** (20170101); **C12N15/113** (20100101)

U.S. Cl.:

CPC **A61K31/713** (20130101); **A61K47/549** (20170801); **C12N15/113** (20130101); C12N2310/14 (20130101); C12N2310/315 (20130101); C12N2310/3515 (20130101)

Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/544,984, filed on Dec. 8, 2021, which is a continuation of U.S. patent application Ser. No. 16/877,586, filed May 19, 2020, now U.S. Pat. No. 11,229,663, issued on Jan. 25, 2022, which is a 35 § U.S.C. 111(a) continuation application which claims the benefit of priority to PCT/US2018/061906, filed on Nov. 20, 2018, which claims the benefit of priority to U.S. Provisional Application No. 62/588,506, filed on Nov. 20, 2017. The entire contents of each of the foregoing applications are incorporated herein by reference.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Dec. 13, 2024, is named 121301_06505_SL.xml and is 5,727,508 bytes in size.

BACKGROUND OF THE INVENTION

[0003] Serum amyloid P component (SAP), also referred to as APCS or pentraxin-2 (PTX-2), is a 25 kDa pentameric protein that is identical to the amyloid P component (AP), but is located in the serum. In humans, SAP is encoded by the APCS gene (amyloid P component, serum). SAP is a glycoprotein that belongs to the pentraxin family of proteins, members of which have a characteristic pentameric organization.

[0004] Under normal conditions, SAP is thought to be synthesized and secreted only in hepatocytes and has a half-life of approximately 24 hours. In some diseases, SAP can also be generated by macrophages and smooth muscle cells such as in the atherosclerotic aortic intima.

[0005] SAP plays a role in protein aggregation and in regulating immune response, as illustrated in FIG. **1**. SAP is best known as a constituent of in vivo pathological amyloid deposits. Amyloid deposits are inherently stable structures consisting of fibrils that are formed when normally soluble proteins assemble to form insoluble fibers predominantly composed of 3-sheet structures in a characteristic cross-0 conformation. The amyloid deposits are deposited extracellularly in the tissues and are thought to have a pathogenic effect.

[0006] SAP accounts for 14% of the dry mass of amyloid deposits. It can be found in all types of amyloid deposits, in the glomerula basement membranes and in elastic fibers in blood vessels, all of which are characterized by the ordered aggregation of normal globular proteins and peptides into insoluble fibers. SAP is thought to contribute to amyloid deposit formation by decorating and stabilizing protein aggregates, thereby preventing proteolytic cleavage and inhibiting fibril removal via the normal protein scavenging mechanisms (see Xi et al., *Int. J. of Cardiol.* 187, 20-26, 2015). SAP was also found to be present in atherosclerotic lesions and plasma levels of SAP were found to be positively associated with cardiovascular disease in the elderly. Further, it was recently demonstrated that SAP deficiency mitigated atherosclerotic lesions in a mouse model of atherosclerosis (Zheng et al., *Atherosclerosis* 244, 179-187, 2016). Thus, SAP may contribute to the pathogenesis of the diseases associated with amyloid deposits, e.g., amyloidosis and Alzheimer's disease, and cardiovascular diseases, e.g., atherosclerotic heart disease.

[0007] Amyloidosis is a rare, serious disease caused by accumulation of amyloid deposits within the extracellular space in the tissues of the body. The amyloid deposits disrupt the normal tissue architecture, damaging the function of tissues and organs and causing disease. In contrast to the normally efficient clearance of abnormal debris from the tissues, amyloid deposits are removed very slowly, if at all. There are many different types of amyloidoses, each caused by formation of amyloid fibrils from different soluble precursor proteins in different patients. About 30 different proteins are known to form amyloid fibrils in humans, and amyloidosis is named and classified according to the identity of the respective fibril protein. Amyloid deposits can be confined to only one part of the body or a single organ system in "local amyloidosis" or they can be widely distributed in organs and tissues throughout the body in "systemic amyloidosis". The clinical manifestations of amyloidosis are accordingly highly variable and confirmation of the presence of amyloid in the tissues can be challenging, so that diagnosis is often delayed. Broad classification of amyloidosis includes primary (systemic AL) amyloidosis, secondary (systemic AA) amyloidosis, dialysis-related amyloidosis (DRA), familial (hereditary FA) amyloidosis, senile systemic amyloidosis (SSA) and organ-specific amyloidosis. Primary amyloidosis (AL) occurs without a known cause, but it has been seen in people with a blood cancer called multiple myeloma. This is the most common type of amyloidosis. "Systemic" means it affects the entire body, and the most commonly affected body parts are the kidney, heart, liver, intestines, and certain nerves. AL stands for "amyloid light chains," which is the type of protein responsible for this type of amyloidosis. Secondary amyloidosis (AA) is is the result of another chronic inflammatory disease, such as lupus, rheumatoid arthritis, tuberculosis, inflammatory bowel disease (Crohn's disease and ulcerative colitis), and certain cancers. It most commonly affects the spleen, kidneys, liver, adrenal gland, and lymph nodes. AA means the amyloid type A protein causes this type of amyloidosis. Dialysis-related amyloidosis (DRA) is more common in older adults and people who have been on dialysis for more than 5 years. This form of amyloidosis is caused by deposits of beta-2 microglobulin that build up in the blood. Deposits can occur in many different tissues, but most commonly affects bones, joints, and tendons.

[0008] Familial, or hereditary, amyloidosis (AF) is a rare form that is passed down through families. It is caused by an abnormal amyloid transthyretin (TTR) protein, which is made in the liver. This protein is responsible for the most common forms of hereditary amyloidosis. Senile systemic amyloidosis (SSA) is caused deposits of normal TTR in the heart and other tissues. It occurs most commonly in older men. Organ-specific amyloidosis is caused by deposits of amyloid protein in single organs, including the skin (cutaneous amyloidosis).

[0009] Symptoms of amyloidosis may be varied and depend on the organ most affected by the deposition of amyloid fibrils. For example, symptoms of cardiac amyloidosis may include shortness of breath, an irregular heartbeat, signs of heart failure; symptoms of renal amyloidosis may include signs of renal failure and high levels of urinary proteins; and symptoms of gastrointestinal amyloidosis may include diarrhea, nausea, stomach pain, decreased appetite and weight loss.

[0010] There is no cure for amyloidosis. Several treatments may be used to slow the progression of the disease or to manage symptoms. Some of the treatments used to manage the symptoms of amyloidosis include chemotherapy, high-dose chemotherapy combined with stem cell transplant; steroids, liver, heart and kidney transplants and diuretics. It is therefore clear that new and effective treatment for amyloid-related diseases, e.g., amyloidosis, are needed.

[0011] Accordingly, there is a need in the art for therapies for subjects having a APCS-associated disease, e.g., amyloidosis, Alzheimer's disease, or a cardiovascular disease, e.g., atherosclerotic heart disease.

SUMMARY OF THE INVENTION

[0012] The present invention provides iRNA compositions which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of an APCS gene. The APCS gene may be within a cell, e.g., a cell within a subject, such as a human. The use of these iRNA agents enables the selective targeted degradation of mRNAs of the corresponding gene (the APCS gene) in mammals.

[0013] The present invention also provides methods and combination therapies for treating a subject having a disease that would benefit from inhibiting or reducing the expression of an APCS

gene, e.g., a APCS-associated disease. An APCS-associated disease may include an amyloid-associated disease, e.g., amyloidosis, such as primary (systemic AL) amyloidosis, secondary (systemic AA) amyloidosis, dialysis-related amyloidosis (DRA), familial (hereditary FA) amyloidosis, senile systemic amyloidosis (SSA) and organ-specific amyloidosis; Alzheimer's disease; diabetes mellitus type 2; Parkinson's disease; transmissible spongiform encephalopathy (such as bovine spongiform encephalopathy); fatal familial insomnia; Huntington's disease; medullary carcinoma of the thyroid; cardiac arrhythmias; isolated atrial amyloidosis; rheumatoid arthritis; aortic medial amyloid; prolactinoma; familial amyloid polyneuropathy; lattice corneal dystrophy; cerebral amyloid angiopathy; cerebral amyloid angiopathy (Icelandic type); sporadic inclusion body myositis; or a cardiovascular disease, e.g., coronary atherosclerotic heart disease.
[0014] Accordingly, in some embodiments, the present invention provides a double stranded ribonucleic acid (RNAi) agent for inhibiting expression of a serum amyloid P component (APCS) gene, comprising a sense strand and an antisense strand, wherein the sense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the nucleotide sequence of SEQ ID NO:1, and the antisense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the nucleotide sequence of SEQ ID NO:5.
[0015] In some embodiments, the present invention also provides a double stranded ribonucleic acid (RNAi) agent for inhibiting expression of a serum amyloid P component (APCS) gene, comprising a sense strand and an antisense strand, the antisense strand comprising a region of complementarity which comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from any one of the antisense sequences listed in any one of Tables 3A, 3B, 4A, 4B, 6, and 7.
[0016] In some aspects, the present invention also provides a double stranded ribonucleic acid (RNAi) agent for inhibiting expression of a serum amyloid P component (APCS) gene, comprising a sense strand and an antisense strand, wherein the antisense strand comprises a region complementary to part of an mRNA encoding SAP, wherein each strand is about 14 to about 30 nucleotides in length, wherein the sense strand is represented by formula (I):

sense: 5′*n*.sub.p-N.sub.a-(XXX).sub.i-N.sub.b-YYY-N.sub.b-(ZZZ).sub.j-N.sub.a-*n*.sub.q3′ (I)
[0017] wherein: [0018] i and j are each independently 0 or 1; [0019] p and q are each independently 0-6; [0020] N.sub.a represents an oligonucleotide sequence comprising 0-25 nucleotides which are either modified or unmodified or combinations thereof; [0021] N.sub.b represents an oligonucleotide sequence comprising 0-10 nucleotides which are either modified or unmodified or combinations thereof; [0022] each n.sub.p and n.sub.q, each of which may or may not be present, independently represents an overhang nucleotide; [0023] XXX, YYY and ZZZ each independently represent one motif of three identical modifications on three consecutive nucleotides; [0024] modifications on N.sub.b differ from the modification on Y; and [0025] wherein the sense strand is conjugated to at least one ligand.
[0026] In some embodiments, i is 0; j is 0; i is 1; j is 1; both i and j are 0; or both i and j are 1.
[0027] In some aspects, the YYY motif occurs at or near the cleavage site of the sense strand.
[0028] In a further embodiment, formula (I) is represented by formula (Ia):

sense: 5′*n*.sub.p-N.sub.a-YYY-N.sub.a-*n*.sub.q3′ (Ia).
[0029] In some aspects of the invention, the sense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the nucleotide sequence of nucleotides 61-83, 94-116, 163-185, 173-195, 194-216, 194-216, 197-219, 199-221, 204-226, 205-227, 237-259, 246-268, 258-280, 271-293, 309-331, 315-337, 316-338, 325-347, 326-348, 327-349, 328-350, 329-351, 330-352, 331-353, 332-354, 336-358, 338-360, 343-365, 353-375, 360-382, 364-386, 365-387, 371-393, 375-397, 380-402, 384-406, 387-409, 391-413, 395-417, 395-417, 396-418, 400-422, 403-425, 451-473, 455-477, 459-481, 527-549, 533-555, 536-558, 564-586, 651-

673, 652-674, 653-675, 707-729, 708-730, 709-731, 710-732, 729-751, 734-756, 739-761, 778-800, 780-802, 784-806, 785-807, 793-815, 794-816, 796-818, 797-819, 798-820, 827-849, 852-874, 856-878, 858-880, 865-887, 866-888, 868-890, 870-892, 878-900, 881-903, 882-904, 883-905, 885-907, 911-933, 916-938 of SEQ ID NO: 1.

[0030] In some embodiments, the antisense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the antisense nucleotide sequence of a duplex selected from the group consisting of AD-75708, AD-75723, AD-75694, AD-75692, AD-75664, AD-75664.2, AD-75679, AD-75659, AD-75662, AD-75680, AD-75687, AD-75657, AD-75699, AD-75727, AD-75731, AD-75728, AD-75737, AD-75696, AD-75718, AD-75676, AD-75663, AD-75669, AD-75666, AD-75735, AD-75686, AD-75736, AD-75674, AD-75717, AD-75706, AD-75719, AD-75688, AD-75734, AD-75724, AD-75711, AD-75703, AD-75721, AD-75712, AD-75697, AD-75726, AD-75730, AD-75732, AD-75733, AD-75729, AD-75685, AD-75673, AD-75691, AD-75670, AD-75739, AD-75738, AD-75722, AD-75714, AD-75681, AD-75668, AD-75693, AD-75677, AD-75690, AD-75716, AD-75682, AD-75720, AD-75725, AD-75695, AD-75665, AD-75661, AD-75658, AD-75700, AD-75698, AD-75672, AD-75684, AD-75667, AD-75678, AD-75660, AD-75701, AD-75707, AD-75675, AD-75671, AD-75683, AD-75689, AD-75715, AD-75705, AD-75704, AD-75713, AD-75702, AD-75709, AD-75710 as listed in Tables 3A, 3B, 4A and 4B.

[0031] In some embodiments, the sense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the nucleotide sequence of nucleotides 53-71; 68-86; 82-100; 90-108; 104-122; 118-136; 132-150; 148-166; 162-180; 170-188; 185-203; 200-218; 214-232; 228-246; 242-260; 248-266; 277-295; 283-301; 294-312; 309-327; 323-341; 352-370; 366-384; 377-395; 385-403; 461-479; 476-494; 491-509; 505-523; 750-768; 758-776; 773-791; 787-805; 854-872; 869-887; or 878-896 of SEQ ID NO: 1.

[0032] In other embodiments, the antisense strand comprises at least 15 contiguous nucleotides differing by no more than 3 nucleotides from the antisense nucleotide sequence of a duplex selected from the group consisting of AD-77752; AD-77753; AD-77754; AD-77755; AD-77756; AD-77757; AD-77758; AD-77759; AD-77760; AD-77761; AD-77762; AD-77763; AD-77764; AD-77765; AD-77766; AD-77767; AD-77769; AD-77770; AD-77771; AD-77772; AD-77773; AD-77775; AD-77776; AD-77777; AD-77778; AD-77783; AD-77784; AD-77785; AD-77786; AD-77804; AD-77805; AD-77806; AD-77807; AD-77812; AD-77813; and AD-77814 as listed in Tables 6 and 7.

[0033] In some embodiments, the sense and antisense strands comprise nucleotide sequences selected from any of the nucleotide sequences in any one of Tables 3A, 3B, 4A, 4B, 6, and 7.

[0034] In some embodiments, the sense and antisense strands consist of nucleotide sequences selected from any of the nucleotide sequences in any one of Tables 3A, 3B, 4A, 4B, 6, and 7.

[0035] In some embodiments, the double stranded RNAi agent comprises at least one modified nucleotide. In some aspects, substantially all of the nucleotides of the sense strand are modified nucleotides; substantially all of the nucleotides of the antisense strand are modified nucleotides; or substantially all of the nucleotides of the sense strand and substantially all of the nucleotides of the antisense strand are modified nucleotides. In a further aspect, all of the nucleotides of the sense strand are modified nucleotides; all of the nucleotides of the antisense strand are modified nucleotides; or all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand are modified nucleotides.

[0036] In some embodiments, at least one of said modified nucleotides is selected from the group consisting of a deoxy-nucleotide, a 3′-terminal deoxy-thymine (dT) nucleotide, a 2′-O-methyl modified nucleotide, a 2′-fluoro modified nucleotide, a 2′-deoxy-modified nucleotide, a locked nucleotide, an unlocked nucleotide, a conformationally restricted nucleotide, a constrained ethyl nucleotide, an abasic nucleotide, a 2′-amino-modified nucleotide, a 2′-O-allyl-modified nucleotide, 2′-C-alkyl-modified nucleotide, 2′-hydroxyl-modified nucleotide, a 2′-methoxyethyl modified nucleotide, a 2′-O-alkyl-modified nucleotide, a morpholino nucleotide, a 5′-vinyl phosphate, a phosphoramidate, a non-natural base comprising nucleotide, a tetrahydropyran modified nucleotide, a 1,5-anhydrohexitol modified nucleotide, a cyclohexenyl modified nucleotide, a nucleotide comprising a phosphorothioate group, a nucleotide comprising a methylphosphonate group, a nucleotide comprising a 5′-phosphate, and a nucleotide comprising a 5′-phosphate mimic. For example, at least one of the modified nucleotides is selected from the group consisting of a 2′-O-methyl modification, a 2′fluoro modification, a 5′-vinyl phosphate, and a 3′-terminal deoxy-thymine (dT) nucleotide.

[0037] In one aspect, the double stranded RNAi agent of the invention comprises at least one phosphorothioate internucleotide linkage.

[0038] In some embodiments, the region of complementarity between the sense strand and the antisense strand is at least 17 nucleotides in length, e.g., 19 to 30 nucleotides in length; 21 nucleotides in length; 21 to 23 nucleotides in length; 19 nucleotides in length, or no more than 30 nucleotides in length.

[0039] In some aspects, each strand is independently 19-30 nucleotides in length, e.g., 19-25 nucleotides in length.

[0040] In one embodiment, the sense strand has a total of 21 nucleotides and the antisense strand has a total of 23 nucleotides.

[0041] In some aspects, the at least one strand comprises a 3′ overhang of at least 1 nucleotide or at least 2 nucleotides.

[0042] In some embodiments, the double stranded RNAi agent of the invention further comprises a ligand. In a further embodiment, the ligand is conjugated to the 3′ end of the sense strand of the double stranded RNAi agent. In one embodiment, the ligand is an N-acetylgalactosamine (GalNAc) derivative, e.g., the ligand is

##STR00001##

[0043] In an embodiment, the double stranded RNAi agent of the invention is conjugated to the ligand as shown in the following schematic

##STR00002##

[0044] and wherein X is O or S. In one embodiment, the X is O.

[0045] In some aspects, the double stranded RNAi agent is selected from the group consisting of of any one of the agents listed in any one of Tables 3A, 3B, 4A, 4B, 6, and 7.

[0046] In some aspects, the present invention also provides a cell containing the double stranded RNAi agent of the invention and a vector encoding at least one strand of the double stranded RNAi agent of the invention.

[0047] In other aspects, the present invention also provides a pharmaceutical composition for inhibiting expression of a serum amyloid P component (APCS) gene comprising the double stranded RNAi agent of the invention. In some embodiments, the RNAi agent is in an unbuffered solution, e.g., in saline or water. In other embodiments, the RNAi agent is with a buffer solution, e.g., a buffer solution comprising acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof. In a specific embodiment, the buffer solution is phosphate buffered saline (PBS).

[0048] In some embodiments, the present invention also provides a method of inhibiting expression of a serum amyloid P component (APCS) gene in a cell. The method includes contacting the cell with an RNAi agent of the invention or a pharmaceutical composition of the invention, thereby inhibiting expression of the SAP gene in the cell. In one aspect, the cell is within a subject. In one aspect, the subject is a human.

[0049] In some embodiments, the APCS expression is inhibited by at least 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or to below the level of detection of APCS expression.

[0050] In one embodiments, the human suffers from an APCS-associated disease, e.g., a disease selected from the group consisting of amyloidosis, Alzheimer's disease, diabetes mellitus type 2, Parkinson's disease, transmissible spongiform encephalopathy, fatal familial insomnia, Huntington's disease, medullary carcinoma of the thyroid, cardiac arrhythmias, isolated atrial amyloidosis, rheumatoid arthritis, aortic medial amyloid, prolactinoma, familial amyloid polyneuropathy, lattice corneal dystrophy, cerebral amyloid angiopathy, cerebral amyloid angiopathy (Icelandic type), sporadic inclusion body myositis, and a cardiovascular disease.

[0051] In one aspect, the present invention provides a method of treating a subject having a disease that would benefit from reduction in serum amyloid P component (APCS) gene expression. The method includes administering to the subject a therapeutically effective amount of the double stranded RNAi agent of the invention, or a pharmaceutical composition of the invention, thereby treating the subject.

[0052] In another aspect, the present invention also provides a method of preventing at least one symptom in a subject having a disease that would benefit from reduction in serum amyloid P component (APCS) gene expression. The method includes administering to the subject a therapeutically effective amount of the double stranded RNAi agent of the invention, or a pharmaceutical composition of the invention, thereby preventing at least one symptom in the subject having a disease that would benefit from reduction in APCS expression.

[0053] In some embodiments, the disease is a APCS-associated disease, e.g., a disease selected from the group consisting of amyloidosis, Alzheimer's disease, diabetes mellitus type 2, Parkinson's disease, transmissible spongiform encephalopathy, fatal familial insomnia Huntington's disease, medullary carcinoma of the thyroid, cardiac arrhythmias, isolated atrial amyloidosis, rheumatoid arthritis, aortic medial amyloid, prolactinoma, familial amyloid polyneuropathy, lattice corneal dystrophy, cerebral amyloid angiopathy, cerebral amyloid angiopathy (Icelandic type), sporadic inclusion body myositis and a cardiovascular disease.

[0054] In one embodiment, the subject is a human.

[0055] In some embodiments, the administration of the RNAi agent to the subject causes a decrease in plasma levels of SAP protein. In other embodiments, the administration of the RNAi agent to the subject causes a decrease in amyloid load in the subject.

[0056] In some aspects, the RNAi agent is administered to the subject at a dose of about 0.01 mg/kg to about 10 mg/kg or about 0.5 mg/kg to about 50 mg/kg. In other aspects, the RNAi agent is administered to the subject subcutaneously or intravenously.

[0057] In some embodiments, the methods of the invention further comprise administering CPHPC and/or an anti-SAP antibody, or antigen-binding fragment there, to the subject.

[0058] In one aspect, the present invention also provides a method of inhibiting the expression of an APCS protein (SAP) in a subject. The method includes administering to the subject a therapeutically effective amount of the double stranded RNAi agent of the invention or a pharmaceutical composition of the invention, thereby inhibiting the expression of APCS in the subject.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0059] FIG. **1** is a schematic overview of the major functions of serum amyloid P component (SAP).

DETAILED DESCRIPTION OF THE INVENTION

[0060] The present invention provides iRNA compositions, which selectively effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of an APCS gene. The gene may be within a cell, e.g., a cell within a subject, such as a human. The use of these iRNAs enables the selective targeted degradation of mRNAs of the APCS gene in mammals.

[0061] The RNAi agents of the invention have been designed to potently and selectively target the corresponding human APCS gene. Without intending to be limited by theory, it is believed that a combination or sub-combination of the foregoing properties and the specific target sites and/or the specific modifications in these RNAi agents confer to the RNAi agents of the invention improved efficacy, stability, potency, durability, and safety.

[0062] The iRNAs of the invention may include an RNA strand (the antisense strand) having a region which is about 30 nucleotides or less in length, e.g., 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length, which region is substantially complementary to at least part of an mRNA transcript of a human APCS gene.

[0063] In certain embodiments, the iRNAs of the invention include an RNA strand (the antisense strand) which can include longer lengths, for example up to 66 nucleotides, e.g., 36-66, 26-36, 25-36, 31-60, 22-43, 27-53 nucleotides in length with a region of at least 19 contiguous nucleotides that is substantially complementary to at least a part of an mRNA transcript of a APCS gene. These iRNAs with the longer length antisense strands preferably include a second RNA strand (the sense strand) of 20-60 nucleotides in length wherein the sense and antisense strands form a duplex of 18-30 contiguous nucleotides.

[0064] It is believed that iRNAs targeting an APCS gene can potently mediate RNAi, resulting in significant inhibition of expression of an APCS gene. Thus, methods and compositions including these iRNAs are useful for treating a subject having a APCS-associated disease or disorder, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.

[0065] Accordingly, the present invention provides methods and combination therapies for treating a subject having a disorder that would benefit from inhibiting or reducing the expression of a APCS gene, e.g., a APCS-associated disease, such as amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease, using iRNA compositions which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of a APCS gene.

[0066] The present invention also provides methods for ameliorating and/or preventing at least one symptom e.g., a symptom associated with formation and/or deposition of amyloid deposits or atherosclerotic lesions, in a subject having a disorder that would benefit from inhibiting or reducing the expression of an APCS gene, e.g., a APCS-associated disease, such as amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.

[0067] The following detailed description discloses how to make and use compositions containing iRNAs to inhibit the expression of an APCS gene, as well as compositions, uses, and methods for treating subjects having diseases and disorders that would benefit from inhibition and/or reduction of the expression of an APCS gene.

I. Definitions

[0068] In order that the present invention may be more readily understood, certain terms are first defined. In addition, it should be noted that whenever a value or range of values of a parameter are recited, it is intended that values and ranges intermediate to the recited values are also intended to be part of this invention.

[0069] The articles “a” and “an” are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element, e.g., a plurality of elements.

[0070] The term “including” is used herein to mean, and is used interchangeably with, the phrase “including but not limited to”.

[0071] The term “or” is used herein to mean, and is used interchangeably with, the term “and/or,” unless context clearly indicates otherwise.

[0072] The term “about” is used herein to mean within the typical ranges of tolerances in the art. For example, “about” can be understood as within about 2 standard deviations from the mean. In

certain embodiments, about means ±10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2% or ±1%. When "about" is present before a series of numbers or a range, it is understood that "about" can modify each of the numbers in the series or range.
[0073] The term "at least" prior to a number or series of numbers is understood to include the number adjacent to the term "at least", and all subsequent numbers or integers that could logically be included, as clear from context. For example, the number of nucleotides in a nucleic acid molecule must be an integer. For example, "at least 18 nucleotides of a 21 nucleotide nucleic acid molecule" means that 18, 19, 20, or 21 nucleotides have the indicated property. When at least is present before a series of numbers or a range, it is understood that "at least" can modify each of the numbers in the series or range.
[0074] As used herein, "no more than" or "less than" is understood as the value adjacent to the phrase and logical lower values or integers, as logical from context, to zero. For example, a duplex with an overhang of "no more than 2 nucleotides" has a 2, 1, or 0 nucleotide overhang. When "no more than" is present before a series of numbers or a range, it is understood that "no more than" can modify each of the numbers in the series or range. As used herein, ranges include both the upper and lower limit.
[0075] As used herein, the term "*Homo sapiens* amyloid P component, serum", used interchangeably with the term "APCS," refers to the well-known gene and polypeptide, also known in the art as SAP. The term "APCS" includes human APCS, the amino acid and nucleotide sequence of which may be found in, for example, GenBank Accession No. NM_001639 (GI:206597534) (SEQ ID NO: 1); cynomolgus monkey APCS, the amino acid and nucleotide sequence of which may be found in, for example, GenBank Accession No. XM_005541312 (GI:982224943) (SEQ ID NO: 2); mouse APCS, the amino acid and nucleotide sequence of which may be found in, for example, GenBank Accession No. NM_011318 (GI:226958496) (SEQ ID NO: 3); and rat APCS, the amino acid and nucleotide sequence of which may be found in, for example, GenBank Accession No. NM_017170 (GI:148747487) (SEQ ID NO: 4).
[0076] The term "APCS," as used herein, also refers to naturally occurring DNA sequence variations of the APCS gene. Numerous sequence variations within the APCS gene have been identified and may be found at, for example, NCBI dbSNP and UniProt (see, e.g., http://www.ncbi.nlm.nih.gov/gene/325, the entire contents of which is incorporated herein by reference as of the date of filing this application.
[0077] As used herein, "target sequence" refers to a contiguous portion of the nucleotide sequence of an mRNA molecule formed during the transcription of a APCS gene, including mRNA that is a product of RNA processing of a primary transcription product. In one embodiment, the target portion of the sequence will be at least long enough to serve as a substrate for iRNA-directed cleavage at or near that portion of the nucleotide sequence of an mRNA molecule formed during the transcription of an APCS gene. In one embodiment, the target sequence is within the protein coding region of an APCS gene. In another embodiment, the target sequence is within the 3′ UTR of an APCS gene.
[0078] The target sequence may be from about 9-36 nucleotides in length, e.g., about 15-30 nucleotides in length. For example, the target sequence can be from about 15-30 nucleotides, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. In some embodiments, the target sequence is about 19 to about 30 nucleotides in length. In other embodiments, the target sequence is about 19 to about 25 nucleotides in length. In still other embodiments, the target sequence is about 19 to about 23 nucleotides in length. In some embodiments, the target sequence is about 21 to about 23 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the

invention.
[0079] As used herein, the term “strand comprising a sequence” refers to an oligonucleotide comprising a chain of nucleotides that is described by the sequence referred to using the standard nucleotide nomenclature.
[0080] A “nucleoside” is a base-sugar combination. The “nucleobase” (also known as “base”) portion of the nucleoside is normally a heterocyclic base moiety. “Nucleotides” are nucleosides that further include a phosphate group covalently linked to the sugar portion of the nucleoside. For those nucleosides that include a pentofuranosyl sugar, the phosphate group can be linked to the 2′, 3′ or 5′ hydroxyl moiety of the sugar.
[0081] “G,” “C,” “A,” “T” and “U” each generally stand for a nucleotide that contains guanine, cytosine, adenine, thymidine and uracil as a base, respectively. However, it will be understood that the term “ribonucleotide” or “nucleotide” can also refer to a modified nucleotide, as further detailed below, or a surrogate replacement moiety (see, e.g., Table 2). The skilled person is well aware that guanine, cytosine, adenine, and uracil can be replaced by other moieties without substantially altering the base pairing properties of an oligonucleotide comprising a nucleotide bearing such replacement moiety. For example, without limitation, a nucleotide comprising inosine as its base can base pair with nucleotides containing adenine, cytosine, or uracil. Hence, nucleotides containing uracil, guanine, or adenine can be replaced in the nucleotide sequences of dsRNA featured in the invention by a nucleotide containing, for example, inosine. In another example, adenine and cytosine anywhere in the oligonucleotide can be replaced with guanine and uracil, respectively to form G-U Wobble base pairing with the target mRNA. Sequences containing such replacement moieties are suitable for the compositions and methods featured in the invention.
[0082] “Polynucleotides,” also referred to as “oligonucleotides,” are formed through the covalent linkage of adjacent nucleosides to one another, to form a linear polymeric oligonucleotide. Within the polynucleotide structure, the phosphate groups are commonly referred to as forming the internucleoside linkages of the polynucleotide.
[0083] The terms “iRNA,” “RNAi agent,” “iRNA agent,” “RNA interference agent” as used interchangeably herein, refer to an agent that contains RNA as that term is defined herein, and which mediates the targeted cleavage of an RNA transcript via an RNA-induced silencing complex (RISC) pathway. iRNA directs the sequence-specific degradation of mRNA through a process known as RNA interference (RNAi). The iRNA modulates, e.g., inhibits, the expression of an APCS gene in a cell, e.g., a cell within a subject, such as a mammalian subject.
[0084] In one embodiment, an RNAi agent of the invention includes a single stranded RNAi that interacts with a target RNA sequence, e.g., an APCS target mRNA sequence, to direct the cleavage of the target RNA. Without wishing to be bound by theory it is believed that long double stranded RNA introduced into cells is broken down into double stranded short interfering RNAs (siRNAs) comprising a sense strand and an antisense strand by a Type III endonuclease known as Dicer (Sharp et al. (2001) *Genes Dev.* 15:485). Dicer, a ribonuclease-III-like enzyme, processes these dsRNA into 19-23 base pair short interfering RNAs with characteristic two base 3′ overhangs (Bernstein, et al., (2001) *Nature* 409:363). These siRNAs are then incorporated into an RNA-induced silencing complex (RISC) where one or more helicases unwind the siRNA duplex, enabling the complementary antisense strand to guide target recognition (Nykanen, et al., (2001) *Cell* 107:309). Upon binding to the appropriate target mRNA, one or more endonucleases within the RISC cleave the target to induce silencing (Elbashir, et al., (2001) *Genes Dev.* 15:188). Thus, in one aspect the invention relates to a single stranded RNA (ssRNA) (the antisense strand of an siRNA duplex) generated within a cell and which promotes the formation of a RISC complex to effect silencing of the target gene, i.e., an APCS gene. Accordingly, the term “siRNA” is also used herein to refer to an RNAi as described above.
[0085] In another embodiment, the RNAi agent may be a single-stranded RNA that is introduced into a cell or organism to inhibit a target mRNA. Single-stranded RNAi agents bind to the RISC

endonuclease, Argonaute 2, which then cleaves the target mRNA. The single-stranded siRNAs are generally 15-30 nucleotides and are chemically modified. The design and testing of single-stranded RNAs are described in U.S. Pat. No. 8,101,348 and in Lima et al., (2012) *Cell* 150: 883-894, the entire contents of each of which are hereby incorporated herein by reference. Any of the antisense nucleotide sequences described herein may be used as a single-stranded siRNA as described herein or as chemically modified by the methods described in Lima et al., (2012) *Cell* 150:883-894.

[0086] In another embodiment, an "iRNA" for use in the compositions, uses, and methods of the invention is a double stranded RNA and is referred to herein as a "double stranded RNAi agent," "double stranded RNA (dsRNA) molecule," "dsRNA agent," or "dsRNA". The term "dsRNA" refers to a complex of ribonucleic acid molecules, having a duplex structure comprising two anti-parallel and substantially complementary nucleic acid strands, referred to as having "sense" and "antisense" orientations with respect to a target RNA, i.e., an APCS gene. In some embodiments of the invention, a double stranded RNA (dsRNA) triggers the degradation of a target RNA, e.g., an mRNA, through a post-transcriptional gene-silencing mechanism referred to herein as RNA interference or RNAi.

[0087] In general, the majority of nucleotides of each strand of a dsRNA molecule are ribonucleotides, but as described in detail herein, each or both strands can also include one or more non-ribonucleotides, e.g., a deoxyribonucleotide and/or a modified nucleotide. In addition, as used in this specification, an "RNAi agent" may include ribonucleotides with chemical modifications. Such modifications may include all types of modifications disclosed herein or known in the art. Ant such modification, as used in an RNAi agent are encompassed by iRNA for the purposes of the specification and claims. In some embodiments, an RNAi agent includes substantial modifications at multiple nucleotides.

[0088] As used herein, the term "modified nucleotide" refers to a nucleotide having, independently, a modified sugar moiety, a modified internucleotide linkage, and/or a modified nucleobase. Thus, the term modified nucleotide encompasses substitutions, additions or removal of, e.g., a functional group or atom, to internucleoside linkages, sugar moieties, or nucleobases. The modifications suitable for use in the agents of the invention include all types of modifications disclosed herein or known in the art. Any such modifications, as used in a siRNA type molecule, are encompassed by "RNAi agent" for the purposes of this specification and claims.

[0089] The duplex region may be of any length that permits specific degradation of a desired target RNA through a RISC pathway, and may range from about 9 to 36 base pairs in length, e.g., about 15-30 base pairs in length, for example, about 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, or 36 base pairs in length, such as about 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the invention.

[0090] The two strands forming the duplex structure may be different portions of one larger RNA molecule, or they may be separate RNA molecules. Where the two strands are part of one larger molecule, and therefore are connected by an uninterrupted chain of nucleotides between the 3′-end of one strand and the 5′-end of the respective other strand forming the duplex structure, the connecting RNA chain is referred to as a "hairpin loop." A hairpin loop can comprise at least one unpaired nucleotide. In some embodiments, the hairpin loop can comprise at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 20, at least 23 or more unpaired nucleotides. In some embodiments, the hairpin loop can be 10 or fewer nucleotides. In some embodiments, the hairpin loop can be 8 or fewer unpaired nucleotides. In some embodiments, the hairpin loop can be 4-10 unpaired nucleotides. In some embodiments, the hairpin loop can be 4-

8 nucleotides.
[0091] Where the two substantially complementary strands of a dsRNA are comprised of separate RNA molecules, those molecules need not, but can be covalently connected. Where the two strands are connected covalently by means other than an uninterrupted chain of nucleotides between the 3′-end of one strand and the 5′-end of the respective other strand forming the duplex structure, the connecting structure is referred to as a "linker." The RNA strands may have the same or a different number of nucleotides. The maximum number of base pairs is the number of nucleotides in the shortest strand of the dsRNA minus any overhangs that are present in the duplex. In addition to the duplex structure, an RNAi may comprise one or more nucleotide overhangs. In one embodiment of the RNAi agent, at least one strand comprises a 3′ overhang of at least 1 nucleotide. In another embodiment, at least one strand comprises a 3′ overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In other embodiments, at least one strand of the RNAi agent comprises a 5′ overhang of at least 1 nucleotide. In certain embodiments, at least one strand comprises a 5′ overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In still other embodiments, both the 3′ and the 5′ end of one strand of the RNAi agent comprise an overhang of at least 1 nucleotide.
[0092] In one embodiment, an RNAi agent of the invention is a dsRNA agent, each strand of which comprises 19-23 nucleotides that interacts with a target RNA sequence, i.e., an APCS target mRNA sequence, to direct the cleavage of the target RNA. Without wishing to be bound by theory, long double stranded RNA introduced into cells is broken down into siRNA by a Type III endonuclease known as Dicer (Sharp et al. (2001) *Genes Dev.* 15:485). Dicer, a ribonuclease-III-like enzyme, processes the dsRNA into 19-23 base pair short interfering RNAs with characteristic two base 3′ overhangs (Bernstein, et al., (2001) *Nature* 409:363). The siRNAs are then incorporated into an RNA-induced silencing complex (RISC) where one or more helicases unwind the siRNA duplex, enabling the complementary antisense strand to guide target recognition (Nykanen, et al., (2001) *Cell* 107:309). Upon binding to the appropriate target mRNA, one or more endonucleases within the RISC cleave the target to induce silencing (Elbashir, et al., (2001) *Genes Dev.* 15:188). In one embodiment, an RNAi agent of the invention is a dsRNA of 24-30 nucleotides that interacts with a target RNA sequence, i.e., an APCS target mRNA sequence, to direct the cleavage of the target RNA.
[0093] As used herein, the term "nucleotide overhang" refers to at least one unpaired nucleotide that protrudes from the duplex structure of an iRNA, e.g., a double-stranded RNA (dsRNA). For example, when a 3′-end of one strand of a dsRNA extends beyond the 5′-end of the other strand, or vice versa, there is a nucleotide overhang. A dsRNA can comprise an overhang of at least one nucleotide; alternatively the overhang can comprise at least two nucleotides, at least three nucleotides, at least four nucleotides, at least five nucleotides or more. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5′-end, 3′-end or both ends of either an antisense or sense strand of a dsRNA. In one embodiment of the dsRNA, at least one strand comprises a 3′ overhang of at least 1 nucleotide. In another embodiment, at least one strand comprises a 3′ overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In other embodiments, at least one strand of the RNAi agent comprises a 5′ overhang of at least 1 nucleotide. In certain embodiments, at least one strand comprises a 5′ overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In still other embodiments, both the 3′ and the 5′ end of one strand of the RNAi agent comprise an overhang of at least 1 nucleotide.
[0094] In certain embodiments, the antisense strand of a dsRNA has a 1-10 nucleotide, e.g., 0-3, 1-3, 2-4, 2-5, 4-10, 5-10, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3′-end and/or the 5′-end. In one embodiment, the sense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4,

5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3′-end and/or the 5′-end. In another embodiment, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate.
[0095] In certain embodiments, the overhang on the sense strand or the antisense strand, or both, can include extended lengths longer than 10 nucleotides, e.g., 1-30 nucleotides, 2-30 nucleotides, 10-30 nucleotides, or 10-15 nucleotides in length. In certain embodiments, an extended overhang is on the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 3′ end of the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 5′ end of the sense strand of the duplex. In certain embodiments, an extended overhang is on the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 3′ end of the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 5′ end of the antisense strand of the duplex. In certain embodiments, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate. In certain embodiments, the overhang includes a self-complementary portion such that the overhang is capable of forming a hairpin structure that is stable under physiological conditions.
[0096] “Blunt” or “blunt end” means that there are no unpaired nucleotides at that end of the double stranded RNAi agent, i.e., no nucleotide overhang. A “blunt ended” RNAi agent is a dsRNA that is double stranded over its entire length, i.e., no nucleotide overhang at either end of the molecule. The RNAi agents of the invention include RNAi agents with nucleotide overhangs at one end (i.e., agents with one overhang and one blunt end) or with nucleotide overhangs at both ends.
[0097] The term “antisense strand” or “guide strand” refers to the strand of an iRNA, e.g., a dsRNA, which includes a region that is substantially complementary to a target sequence, e.g., an APCS mRNA. As used herein, the term “region of complementarity” refers to the region on the antisense strand that is substantially complementary to a sequence, for example a target sequence, e.g., an APCS nucleotide sequence, as defined herein. Where the region of complementarity is not fully complementary to the target sequence, the mismatches can be in the internal or terminal regions of the molecule. Generally, the most tolerated mismatches are in the terminal regions, e.g., within 5, 4, 3, 2, or 1 nucleotides of the 5′- and/or 3′-terminus of the iRNA. In one embodiment, a double stranded RNAi agent of the invention includes a nucleotide mismatch in the antisense strand. In another embodiment, a double stranded RNAi agent of the invention includes a nucleotide mismatch in the sense strand. In one embodiment, the nucleotide mismatch is, for example, within 5, 4, 3, 2, or 1 nucleotides from the 3′-terminus of the iRNA. In another embodiment, the nucleotide mismatch is, for example, in the 3′-terminal nucleotide of the iRNA.
[0098] The term “sense strand” or “passenger strand” as used herein, refers to the strand of an iRNA that includes a region that is substantially complementary to a region of the antisense strand as that term is defined herein.
[0099] As used herein, the term “cleavage region” refers to a region that is located immediately adjacent to the cleavage site. The cleavage site is the site on the target at which cleavage occurs. In some embodiments, the cleavage region comprises three bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage region comprises two bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage site specifically occurs at the site bound by nucleotides 10 and 11 of the antisense strand, and the cleavage region comprises nucleotides 11, 12 and 13.
[0100] As used herein, and unless otherwise indicated, the term “complementary,” when used to describe a first nucleotide sequence in relation to a second nucleotide sequence, refers to the ability of an oligonucleotide or polynucleotide comprising the first nucleotide sequence to hybridize and form a duplex structure under certain conditions with an oligonucleotide or polynucleotide comprising the second nucleotide sequence, as will be understood by the skilled person. Such conditions can, for example, be stringent conditions, where stringent conditions can include: 400 mM NaCl, 40 mM PIPES pH 6.4, 1 mM EDTA, 50° C. or 70° C. for 12-16 hours followed by washing (see, e.g., “Molecular Cloning: A Laboratory Manual, Sambrook, et al. (1989) Cold Spring

Harbor Laboratory Press). Other conditions, such as physiologically relevant conditions as can be encountered inside an organism, can apply. The skilled person will be able to determine the set of conditions most appropriate for a test of complementarity of two sequences in accordance with the ultimate application of the hybridized nucleotides.

[0101] Complementary sequences within an iRNA, e.g., within a dsRNA as described herein, include base-pairing of the oligonucleotide or polynucleotide comprising a first nucleotide sequence to an oligonucleotide or polynucleotide comprising a second nucleotide sequence over the entire length of one or both nucleotide sequences. Such sequences can be referred to as "fully complementary" with respect to each other herein. However, where a first sequence is referred to as "substantially complementary" with respect to a second sequence herein, the two sequences can be fully complementary, or they can form one or more, but generally not more than 5, 4, 3 or 2 mismatched base pairs upon hybridization for a duplex up to 30 base pairs, while retaining the ability to hybridize under the conditions most relevant to their ultimate application, e.g., inhibition of gene expression via a RISC pathway. However, where two oligonucleotides are designed to form, upon hybridization, one or more single stranded overhangs, such overhangs shall not be regarded as mismatches with regard to the determination of complementarity. For example, a dsRNA comprising one oligonucleotide 21 nucleotides in length and another oligonucleotide 23 nucleotides in length, wherein the longer oligonucleotide comprises a sequence of 21 nucleotides that is fully complementary to the shorter oligonucleotide, can yet be referred to as "fully complementary" for the purposes described herein.

[0102] "Complementary" sequences, as used herein, can also include, or be formed entirely from, non-Watson-Crick base pairs and/or base pairs formed from non-natural and modified nucleotides, in so far as the above requirements with respect to their ability to hybridize are fulfilled. Such non-Watson-Crick base pairs include, but are not limited to, G:U Wobble or Hoogstein base pairing.

[0103] The terms "complementary," "fully complementary" and "substantially complementary" herein can be used with respect to the base matching between the sense strand and the antisense strand of a dsRNA, or between the antisense strand of an iRNA agent and a target sequence, as will be understood from the context of their use.

[0104] As used herein, a polynucleotide that is "substantially complementary to at least part of" a messenger RNA (mRNA) refers to a polynucleotide that is substantially complementary to a contiguous portion of the mRNA of interest (i.e., an APCS gene). For example, a polynucleotide is complementary to at least a part of an APCS mRNA if the sequence is substantially complementary to a non-interrupted portion of an mRNA encoding APCS.

[0105] Accordingly, in some embodiments, the sense strand polynucleotides and the antisense polynucleotides disclosed herein are fully complementary to the target APCS gene sequence.

[0106] In one embodiment, the antisense polynucleotides disclosed herein are fully complementary to the target APCS sequence. In other embodiments, the antisense polynucleotides disclosed herein are substantially complementary to the target APCS sequence and comprise a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to the equivalent region of the nucleotide sequence of any one of SEQ ID NO:1, or a fragment of any one of SEQ ID NO:1, such as about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about % 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

[0107] In other embodiments, the antisense polynucleotides disclosed herein are substantially complementary to the target APCS sequence and comprise a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to any one of the sense strand nucleotide sequences in any one of Tables 3A, 3B, 4A, 4B, 6, and 7, or a fragment of any one of the sense strand nucleotide sequences in any one of Tables 3A, 3B, 4A, 4B, 6, and 7, such as about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about % 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

[0108] In one embodiment, an RNAi agent of the invention includes a sense strand that is substantially complementary to an antisense polynucleotide which, in turn, is complementary to a target APCS sequence and comprises a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to any one of the sense strand nucleotide sequences in any one of Tables 3A, 3B, 4A, 4B, 6, and 7, or a fragment of any one of the sense strand nucleotide sequences in any one of Tables 3A, 3B, 4A, 4B, 6, and 7, such as about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about % 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

[0109] In one aspect of the invention, an agent for use in the methods and compositions of the invention is a single-stranded antisense RNA molecule that inhibits a target mRNA via an antisense inhibition mechanism. The single-stranded antisense RNA molecule is complementary to a sequence within the target mRNA. The single-stranded antisense oligonucleotides can inhibit translation in a stoichiometric manner by base pairing to the mRNA and physically obstructing the translation machinery, see Dias, N. et al., (2002) *Mol Cancer Ther* 1:347-355. The single-stranded antisense RNA molecule may be about 15 to about 30 nucleotides in length and have a sequence that is complementary to a target sequence. For example, the single-stranded antisense RNA molecule may comprise a sequence that is at least about 15, 16, 17, 18, 19, 20, or more contiguous nucleotides from any one of the antisense sequences described herein.

[0110] As used herein, the term “APCS-associated disease” is a disease or disorder that is caused by, or associated with the expression of the APCS gene. The term “APCS-associated disease” includes a disease, disorder or condition that would benefit from reduction in APCS expression. An APCS-associated disease includes amyloid-associated diseases, e.g., diseases that are characterized by formation of amyloid deposits. Exemplary amyloid-associated diseases include, but are not limited to, amyloidosis, Alzheimer's disease; diabetes mellitus type 2; Parkinson's disease; transmissible spongiform encephalopathy (such as bovine spongiform encephalopathy); fatal familial insomnia; Huntington's disease; medullary carcinoma of the thyroid; cardiac arrhythmias; isolated atrial amyloidosis; rheumatoid arthritis; aortic medial amyloid; prolactinoma; familial amyloid polyneuropathy; lattice corneal dystrophy; cerebral amyloid angiopathy; cerebral amyloid angiopathy (Icelandic type); sporadic inclusion body myositis.

[0111] In one embodiment, the amyloid-associated disease is amyloidosis, e.g., primary (systemic AL) amyloidosis, secondary (systemic AA) amyloidosis, dialysis-related amyloidosis (DRA), familial (hereditary FA) amyloidosis, senile systemic amyloidosis (SSA) and organ-specific amyloidosis.

[0112] An APCS-associated disease may also be a cardiovascular disease, e.g., a coronary atherosclerotic heard disease.

[0113] “Therapeutically effective amount,” as used herein, is intended to include the amount of an RNAi agent that, when administered to a subject having a APCS-associated disease, is sufficient to effect treatment of the disease (e.g., by diminishing, ameliorating or maintaining the existing disease or one or more symptoms of disease). The “therapeutically effective amount” may vary depending on the RNAi agent or antibody, or antigen-binding fragment thereof, how the agent is administered, the disease and its severity and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the subject to be treated.

[0114] “Prophylactically effective amount,” as used herein, is intended to include the amount of an iRNA agent that, when administered to a subject having a APCS-associated disease but not yet (or currently) experiencing or displaying symptoms of the disease, and/or a subject at risk of developing a APCS-associated disease, e.g., a subject having multiple myeloma, a subject on kidney dialysis or a subject with family history of amyloidosis, is sufficient to prevent or ameliorate the disease or one or more symptoms of the disease. Ameliorating the disease includes slowing the course of the disease or reducing the severity of later-developing disease. The "prophylactically effective amount" may vary depending on the iRNA agent, how the agent is administered, the degree of risk of disease, and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the patient to be treated.

[0115] A "therapeutically effective amount" or "prophylactically effective amount" also includes an amount of an RNAi agent that produces some desired local or systemic effect at a reasonable benefit/risk ratio applicable to any treatment. iRNA agents employed in the methods of the present invention may be administered in a sufficient amount to produce a reasonable benefit/risk ratio applicable to such treatment.

[0116] As used herein, a "subject" is an animal, such as a mammal, including a primate (such as a human, a non-human primate, e.g., a monkey, and a chimpanzee), a non-primate (such as a cow, a pig, a camel, a llama, a horse, a goat, a rabbit, a sheep, a hamster, a guinea pig, a cat, a dog, a rat, a mouse, a horse, and a whale), or a bird (e.g., a duck or a goose). In an embodiment, the subject is a human, such as a human being treated or assessed for a disease, disorder or condition that would benefit from reduction in APCS expression; a human at risk for a disease, disorder or condition that would benefit from reduction in APCS expression; a human having a disease, disorder or condition that would benefit from reduction in APCS expression; and/or human being treated for a disease, disorder or condition that would benefit from reduction in APCS expression as described herein.

II. iRNAs of the Invention

[0117] The present invention provides iRNAs which inhibit the expression of an APCS gene. In one embodiment, the iRNA agent includes double stranded ribonucleic acid (dsRNA) molecules for inhibiting the expression of an APCS gene in a cell, such as a cell within a subject, e.g., a mammal, such as a human having a APCS-associated disease as described herein, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease. The dsRNA includes an antisense strand having a region of complementarity which is complementary to at least a part of an mRNA formed in the expression of a target gene, i.e., APCS gene. The region of complementarity is about 30 nucleotides or less in length (e.g., about 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, or 18 nucleotides or less in length). Upon contact with a cell expressing the target gene, the iRNA selectively inhibits the expression of the target gene (e.g., a human, a primate, a non-primate, or a bird APCS gene) by at least about 10%, by at least 30%, preferably at least 50% as assayed by, for example, a PCR or branched DNA (bDNA)-based method, or by a protein-based method, such as by immunofluorescence analysis, using, for example, western blotting or flow cytometric techniques. In preferred embodiments, inhibition of expression is determined by the luciferase assay provided in the examples. For in vitro assessment of activity, percent inhibition is determined using the methods provided in Example 2 at a single dose at a 10 nM duplex final concentration. For in vivo studies, the level after treatment can be compared to, for example, an appropriate historical control or a pooled population sample control to determine the level of reduction, e.g., when a baseline value is no available for the subject.

[0118] An RNAi agent is a dsRNA that includes two RNA strands that are complementary and hybridize to form a duplex structure under conditions in which the dsRNA will be used. One strand of a dsRNA (the antisense strand) includes a region of complementarity that is substantially complementary, and generally fully complementary, to a target sequence. The target sequence can be derived from the sequence of an mRNA formed during the expression of an APCS gene. The other strand (the sense strand) includes a region that is complementary to the antisense strand, such that the two strands hybridize and form a duplex structure when combined under suitable conditions. As described elsewhere herein and as known in the art, the complementary sequences of a dsRNA can also be contained as self-complementary regions of a single nucleic acid molecule, as opposed to being on separate oligonucleotides.

[0119] Generally, the duplex structure is between 15 and 30 base pairs in length, e.g., between, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-

29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the invention.

[0120] Similarly, the region of complementarity to the target sequence is between 15 and 30 nucleotides in length, e.g., between 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the invention.

[0121] In some embodiments, the sense and antisense strands of the dsRNA are each independently about 15 to about 30 nucleotides in length, or about 25 to about 30 nucleotides in length, e.g., each strand is independently between 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. In one embodiment, an RNAi agent of the invention is a dsRNA of 24-30 nucleotides that interacts with a target RNA sequence, i.e., an APCS target mRNA sequence, to direct the cleavage of the target RNA. In general, the dsRNA is long enough to serve as a substrate for the Dicer enzyme. For example, it is well-known in the art that dsRNAs longer than about 21-23 nucleotides in length may serve as substrates for Dicer. As the ordinarily skilled person will also recognize, the region of an RNA targeted for cleavage will most often be part of a larger RNA molecule, often an mRNA molecule. Where relevant, a “part” of an mRNA target is a contiguous sequence of an mRNA target of sufficient length to allow it to be a substrate for RNAi-directed cleavage (i.e., cleavage through a RISC pathway).

[0122] One of skill in the art will also recognize that the duplex region is a primary functional portion of a dsRNA, e.g., a duplex region of about 9 to 36 base pairs, e.g., about 10-36, 11-36, 12-36, 13-36, 14-36, 15-36, 9-35, 10-35, 11-35, 12-35, 13-35, 14-35, 15-35, 9-34, 10-34, 11-34, 12-34, 13-34, 14-34, 15-34, 9-33, 10-33, 11-33, 12-33, 13-33, 14-33, 15-33, 9-32, 10-32, 11-32, 12-32, 13-32, 14-32, 15-32, 9-31, 10-31, 11-31, 12-31, 13-32, 14-31, 15-31, 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs. Thus, in one embodiment, to the extent that it becomes processed to a functional duplex, of e.g., 15-30 base pairs, that targets a desired RNA for cleavage, an RNA molecule or complex of RNA molecules having a duplex region greater than 30 base pairs is a dsRNA. Thus, an ordinarily skilled artisan will recognize that in one embodiment, a miRNA is a dsRNA. In another embodiment, a dsRNA is not a naturally occurring miRNA. In another embodiment, an iRNA agent useful to target APCS expression is not generated in the target cell by cleavage of a larger dsRNA.

[0123] A RNAi agent as described herein can further include one or more single-stranded nucleotide overhangs e.g., 1, 2, 3, or 4 nucleotides. RNAi agent having at least one nucleotide overhang can have unexpectedly superior inhibitory properties relative to their blunt-ended counterparts. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5′-end, 3′-end or both ends of either an antisense or sense strand of the RNAi agent. In certain embodiments, longer, extended overhangs are possible.

[0124] A dsRNA can be synthesized by standard methods known in the art as further discussed below, e.g., by use of an automated DNA synthesizer, such as are commercially available from, for example, Biosearch, Applied Biosystems, Inc.

[0125] RNAi agents of the invention may be prepared using a two-step procedure. First, the individual strands of the double stranded RNA molecule are prepared separately. Then, the component strands are annealed. The individual strands of the iRNA agent can be prepared using solution-phase or solid-phase organic synthesis or both. Organic synthesis offers the advantage that the oligonucleotide strands comprising unnatural or modified nucleotides can be easily prepared. Single-stranded oligonucleotides of the invention can be prepared using solution-phase or solid-phase organic synthesis or both.

[0126] In one aspect, an RNAi agent of the invention includes at least two nucleotide sequences, a sense sequence and an anti-sense sequence.

[0127] In one embodiment, the sense strand is selected from the group of sequences provided in any one of Tables 3A, 3B, 4A, 4B, 6, and 7, and the corresponding antisense strand of the sense strand is selected from the group of sequences of any one of Tables 3A, 3B, 4A, 4B, 6, and 7. In this aspect, one of the two sequences is complementary to the other of the two sequences, with one of the sequences being substantially complementary to a sequence of an mRNA generated in the expression of an APCS gene. As such, in this aspect, a dsRNA will include two oligonucleotides, where one oligonucleotide is described as the sense strand in any one of Tables 3A, 3B, 4A, 4B, 6, and 7 and the second oligonucleotide is described as the corresponding antisense strand of the sense strand in any one of Tables 3A, 3B, 4A, 4B, 6, and 7. In one embodiment, the substantially complementary sequences of the RNAi agent are contained on separate oligonucleotides. In another embodiment, the substantially complementary sequences of the RNAi agent are contained on a single oligonucleotide.

[0128] It will be understood that, although some of the sequences in Tables 3A, 3B, 4A, 4B, 6, and 7 are described as modified and/or conjugated sequences, the RNA of the iRNA of the invention e.g., a dsRNA of the invention, may comprise any one of the sequences set forth in Tables 3A, 3B, 4A, 4B, 6, and 7 that is un-modified, un-conjugated, and/or modified and/or conjugated differently than described therein.

[0129] The skilled person is well aware that dsRNAs having a duplex structure of between about 20 and 23 base pairs, e.g., 21, base pairs have been hailed as particularly effective in inducing RNA interference (Elbashir et al., *EMBO* 2001, 20:6877-6888). However, others have found that shorter or longer RNA duplex structures can also be effective (Chu and Rana (2007) *RNA* 14:1714-1719; Kim et al. (2005) *Nat Biotech* 23:222-226). In the embodiments described above, by virtue of the nature of the oligonucleotide sequences provided in any one of Tables 3A, 3B, 4A, 4B, 6, and 7, the RNAi agents described herein can include at least one strand of a length of minimally 21 nucleotides. It can be reasonably expected that shorter duplexes having one of the sequences of any one of Tables 3A, 3B, 4A, 4B, 6, and 7 minus only a few nucleotides on one or both ends can be similarly effective as compared to the dsRNAs described above. Hence, dsRNAs having a sequence of at least 15, 16, 17, 18, 19, 20, or more contiguous nucleotides derived from one of the sequences of any one of Tables 3A, 3B, 4A, 4B, 6, and 7, and differing in their ability to inhibit the expression of the target gene by not more than about 5, 10, 15, 20, 25, or 30% inhibition from a dsRNA comprising the full sequence, are contemplated to be within the scope of the present invention.

[0130] In addition, the RNAs provided in any one of Tables 3A, 3B, 4A, 4B, 6, and 7 identify a site(s) in an APCS transcript that is susceptible to RISC-mediated cleavage. As such, the present invention further features iRNAs that target within one of these sites. As used herein, an iRNA is said to target within a particular site of an RNA transcript if the iRNA promotes cleavage of the transcript anywhere within that particular site. Such an iRNA will generally include at least about 15 contiguous nucleotides from one of the sequences provided in any one of Tables 3A, 3B, 4A, 4B, 6, and 7 coupled to additional nucleotide sequences taken from the region contiguous to the selected sequence in the target gene.

[0131] While a target sequence is generally about 15-30 nucleotides in length, there is wide variation in the suitability of particular sequences in this range for directing cleavage of any given target RNA. Various software packages and the guidelines set out herein provide guidance for the identification of optimal target sequences for any given gene target, but an empirical approach can also be taken in which a “window” or “mask” of a given size (as a non-limiting example, 21 nucleotides) is literally or figuratively (including, e.g., in silico) placed on the target RNA sequence to identify sequences in the size range that can serve as target sequences. By moving the sequence “window” progressively one nucleotide upstream or downstream of an initial target sequence location, the next potential target sequence can be identified, until the complete set of possible sequences is identified for any given target size selected. This process, coupled with systematic synthesis and testing of the identified sequences (using assays as described herein or as known in the art) to identify those sequences that perform optimally can identify those RNA sequences that, when targeted with an RNAi agent, mediate the best inhibition of target gene expression. Thus, while the sequences identified, for example, in any one of Tables 3A, 3B, 4A, 4B, 6, and 7 represent effective target sequences, it is contemplated that further optimization of inhibition efficiency can be achieved by progressively “walking the window” one nucleotide upstream or downstream of the given sequences to identify sequences with equal or better inhibition characteristics.

[0132] Further, it is contemplated that for any sequence identified, e.g., in any one of Tables 3A, 3B, 4A, 4B, 6, and 7, further optimization could be achieved by systematically either adding or removing nucleotides to generate longer or shorter sequences and testing those sequences generated by walking a window of the longer or shorter size up or down the target RNA from that point. Again, coupling this approach to generating new candidate targets with testing for effectiveness of iRNAs based on those target sequences in an inhibition assay as known in the art and/or as described herein can lead to further improvements in the efficiency of inhibition. Further still, such optimized sequences can be adjusted by, e.g., the introduction of modified nucleotides as described herein or as known in the art, addition or changes in overhang, or other modifications as known in the art and/or discussed herein to further optimize the molecule (e.g., increasing serum stability or circulating half-life, increasing thermal stability, enhancing transmembrane delivery, targeting to a particular location or cell type, increasing interaction with silencing pathway enzymes, increasing release from endosomes) as an expression inhibitor.

[0133] An iRNA as described herein can contain one or more mismatches to the target sequence. In one embodiment, an iRNA as described herein contains no more than 3 mismatches. If the antisense strand of the iRNA contains mismatches to a target sequence, it is preferable that the area of mismatch is not located in the center of the region of complementarity. If the antisense strand of the iRNA contains mismatches to the target sequence, it is preferable that the mismatch be restricted to be within the last 5 nucleotides from either the 5′- or 3′-end of the region of complementarity. For example, for a 23 nucleotide iRNA agent the strand which is complementary to a region of, e.g., an APCS gene, generally does not contain any mismatch within the central 13 nucleotides. The methods described herein or methods known in the art can be used to determine whether an iRNA containing a mismatch to a target sequence is effective in inhibiting the expression of a target gene, e.g., an APCS gene. Consideration of the efficacy of iRNAs with mismatches in inhibiting expression of a target gene is important, especially if the particular region of complementarity in a target gene is known to have polymorphic sequence variation within the population.

III. Modified iRNAs of the Invention

[0134] In one embodiment, the RNA of the iRNA of the invention e.g., a dsRNA, is un-modified, and does not comprise, e.g., chemical modifications and/or conjugations known in the art and described herein. In another embodiment, the RNA of an iRNA of the invention, e.g., a dsRNA, is chemically modified to enhance stability or other beneficial characteristics. In certain embodiments of the invention, substantially all of the nucleotides of an iRNA of the invention are modified. In other embodiments of the invention, all of the nucleotides of an iRNA of the invention are modified iRNAs of the invention in which "substantially all of the nucleotides are modified" are largely but not wholly modified and can include not more than 5, 4, 3, 2, or 1 unmodified nucleotides.

[0135] The nucleic acids featured in the invention can be synthesized and/or modified by methods well established in the art, such as those described in "Current protocols in nucleic acid chemistry," Beaucage, S. L. et al. (Edrs.), John Wiley & Sons, Inc., New York, NY, USA, which is hereby incorporated herein by reference. Modifications include, for example, end modifications, e.g., 5′-end modifications (phosphorylation, conjugation, inverted linkages) or 3′-end modifications (conjugation, DNA nucleotides, inverted linkages, etc.); base modifications, e.g., replacement with stabilizing bases, destabilizing bases, or bases that base pair with an expanded repertoire of partners, removal of bases (abasic nucleotides), or conjugated bases; sugar modifications (e.g., at the 2′-position or 4′-position) or replacement of the sugar; and/or backbone modifications, including modification or replacement of the phosphodiester linkages. Specific examples of iRNA compounds useful in the embodiments described herein include, but are not limited to RNAs containing modified backbones or no natural internucleoside linkages. RNAs having modified backbones include, among others, those that do not have a phosphorus atom in the backbone. For the purposes of this specification, and as sometimes referenced in the art, modified RNAs that do not have a phosphorus atom in their internucleoside backbone can also be considered to be oligonucleosides. In some embodiments, a modified iRNA will have a phosphorus atom in its internucleoside backbone.

[0136] Modified RNA backbones include, for example, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates including 3′-alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates including 3′-amino phosphoramidate and aminoalkylphosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates having normal 3′-5′ linkages, 2′-5′-linked analogs of these, and those having inverted polarity wherein the adjacent pairs of nucleoside units are linked 3′-5′ to 5′-3′ or 2′-5′ to 5′-2′. Various salts, mixed salts and free acid forms are also included. In some embodiments of the invention, the dsRNA agents of the invention are in a free acid form. In other embodiments of the invention, the dsRNA agents of the invention are in a salt form. In one embodiment, the dsRNA agents of the invention are in a sodium salt form. In certain embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for substantially all of the phosphodiester and/or phosphorothiotate groups present in the agent. Agents in which substantially all of the phosphodiester and/or phosphorothioate linkages have a sodium counterion include not more than 5, 4, 3, 2, or 1 phosphodiester and/or phosphorothioate linkages without a sodium counterion. In some embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for all of the phosphodiester and/or phosphorothiotate groups present in the agent.

[0137] Representative U.S. patents that teach the preparation of the above phosphorus-containing linkages include, but are not limited to, U.S. Pat. Nos. 3,687,808; 4,469,863; 4,476,301; 5,023,243; 5,177,195; 5,188,897; 5,264,423; 5,276,019; 5,278,302; 5,286,717; 5,321,131; 5,399,676; 5,405,939; 5,453,496; 5,455,233; 5,466,677; 5,476,925; 5,519,126; 5,536,821; 5,541,316; 5,550,111; 5,563,253; 5,571,799; 5,587,361; 5,625,050; 6,028,188; 6,124,445; 6,160,109; 6,169,170; 6,172,209; 6,239,265; 6,277,603; 6,326,199; 6,346,614; 6,444,423; 6,531,590; 6,534,639; 6,608,035; 6,683,167; 6,858,715; 6,867,294; 6,878,805; 7,015,315; 7,041,816; 7,273,933; 7,321,029; and U.S. Pat. No. RE39,464, the entire contents of each of which are hereby

incorporated herein by reference.
[0138] Modified RNA backbones that do not include a phosphorus atom therein have backbones that are formed by short chain alkyl or cycloalkyl internucleoside linkages, mixed heteroatoms and alkyl or cycloalkyl internucleoside linkages, or one or more short chain heteroatomic or heterocyclic internucleoside linkages. These include those having morpholino linkages (formed in part from the sugar portion of a nucleoside); siloxane backbones; sulfide, sulfoxide and sulfone backbones; formacetyl and thioformacetyl backbones; methylene formacetyl and thioformacetyl backbones; alkene containing backbones; sulfamate backbones; methyleneimino and methylenehydrazino backbones; sulfonate and sulfonamide backbones; amide backbones; and others having mixed N, O, S and CH.sub.2 component parts.
[0139] Representative U.S. patents that teach the preparation of the above oligonucleosides include, but are not limited to, U.S. Pat. Nos. 5,034,506; 5,166,315; 5,185,444; 5,214,134; 5,216,141; 5,235,033; 5,64,562; 5,264,564; 5,405,938; 5,434,257; 5,466,677; 5,470,967; 5,489,677; 5,541,307; 5,561,225; 5,596,086; 5,602,240; 5,608,046; 5,610,289; 5,618,704; 5,623,070; 5,663,312; 5,633,360; 5,677,437; and, 5,677,439, the entire contents of each of which are hereby incorporated herein by reference.
[0140] In other embodiments, suitable RNA mimetics are contemplated for use in iRNAs, in which both the sugar and the internucleoside linkage, i.e., the backbone, of the nucleotide units are replaced with novel groups. The base units are maintained for hybridization with an appropriate nucleic acid target compound. One such oligomeric compound, an RNA mimetic that has been shown to have excellent hybridization properties, is referred to as a peptide nucleic acid (PNA). In PNA compounds, the sugar backbone of an RNA is replaced with an amide containing backbone, in particular an aminoethylglycine backbone. The nucleobases are retained and are bound directly or indirectly to aza nitrogen atoms of the amide portion of the backbone. Representative U.S. patents that teach the preparation of PNA compounds include, but are not limited to, U.S. Pat. Nos. 5,539,082; 5,714,331; and 5,719,262, the entire contents of each of which are hereby incorporated herein by reference. Additional PNA compounds suitable for use in the iRNAs of the invention are described in, for example, in Nielsen et al., *Science*, 1991, 254, 1497-1500.
[0141] Some embodiments featured in the invention include RNAs with phosphorothioate backbones and oligonucleosides with heteroatom backbones, and in particular —CH.sub.2—NH—CH.sub.2—, —CH.sub.2—N(CH.sub.3)—O—CH.sub.2— [known as a methylene (methylimino) or MMI backbone], —CH.sub.2—O—N(CH.sub.3)—CH.sub.2—, —CH.sub.2—N(CH.sub.3)—N(CH.sub.3)—CH.sub.2— and —N(CH.sub.3)—CH.sub.2—CH.sub.2— [wherein the native phosphodiester backbone is represented as —O—P—O—CH.sub.2—] of the above-referenced U.S. Pat. No. 5,489,677, and the amide backbones of the above-referenced U.S. Pat. No. 5,602,240. In some embodiments, the RNAs featured herein have morpholino backbone structures of the above-referenced U.S. Pat. No. 5,034,506.
[0142] Modified RNAs can also contain one or more substituted sugar moieties. The iRNAs, e.g., dsRNAs, featured herein can include one of the following at the 2′-position: OH; F; O—, S—, or N-alkyl; O—, S—, or N-alkenyl; O-, S- or N-alkynyl; or O-alkyl-O-alkyl, wherein the alkyl, alkenyl and alkynyl can be substituted or unsubstituted C.sub.1 to C.sub.10 alkyl or C.sub.2 to C.sub.10 alkenyl and alkynyl. Exemplary suitable modifications include O[(CH.sub.2).sub.nO].sub.mCH.sub.3, O(CH.sub.2).sub.nOCH.sub.3, O(CH.sub.2).sub.nNH.sub.2, O(CH.sub.2).sub.nCH.sub.3, O(CH.sub.2).sub.nONH.sub.2, and O(CH.sub.2).sub.nON[(CH.sub.2).sub.nCH.sub.3)].sub.2, where n and m are from 1 to about 10. In other embodiments, dsRNAs include one of the following at the 2′ position: C.sub.1 to C.sub.10 lower alkyl, substituted lower alkyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH.sub.3, OCN, Cl, Br, CN, CF.sub.3, OCF.sub.3, SOCH.sub.3, SO.sub.2CH.sub.3, ONO.sub.2, NO.sub.2, N.sub.3, NH.sub.2, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving the

pharmacokinetic properties of an iRNA, or a group for improving the pharmacodynamic properties of an iRNA, and other substituents having similar properties. In some embodiments, the modification includes a 2′-methoxyethoxy (2′-O—CH.sub.2CH.sub.2OCH.sub.3, also known as 2′-O-(2-methoxyethyl) or 2′-MOE) (Martin et al., *Helv. Chim. Acta*, 1995, 78:486-504) i.e., an alkoxy-alkoxy group. Another exemplary modification is 2′-dimethylaminooxyethoxy, i.e., a O(CH.sub.2).sub.2ON(CH.sub.3).sub.2 group, also known as 2′-DMAOE, as described in examples herein below, and 2′-dimethylaminoethoxyethoxy (also known in the art as 2′-O-dimethylaminoethoxyethyl or 2′-DMAEOE), i.e., 2′-O—CH.sub.2—O—CH.sub.2—N(CH.sub.2).sub.2.

[0143] Other modifications include 2′-methoxy (2′-OCH.sub.3), 2′-aminopropoxy (2′-OCH.sub.2CH.sub.2CH.sub.2NH.sub.2) and 2′-fluoro (2′-F). Similar modifications can also be made at other positions on the RNA of an iRNA, particularly the 3′ position of the sugar on the 3′ terminal nucleotide or in 2′-5′ linked dsRNAs and the 5′ position of 5′ terminal nucleotide. iRNAs can also have sugar mimetics such as cyclobutyl moieties in place of the pentofuranosyl sugar. Representative U.S. patents that teach the preparation of such modified sugar structures include, but are not limited to, U.S. Pat. Nos. 4,981,957; 5,118,800; 5,319,080; 5,359,044; 5,393,878; 5,446,137; 5,466,786; 5,514,785; 5,519,134; 5,567,811; 5,576,427; 5,591,722; 5,597,909; 5,610,300; 5,627,053; 5,639,873; 5,646,265; 5,658,873; 5,670,633; and 5,700,920, certain of which are commonly owned with the instant application. The entire contents of each of the foregoing are hereby incorporated herein by reference.

[0144] The RNA of an iRNA of the invention can also include nucleobase (often referred to in the art simply as “base”) modifications or substitutions. As used herein, “unmodified” or “natural” nucleobases include the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U). Modified nucleobases include other synthetic and natural nucleobases such as deoxy-thymine (dT), 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl uracil and cytosine, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl anal other 8-substituted adenines and guanines, 5-halo, particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-daazaadenine and 3-deazaguanine and 3-deazaadenine. Further nucleobases include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in Modified Nucleosides in Biochemistry, Biotechnology and Medicine, Herdewijn, P. ed. Wiley-VCH, 2008; those disclosed in The Concise Encyclopedia Of Polymer Science And Engineering, pages 858-859, Kroschwitz, J. L, ed. John Wiley & Sons, 1990, these disclosed by Englisch et al., Angewandte Chemie, International Edition, 1991, 30, 613, and those disclosed by Sanghvi, Y S., Chapter 15, dsRNA Research and Applications, pages 289-302, Crooke, S. T. and Lebleu, B., Ed., CRC Press, 1993. Certain of these nucleobases are particularly useful for increasing the binding affinity of the oligomeric compounds featured in the invention. These include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and 0-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine. 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi, Y. S., Crooke, S. T. and Lebleu, B., Eds., dsRNA Research and Applications, CRC Press, Boca Raton, 1993, pp. 276-278) and are exemplary base substitutions, even more particularly when combined with 2′-O-methoxyethyl sugar modifications.

[0145] Representative U.S. patents that teach the preparation of certain of the above noted modified nucleobases as well as other modified nucleobases include, but are not limited to, the above noted U.S. Pat. Nos. 3,687,808, 4,845,205; 5,130,30; 5,134,066; 5,175,273; 5,367,066; 5,432,272; 5,457,187; 5,459,255; 5,484,908; 5,502,177; 5,525,711; 5,552,540; 5,587,469; 5,594,121, 5,596,091; 5,614,617; 5,681,941; 5,750,692; 6,015,886; 6,147,200; 6,166,197; 6,222,025; 6,235,887; 6,380,368; 6,528,640; 6,639,062; 6,617,438; 7,045,610; 7,427,672; and 7,495,088, the entire contents of each of which are hereby incorporated herein by reference.

[0146] The RNA of an iRNA can also be modified to include one or more locked nucleic acids (LNA). A locked nucleic acid is a nucleotide having a modified ribose moiety in which the ribose moiety comprises an extra bridge connecting the 2′ and 4′ carbons. This structure effectively "locks" the ribose in the 3′-endo structural conformation. The addition of locked nucleic acids to siRNAs has been shown to increase siRNA stability in serum, and to reduce off-target effects (Elmen, J. et al., (2005) *Nucleic Acids Research* 33(1):439-447; Mook, O R. et al., (2007) *Mol Canc Ther* 6(3):833-843; Grunweller, A. et al., (2003) *Nucleic Acids Research* 31(12):3185-3193).

[0147] In some embodiments, the iRNA of the invention comprises one or more monomers that are UNA (unlocked nucleic acid) nucleotides. UNA is unlocked acyclic nucleic acid, wherein any of the bonds of the sugar has been removed, forming an unlocked "sugar" residue. In one example, UNA also encompasses monomer with bonds between C1′-C4′ have been removed (i.e., the covalent carbon-oxygen-carbon bond between the C1′ and C4′ carbons). In another example, the C2′-C3′ bond (i.e. the covalent carbon-carbon bond between the C2′ and C3′ carbons) of the sugar has been removed (see *Nuc. Acids Symp. Series*, 52, 133-134 (2008) and Fluiter et al., *Mol. Biosyst.*, 2009, 10, 1039 hereby incorporated by reference).

[0148] The RNA of an iRNA can also be modified to include one or more bicyclic sugar moities. A "bicyclic sugar" is a furanosyl ring modified by the bridging of two atoms. A"bicyclic nucleoside" ("BNA") is a nucleoside having a sugar moiety comprising a bridge connecting two carbon atoms of the sugar ring, thereby forming a bicyclic ring system. In certain embodiments, the bridge connects the 4′-carbon and the 2′-carbon of the sugar ring. Thus, in some embodiments an agent of the invention may include one or more locked nucleic acids (LNA). A locked nucleic acid is a nucleotide having a modified ribose moiety in which the ribose moiety comprises an extra bridge connecting the 2′ and 4′ carbons. In other words, an LNA is a nucleotide comprising a bicyclic sugar moiety comprising a 4′-CH2-O-2′ bridge. This structure effectively "locks" the ribose in the 3′-endo structural conformation. The addition of locked nucleic acids to siRNAs has been shown to increase siRNA stability in serum, and to reduce off-target effects (Elmen, J. et al., (2005) *Nucleic Acids Research* 33(1):439-447; Mook, O R. et al., (2007) *Mol Canc Ther* 6(3):833-843; Grunweller, A. et al., (2003) *Nucleic Acids Research* 31(12):3185-3193). Examples of bicyclic nucleosides for use in the polynucleotides of the invention include without limitation nucleosides comprising a bridge between the 4′ and the 2′ ribosyl ring atoms. In certain embodiments, the antisense polynucleotide agents of the invention include one or more bicyclic nucleosides comprising a 4′ to 2′ bridge. Examples of such 4′ to 2′ bridged bicyclic nucleosides, include but are not limited to 4′-(CH2)O-2′ (LNA); 4′-(CH2)S-2′; 4′-(CH2)2-O-2′ (ENA); 4′-CH(CH3)-O-2′ (also referred to as "constrained ethyl" or "cEt") and 4′-CH(CH2OCH3)-O-2′ (and analogs thereof; see, e.g., U.S. Pat. No. 7,399,845); 4′-C(CH3)(CH3)-O-2′ (and analogs thereof; see e.g., U.S. Pat. No. 8,278,283); 4′-CH2-N(OCH3)-2′ (and analogs thereof; see e.g., U.S. Pat. No. 8,278,425); 4′-CH2-O—N(CH3)-2′ (see, e.g., U.S. Patent Publication No. 2004/0171570); 4′-CH2 N(R)O-2′, wherein R is H, C1-C12 alkyl, or a protecting group (see, e.g., U.S. Pat. No. 7,427,672); 4′-CH2-C(H)(CH3)-2′ (see, e.g., Chattopadhyaya et al., *J. Org.* 30 *Chem.*, 2009, 74, 118-134); and 4′-CH2-C(═CH2)-2′ (and analogs thereof; see, e.g., U.S. Pat. No. 8,278,426). The entire contents of each of the foregoing are hereby incorporated herein by reference.

[0149] Additional representative U.S. Patents and US Patent Publications that teach the preparation of locked nucleic acid nucleotides include, but are not limited to, the following: U.S. Pat. Nos. 6,268,490; 6,525,191; 6,670,461; 6,770,748; 6,794,499; 6,998,484; 7,053,207; 7,034,133; 7,084,125; 7,399,845; 7,427,672; 7,569,686; 7,741,457; 8,022,193; 8,030,467; 8,278,425; 8,278,426; 8,278,283; US 2008/0039618; and US 2009/0012281, the entire contents of each of which are hereby incorporated herein by reference.

[0150] Any of the foregoing bicyclic nucleosides can be prepared having one or more stereochemical sugar configurations including for example α-L-ribofuranose and j-D-ribofuranose (see WO 99/14226).
[0151] The RNA of an iRNA can also be modified to include one or more constrained ethyl nucleotides. As used herein, a “constrained ethyl nucleotide” or “cEt” is a locked nucleic acid comprising a bicyclic sugar moiety comprising a 4′-CH(CH3)-O-2′ bridge. In one embodiment, a constrained ethyl nucleotide is in the S conformation referred to herein as “S-cEt.”
[0152] An iRNA of the invention may also include one or more “conformationally restricted nucleotides” (“CRN”). CRN are nucleotide analogs with a linker connecting the C2′ and C4′ carbons of ribose or the C3 and —C5′ carbons of ribose. CRN lock the ribose ring into a stable conformation and increase the hybridization affinity to mRNA. The linker is of sufficient length to place the oxygen in an optimal position for stability and affinity resulting in less ribose ring puckering.
[0153] Representative publications that teach the preparation of certain of the above noted CRN include, but are not limited to, US Patent Publication No. 2013/0190383; and PCT publication WO 2013/036868, the entire contents of each of which are hereby incorporated herein by reference.
[0154] Potentially stabilizing modifications to the ends of RNA molecules can include N-(acetylaminocaproyl)-4-hydroxyprolinol (Hyp-C6-NHAc), N-(caproyl-4-hydroxyprolinol (Hyp-C6), N-(acetyl-4-hydroxyprolinol (Hyp-NHAc), thymidine-2′-O-deoxythymidine (ether), N-(aminocaproyl)-4-hydroxyprolinol (Hyp-C6-amino), 2-docosanoyl-uridine-3″-phosphate, inverted base dT(idT) and others. Disclosure of this modification can be found in PCT Publication No. WO 2011/005861.
[0155] Other modifications of the nucleotides of an iRNA of the invention include a 5′ phosphate or 5′ phosphate mimic, e.g., a 5′-terminal phosphate or phosphate mimic on the antisense strand of an RNAi agent. Suitable phosphate mimics are disclosed in, for example U.S. Patent Publication No. 2012/0157511, the entire contents of which are incorporated herein by reference.
A. Modified iRNAs Comprising Motifs of the Invention
[0156] In certain aspects of the invention, the double stranded RNAi agents of the invention include agents with chemical modifications as disclosed, for example, in U.S. Provisional Application No. 61/561,710, filed on Nov. 18, 2011, or in PCT/US2012/065691, filed on Nov. 16, 2012, the entire contents of each of which are incorporated herein by reference.
[0157] As shown herein, in Provisional Application No. 61/561,710, and in PCT/US2012/065691, a superior result may be obtained by introducing one or more motifs of three identical modifications on three consecutive nucleotides into a sense strand and/or antisense strand of a RNAi agent, particularly at or near the cleavage site. In some embodiments, the sense strand and antisense strand of the RNAi agent may otherwise be completely modified. The introduction of these motifs interrupts the modification pattern, if present, of the sense and/or antisense strand. The RNAi agent may be optionally conjugated with a GalNAc derivative ligand, for instance on the sense strand. The resulting RNAi agents present superior gene silencing activity.
[0158] More specifically, it has been surprisingly discovered that when the sense strand and antisense strand of the double stranded RNAi agent are modified to have one or more motifs of three identical modifications on three consecutive nucleotides at or near the cleavage site of at least one strand of an RNAi agent, the gene silencing activity of the RNAi agent was superiorly enhanced.
[0159] In one embodiment, the RNAi agent is a double ended bluntmer of 19 nucleotides in length, wherein the sense strand contains at least one motif of three 2′-F modifications on three consecutive nucleotides at positions 7, 8, 9 from the 5′ end. The antisense strand contains at least one motif of three 2′-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5′ end.
[0160] In another embodiment, the RNAi agent is a double ended bluntmer of 20 nucleotides in

length, wherein the sense strand contains at least one motif of three 2′-F modifications on three consecutive nucleotides at positions 8, 9, 10 from the 5′ end. The antisense strand contains at least one motif of three 2′-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5′ end.

[0161] In yet another embodiment, the RNAi agent is a double ended bluntmer of 21 nucleotides in length, wherein the sense strand contains at least one motif of three 2′-F modifications on three consecutive nucleotides at positions 9, 10, 11 from the 5′ end. The antisense strand contains at least one motif of three 2′-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5′ end.

[0162] In one embodiment, the RNAi agent comprises a 21 nucleotide sense strand and a 23 nucleotide antisense strand, wherein the sense strand contains at least one motif of three 2′-F modifications on three consecutive nucleotides at positions 9, 10, 11 from the 5′ end; the antisense strand contains at least one motif of three 2′-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5′ end, wherein one end of the RNAi agent is blunt, while the other end comprises a 2 nucleotide overhang. Preferably, the 2 nucleotide overhang is at the 3′-end of the antisense strand. When the 2 nucleotide overhang is at the 3′-end of the antisense strand, there may be two phosphorothioate internucleotide linkages between the terminal three nucleotides, wherein two of the three nucleotides are the overhang nucleotides, and the third nucleotide is a paired nucleotide next to the overhang nucleotide. In one embodiment, the RNAi agent additionally has two phosphorothioate internucleotide linkages between the terminal three nucleotides at both the 5′-end of the sense strand and at the 5′-end of the antisense strand. In one embodiment, every nucleotide in the sense strand and the antisense strand of the RNAi agent, including the nucleotides that are part of the motifs are modified nucleotides. In one embodiment each residue is independently modified with a 2′-O-methyl or 3′-fluoro, e.g., in an alternating motif. Optionally, the RNAi agent further comprises a ligand (preferably GalNAc3).

[0163] In one embodiment, the RNAi agent comprises sense and antisense strands, wherein the RNAi agent comprises a first strand having a length which is at least 25 and at most 29 nucleotides and a second strand having a length which is at most 30 nucleotides with at least one motif of three 2′-O-methyl modifications on three consecutive nucleotides at position 11, 12, 13 from the 5′ end; wherein the 3′ end of the first strand and the 5′ end of the second strand form a blunt end and the second strand is 1-4 nucleotides longer at its 3′ end than the first strand, wherein the duplex region region which is at least 25 nucleotides in length, and the second strand is sufficiently complementary to a target mRNA along at least 19 nucleotide of the second strand length to reduce target gene expression when the RNAi agent is introduced into a mammalian cell, and wherein dicer cleavage of the RNAi agent preferentially results in an siRNA comprising the 3′ end of the second strand, thereby reducing expression of the target gene in the mammal. Optionally, the RNAi agent further comprises a ligand.

[0164] In one embodiment, the sense strand of the RNAi agent contains at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at the cleavage site in the sense strand.

[0165] In one embodiment, the antisense strand of the RNAi agent can also contain at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at or near the cleavage site in the antisense strand

[0166] For an RNAi agent having a duplex region of 17-23 nucleotides in length, the cleavage site of the antisense strand is typically around the 10, 11 and 12 positions from the 5′-end. Thus the motifs of three identical modifications may occur at the 9, 10, 11 positions; 10, 11, 12 positions; 11, 12, 13 positions; 12, 13, 14 positions; or 13, 14, 15 positions of the antisense strand, the count starting from the 1.sup.st nucleotide from the 5′-end of the antisense strand, or, the count starting from the 1.sup.st paired nucleotide within the duplex region from the 5′-end of the antisense strand. The cleavage site in the antisense strand may also change according to the length of the duplex

region of the RNAi from the 5′-end.
[0167] The sense strand of the RNAi agent may contain at least one motif of three identical modifications on three consecutive nucleotides at the cleavage site of the strand; and the antisense strand may have at least one motif of three identical modifications on three consecutive nucleotides at or near the cleavage site of the strand. When the sense strand and the antisense strand form a dsRNA duplex, the sense strand and the antisense strand can be so aligned that one motif of the three nucleotides on the sense strand and one motif of the three nucleotides on the antisense strand have at least one nucleotide overlap, i.e., at least one of the three nucleotides of the motif in the sense strand forms a base pair with at least one of the three nucleotides of the motif in the antisense strand. Alternatively, at least two nucleotides may overlap, or all three nucleotides may overlap.
[0168] In one embodiment, the sense strand of the RNAi agent may contain more than one motif of three identical modifications on three consecutive nucleotides. The first motif may occur at or near the cleavage site of the strand and the other motifs may be a wing modification. The term “wing modification” herein refers to a motif occurring at another portion of the strand that is separated from the motif at or near the cleavage site of the same strand. The wing modification is either adjacent to the first motif or is separated by at least one or more nucleotides. When the motifs are immediately adjacent to each other then the chemistry of the motifs are distinct from each other and when the motifs are separated by one or more nucleotide than the chemistries can be the same or different. Two or more wing modifications may be present. For instance, when two wing modifications are present, each wing modification may occur at one end relative to the first motif which is at or near cleavage site or on either side of the lead motif.
[0169] Like the sense strand, the antisense strand of the RNAi agent may contain more than one motifs of three identical modifications on three consecutive nucleotides, with at least one of the motifs occurring at or near the cleavage site of the strand. This antisense strand may also contain one or more wing modifications in an alignment similar to the wing modifications that may be present on the sense strand.
[0170] In one embodiment, the wing modification on the sense strand or antisense strand of the RNAi agent typically does not include the first one or two terminal nucleotides at the 3′-end, 5′-end or both ends of the strand.
[0171] In another embodiment, the wing modification on the sense strand or antisense strand of the RNAi agent typically does not include the first one or two paired nucleotides within the duplex region at the 3′-end, 5′-end or both ends of the strand.
[0172] When the sense strand and the antisense strand of the RNAi agent each contain at least one wing modification, the wing modifications may fall on the same end of the duplex region, and have an overlap of one, two or three nucleotides.
[0173] When the sense strand and the antisense strand of the RNAi agent each contain at least two wing modifications, the sense strand and the antisense strand can be so aligned that two modifications each from one strand fall on one end of the duplex region, having an overlap of one, two or three nucleotides; two modifications each from one strand fall on the other end of the duplex region, having an overlap of one, two or three nucleotides; two modifications one strand fall on each side of the lead motif, having an overlap of one, two or three nucleotides in the duplex region.
[0174] In one embodiment, every nucleotide in the sense strand and antisense strand of the RNAi agent, including the nucleotides that are part of the motifs, may be modified. Each nucleotide may be modified with the same or different modification which can include one or more alteration of one or both of the non-linking phosphate oxygens and/or of one or more of the linking phosphate oxygens; alteration of a constituent of the ribose sugar, e.g., of the 2′ hydroxyl on the ribose sugar; wholesale replacement of the phosphate moiety with “dephospho” linkers; modification or replacement of a naturally occurring base; and replacement or modification of the ribose-phosphate backbone.
[0175] As nucleic acids are polymers of subunits, many of the modifications occur at a position

which is repeated within a nucleic acid, e.g., a modification of a base, or a phosphate moiety, or a non-linking O of a phosphate moiety. In some cases the modification will occur at all of the subject positions in the nucleic acid but in many cases it will not. By way of example, a modification may only occur at a 3′ or 5′ terminal position, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand. A modification may occur in a double strand region, a single strand region, or in both. A modification may occur only in the double strand region of an RNA or may only occur in a single strand region of a RNA. For example, a phosphorothioate modification at a non-linking O position may only occur at one or both termini, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand, or may occur in double strand and single strand regions, particularly at termini. The 5′ end or ends can be phosphorylated.

[0176] It may be possible, e.g., to enhance stability, to include particular bases in overhangs, or to include modified nucleotides or nucleotide surrogates, in single strand overhangs, e.g., in a 5′ or 3′ overhang, or in both. For example, it can be desirable to include purine nucleotides in overhangs. In some embodiments all or some of the bases in a 3′ or 5′ overhang may be modified, e.g., with a modification described herein. Modifications can include, e.g., the use of modifications at the 2′ position of the ribose sugar with modifications that are known in the art, e.g., the use of deoxyribonucleotides, 2′-deoxy-2′-fluoro (2′-F) or 2′-O-methyl modified instead of the ribosugar of the nucleobase, and modifications in the phosphate group, e.g., phosphorothioate modifications. Overhangs need not be homologous with the target sequence.

[0177] In one embodiment, each residue of the sense strand and antisense strand is independently modified with LNA, HNA, CeNA, 2′-methoxyethyl, 2′-O-methyl, 2′-O-allyl, 2′-C-allyl, 2′-deoxy, 2′-hydroxyl, or 2′-fluoro. The strands can contain more than one modification. In one embodiment, each residue of the sense strand and antisense strand is independently modified with 2′-O-methyl or 2′-fluoro.

[0178] At least two different modifications are typically present on the sense strand and antisense strand. Those two modifications may be the 2′-O-methyl or 2′-fluoro modifications, or others.

[0179] In one embodiment, the N.sub.a and/or N.sub.b comprise modifications of an alternating pattern. The term “alternating motif” as used herein refers to a motif having one or more modifications, each modification occurring on alternating nucleotides of one strand. The alternating nucleotide may refer to one per every other nucleotide or one per every three nucleotides, or a similar pattern. For example, if A, B and C each represent one type of modification to the nucleotide, the alternating motif can be “ABABABABABAB . . . ,” “AABBAABBAABB . . . ,” “AABAABAABAAB . . . ,” “AAABAAABAAAB . . . ,” “AAABBBAAABBB . . . ,” or “ABCABCABCABC . . . ,” etc.

[0180] The type of modifications contained in the alternating motif may be the same or different. For example, if A, B, C, D each represent one type of modification on the nucleotide, the alternating pattern, i.e., modifications on every other nucleotide, may be the same, but each of the sense strand or antisense strand can be selected from several possibilities of modifications within the alternating motif such as “ABABAB . . . ”, “ACACAC . . . ” “BDBDBD . . . ” or “CDCDCD . . . ,” etc.

[0181] In one embodiment, the RNAi agent of the invention comprises the modification pattern for the alternating motif on the sense strand relative to the modification pattern for the alternating motif on the antisense strand is shifted. The shift may be such that the modified group of nucleotides of the sense strand corresponds to a differently modified group of nucleotides of the antisense strand and vice versa. For example, the sense strand when paired with the antisense strand in the dsRNA duplex, the alternating motif in the sense strand may start with “ABABAB” from 5′-3′ of the strand and the alternating motif in the antisense strand may start with “BABABA” from 5′-3′ of the strand within the duplex region. As another example, the alternating motif in the sense strand may start with “AABBAABB” from 5′-3′ of the strand and the alternating motif in the antisense strand may start with “BBAABBAA” from 5′-3′ of the strand within the duplex region, so that there is a complete or partial shift of the modification patterns between the sense strand and the antisense strand.

[0182] In one embodiment, the RNAi agent comprises the pattern of the alternating motif of 2′-O-methyl modification and 2′-F modification on the sense strand initially has a shift relative to the pattern of the alternating motif of 2′-O-methyl modification and 2′-F modification on the antisense strand initially, i.e., the 2′-O-methyl modified nucleotide on the sense strand base pairs with a 2′-F modified nucleotide on the antisense strand and vice versa. The 1 position of the sense strand may start with the 2′-F modification, and the 1 position of the antisense strand may start with the 2′-O-methyl modification.

[0183] The introduction of one or more motifs of three identical modifications on three consecutive nucleotides to the sense strand and/or antisense strand interrupts the initial modification pattern present in the sense strand and/or antisense strand. This interruption of the modification pattern of the sense and/or antisense strand by introducing one or more motifs of three identical modifications on three consecutive nucleotides to the sense and/or antisense strand surprisingly enhances the gene silencing activity to the target gene.

[0184] In one embodiment, when the motif of three identical modifications on three consecutive nucleotides is introduced to any of the strands, the modification of the nucleotide next to the motif is a different modification than the modification of the motif. For example, the portion of the sequence containing the motif is “ . . . N.sub.aYYYN.sub.b . . . ,” where “Y” represents the modification of the motif of three identical modifications on three consecutive nucleotide, and “N.sub.a” and “N.sub.b” represent a modification to the nucleotide next to the motif “YYY” that is different than the modification of Y, and where N.sub.a and N.sub.b can be the same or different modifications. Alternatively, N.sub.a and/or N.sub.b may be present or absent when there is a wing modification present.

[0185] The RNAi agent may further comprise at least one phosphorothioate or methylphosphonate internucleotide linkage. The phosphorothioate or methylphosphonate internucleotide linkage modification may occur on any nucleotide of the sense strand or antisense strand or both strands in any position of the strand. For instance, the internucleotide linkage modification may occur on every nucleotide on the sense strand and/or antisense strand; each internucleotide linkage modification may occur in an alternating pattern on the sense strand and/or antisense strand; or the sense strand or antisense strand may contain both internucleotide linkage modifications in an alternating pattern. The alternating pattern of the internucleotide linkage modification on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the internucleotide linkage modification on the sense strand may have a shift relative to the alternating pattern of the internucleotide linkage modification on the antisense strand.

[0186] In one embodiment, the RNAi comprises a phosphorothioate or methylphosphonate internucleotide linkage modification in the overhang region. For example, the overhang region may contain two nucleotides having a phosphorothioate or methylphosphonate internucleotide linkage between the two nucleotides. Internucleotide linkage modifications also may be made to link the overhang nucleotides with the terminal paired nucleotides within the duplex region. For example, at least 2, 3, 4, or all the overhang nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkage, and optionally, there may be additional phosphorothioate or methylphosphonate internucleotide linkages linking the overhang nucleotide with a paired nucleotide that is next to the overhang nucleotide. For instance, there may be at least two phosphorothioate internucleotide linkages between the terminal three nucleotides, in which two of the three nucleotides are overhang nucleotides, and the third is a paired nucleotide next to the overhang nucleotide. These terminal three nucleotides may be at the 3′-end of the antisense strand, the 3′-end of the sense strand, the 5′-end of the antisense strand, and/or the 5′ end of the antisense strand.

[0187] In one embodiment, the 2 nucleotide overhang is at the 3′-end of the antisense strand, and there are two phosphorothioate internucleotide linkages between the terminal three nucleotides, wherein two of the three nucleotides are the overhang nucleotides, and the third nucleotide is a paired nucleotide next to the overhang nucleotide. Optionally, the RNAi agent may additionally have two phosphorothioate internucleotide linkages between the terminal three nucleotides at both the 5′-end of the sense strand and at the 5′-end of the antisense strand.
[0188] In one embodiment, the RNAi agent comprises mismatch(es) with the target, within the duplex, or combinations thereof. A “mismatch” may be non-canonical base pairing or other than canonical pairing of nucleotides. The mismatch may occur in the overhang region or the duplex region. The base pair may be ranked on the basis of their propensity to promote dissociation or melting (e.g., on the free energy of association or dissociation of a particular pairing, the simplest approach is to examine the pairs on an individual pair basis, though next neighbor or similar analysis can also be used). In terms of promoting dissociation: A:U is preferred over G:C; G:U is preferred over G:C; and I:C is preferred over G:C (I=inosine). Mismatches, e.g., non-canonical or other than canonical pairings (as described elsewhere herein) are preferred over canonical (A:T, A:U, G:C) pairings; and pairings which include a universal base are preferred over canonical pairings. A “universal base” is a base that exhibits the ability to replace any of the four normal bases (G, C, A, and U) without significantly destabilizing neighboring base-pair interactions or disrupting the expected functional biochemical utility of the modified oligonucleotide. Non-limiting examples of universal bases include 2′-deoxyinosine (hypoxanthine deoxynucleotide) or its derivatives, nitroazole analogues, and hydrophobic aromatic non-hydrogen-bonding bases.
[0189] In one embodiment, the RNAi agent comprises at least one of the first 1, 2, 3, 4, or 5 base pairs within the duplex regions from the 5′-end of the antisense strand independently selected from the group of: A:U, G:U, I:C, and mismatched pairs, e.g., non-canonical or other than canonical pairings or pairings which include a universal base, to promote the dissociation of the antisense strand at the 5′-end of the duplex.
[0190] In one embodiment, the nucleotide at the 1 position within the duplex region from the 5′-end in the antisense strand is selected from the group consisting of A, dA, dU, U, and dT. Alternatively, at least one of the first 1, 2 or 3 base pair within the duplex region from the 5′-end of the antisense strand is an AU base pair. For example, the first base pair within the duplex region from the 5′-end of the antisense strand is an AU base pair.
[0191] In another embodiment, the nucleotide at the 3′-end of the sense strand is deoxy-thymine (dT). In another embodiment, the nucleotide at the 3′-end of the antisense strand is deoxy-thymine (dT). In one embodiment, there is a short sequence of deoxy-thymine nucleotides, for example, two dT nucleotides on the 3′-end of the sense and/or antisense strand.
[0192] In one embodiment, the sense strand sequence may be represented by formula (I):

5′*n*.sub.p-N.sub.a-(XXX).sub.i-N.sub.b-YYY-N.sub.b-(ZZZ).sub.j-N.sub.a-*n*.sub.q3′ (I) [0193] wherein: [0194] i and j are each independently 0 or 1; [0195] p and q are each independently 0-6; [0196] each N.sub.a independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; [0197] each N.sub.b independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; [0198] each n.sub.p and n.sub.q independently represent an overhang nucleotide; [0199] wherein Nb and Y do not have the same modification; and [0200] XXX, YYY and ZZZ each independently represent one motif of three identical modifications on three consecutive nucleotides. Preferably YYY is all 2′-F modified nucleotides.
[0201] In one embodiment, the N.sub.a and/or N.sub.b comprise modifications of alternating pattern.
[0202] In one embodiment, the YYY motif occurs at or near the cleavage site of the sense strand. For example, when the RNAi agent has a duplex region of 17-23 nucleotides in length, the YYY

motif can occur at or the vicinity of the cleavage site (e.g.: can occur at positions 6, 7, 8, 7, 8, 9, 8, 9, 10, 9, 10, 11, 10, 11, 12 or 11, 12, 13) of [0203] the sense strand, the count starting from the 1.sup.st nucleotide, from the 5′-end; or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5′-end.
[0204] In one embodiment, i is 1 and j is 0, or i is 0 and j is 1, or both i and j are 1. The sense strand can therefore be represented by the following formulas:

5′*n*.sub.p-N.sub.a-YYY-N.sub.b-ZZZ-N.sub.a-*n*.sub.q3′ (Ib);

5′*n*.sub.p-N.sub.a-XXX-N.sub.b-YYY-N.sub.a-*n*.sub.q3′ (Ic); or

5′*n*.sub.p-N.sub.a-XXX-N.sub.b-YYY-N.sub.b-ZZZ-N.sub.a-*n*.sub.q3′ (Id).
[0205] When the sense strand is represented by formula (Ib), N.sub.b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.
[0206] When the sense strand is represented as formula (Ic), N.sub.b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.
[0207] When the sense strand is represented as formula (Id), each N.sub.b independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Preferably, N.sub.b is 0, 1, 2, 3, 4, 5 or 6 Each N.sub.a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.
[0208] Each of X, Y and Z may be the same or different from each other.
[0209] In other embodiments, i is 0 and j is 0, and the sense strand may be represented by the formula:

5′*n*.sub.p-N.sub.a-YYY-N.sub.a-*n*.sub.q3′ (Ia).
[0210] When the sense strand is represented by formula (Ia), each N.sub.a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.
[0211] In one embodiment, the antisense strand sequence of the RNAi may be represented by formula (II):

5′*n*.sub.q′-N.sub.a′-(Z′Z′Z′).sub.k-N.sub.b′-Y′Y′Y′-N.sub.b′-(X′X′X′).sub.l-N′.sub.a-*n*.sub.p′3′ (II) [0212] wherein: [0213] k and l are each independently 0 or 1; [0214] p′ and q′ are each independently 0-6; [0215] each N.sub.a′ independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; [0216] each N.sub.b′ independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; [0217] each n.sub.p′ and n.sub.q′ independently represent an overhang nucleotide; [0218] wherein N.sub.b′ and Y′ do not have the same modification; [0219] and [0220] X′X′X′, Y′Y′Y′ and Z′Z′Z′ each independently represent one motif of three identical modifications on three consecutive nucleotides.
[0221] In one embodiment, the N.sub.a′ and/or N.sub.b′ comprise modifications of alternating pattern.
[0222] The Y′Y′Y′ motif occurs at or near the cleavage site of the antisense strand. For example, when the RNAi agent has a duplex region of 17-23 nucleotide in length, the Y′Y′Y′ motif can occur at positions 9, 10, 11; 10, 11, 12; 11, 12, 13; 12, 13, 14; or 13, 14, 15 of the antisense strand, with the count starting from the 1.sup.st nucleotide, from the 5′-end; or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5′-end. Preferably, the Y′Y′Y′

motif occurs at positions 11, 12, 13.

[0223] In one embodiment, Y′Y′Y′ motif is all 2′-OMe modified nucleotides.

[0224] In one embodiment, k is 1 and l is 0, or k is 0 and l is 1, or both k and l are 1.

[0225] The antisense strand can therefore be represented by the following formulas:

5′*n*.sub.q′-N.sub.a′-Z′Z′Z′-N.sub.b′-Y′Y′Y′-N.sub.a′-*n*.sub.p′3′ (IIb);

5′*n*.sub.q′-N.sub.a′-Y′Y′Y′-N.sub.b′-X′X′X′-*n*.sub.p,3′ (IIc); or

5′*n*.sub.q′-N.sub.a′-Z′Z′Z′-N.sub.b′-Y′Y′Y′-N.sub.b′-X′X′X′-N.sub.a′-*n*.sub.p,3′ (IId).

[0226] When the antisense strand is represented by formula (IIb), N.sub.b′ represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a′ independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0227] When the antisense strand is represented as formula (IIc), N.sub.b′ represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a′ independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0228] When the antisense strand is represented as formula (IId), each N.sub.b′ independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a′ independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Preferably, N.sub.b is 0, 1, 2, 3, 4, 5 or 6.

[0229] In other embodiments, k is 0 and 1 is 0 and the antisense strand may be represented by the formula:

5′*n*.sub.p′-N.sub.a′-Y′Y′Y′-N.sub.a′-*n*.sub.q′3′ (Ia).

[0230] When the antisense strand is represented as formula (IIa), each N.sub.a′ independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

[0231] Each of X′, Y′ and Z′ may be the same or different from each other.

[0232] Each nucleotide of the sense strand and antisense strand may be independently modified with LNA, HNA, CeNA, 2′-methoxyethyl, 2′-O-methyl, 2′-O-allyl, 2′-C-allyl, 2′-hydroxyl, or 2′-fluoro. For example, each nucleotide of the sense strand and antisense strand is independently modified with 2′-O-methyl or 2′-fluoro. Each X, Y, Z, X′, Y′ and Z′, in particular, may represent a 2′-O-methyl modification or a 2′-fluoro modification.

[0233] In one embodiment, the sense strand of the RNAi agent may contain YYY motif occurring at 9, 10 and 11 positions of the strand when the duplex region is 21 nt, the count starting from the 1.sup.st nucleotide from the 5′-end, or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5′-end; and Y represents 2′-F modification. The sense strand may additionally contain XXX motif or ZZZ motifs as wing modifications at the opposite end of the duplex region; and XXX and ZZZ each independently represents a 2′-OMe modification or 2′-F modification.

[0234] In one embodiment the antisense strand may contain Y′Y′Y′ motif occurring at positions 11, 12, 13 of the strand, the count starting from the 1.sup.st nucleotide from the 5′-end, or optionally, the count starting at the 1.sup.st paired nucleotide within the duplex region, from the 5′-end; and Y′ represents 2′-O-methyl modification. The antisense strand may additionally contain X′X′X′ motif or Z′Z′Z′ motifs as wing modifications at the opposite end of the duplex region; and X′X′X′ and Z′Z′Z′ each independently represents a 2′-OMe modification or 2′-F modification.

[0235] The sense strand represented by any one of the above formulas (Ia), (Ib), (Ic), and (Id) forms a duplex with a antisense strand being represented by any one of formulas (IIa), (IIb), (IIc), and (IId), respectively.

[0236] Accordingly, the RNAi agents for use in the methods of the invention may comprise a sense strand and an antisense strand, each strand having 14 to 30 nucleotides, the RNAi duplex represented by formula (III):

sense: 5′*n*.sub.p-N.sub.a-(XXX).sub.i-N.sub.b-YYY-N.sub.b-(ZZZ).sub.j-N.sub.a-*n*.sub.q3′

antisense: 3′*n*.sub.p′-N.sub.a′-(X′X′X′).sub.k-N.sub.b′-Y′Y′Y′-N.sub.b′-(Z′Z′Z′).sub.l-N.sub.a′-*n*.sub.q′5′ (III) [0237] wherein: [0238] i, j, k, and 1 are each independently 0 or 1; [0239] p, p′, q, and q′ are each independently 0-6; [0240] each N.sub.a and N.sub.a′ independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; [0241] each N.sub.b and N.sub.b′ independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; [0242] wherein [0243] each n.sub.p′, n.sub.p, n.sub.q′, and n.sub.q, each of which may or may not be present, independently represents an overhang nucleotide; and [0244] XXX, YYY, ZZZ, X′X′X′, Y′Y′Y′, and Z′Z′Z′ each independently represent one motif of three identical modifications on three consecutive nucleotides.
[0245] In one embodiment, i is 0 and j is 0; or i is 1 and j is 0; or i is 0 and j is 1; or both i and j are 0; or both i and j are 1. In another embodiment, k is 0 and l is 0; or k is 1 and l is 0; k is 0 and l is 1; or both k and l are 0; or both k and l are 1.
[0246] Exemplary combinations of the sense strand and antisense strand forming a RNAi duplex include the formulas below:

5′*n*.sub.p-N.sub.a-YYY-N.sub.a-*n*.sub.q3′

3′*n*.sub.p′-N.sub.a′-Y′Y′Y′-N.sub.a′*n*.sub.q′5′ (IIIa)

5′*n*.sub.p-N.sub.a-YYY-N.sub.b-ZZZ-N.sub.a-*n*.sub.q3′

3′*n*.sub.p′-N.sub.a′-Y′Y′Y′-N.sub.b′-Z′Z′Z′-N.sub.a′*n*.sub.q′5′ (IIIb)

5′*n*.sub.p-N.sub.a-XXX-N.sub.b-YYY-N.sub.a-*n*.sub.q3′

3′*n*.sub.p′-N.sub.a′-X′X′X′-N.sub.b′-Y′Y′Y′-N.sub.a′-*n*.sub.q′5′ (IIIc)

5′*n*.sub.p-N.sub.a-XXX-N.sub.b-YYY-N.sub.b-ZZZ-N.sub.a-*n*.sub.q3′

3′*n*.sub.p′-N.sub.a′-X′X′X′-N.sub.b′-Y′Y′Y′-N.sub.b′-Z′Z′Z′-N.sub.a-*n*.sub.q5′ (IIId)

5′-N.sub.a-YYY-N.sub.a-3′

3′*n*.sub.p′-N.sub.a′-Y′Y′Y′-N.sub.a′5′ (IIIe)
[0247] When the RNAi agent is represented by formula (IIIa), each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.
[0248] When the RNAi agent is represented by formula (IIIb), each N.sub.b independently represents an oligonucleotide sequence comprising 1-10, 1-7, 1-5 or 1-4 modified nucleotides. Each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.
[0249] When the RNAi agent is represented as formula (IIIc), each N.sub.b, N.sub.b′ independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a independently represents an oligonucleotide sequence comprising 2-20,

2-15, or 2-10 modified nucleotides.

[0250] When the RNAi agent is represented as formula (IIId), each N.sub.b, N.sub.b′ independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a, N.sub.a′ independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Each of N.sub.a, N.sub.a′, N.sub.b and N.sub.b independently comprises modifications of alternating pattern.

[0251] When the RNAi agent is represented as formula (IIId), each N.sub.b, N.sub.b′ independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N.sub.a, N.sub.a′ independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Each of N.sub.a, N.sub.a′, N.sub.b and N.sub.b independently comprises modifications of alternating pattern.

[0252] When the RNAi agent is represented as formula (IIIe), each N.sub.a and N.sub.a′ independently represents an oligonucleotide sequence comprising 0-25 nucleotides which are either modified or unmodified or combinations thereof, each sequence comprising at least two differently modified nucleotides.

[0253] Each of X, Y and Z in formulas (III), (IIIa), (IIIb), (IIIc), (IIId), and (IIIe) may be the same or different from each other.

[0254] When the RNAi agent is represented by formula (III), (IIIa), (IIIb), (IIIc), (IIId), and (IIIe) at least one of the Y nucleotides may form a base pair with one of the Y′ nucleotides. Alternatively, at least two of the Y nucleotides form base pairs with the corresponding Y′ nucleotides; or all three of the Y nucleotides all form base pairs with the corresponding Y′ nucleotides.

[0255] When the RNAi agent is represented by formula (IIIb) or (IIId), at least one of the Z nucleotides may form a base pair with one of the Z′ nucleotides. Alternatively, at least two of the Z nucleotides form base pairs with the corresponding Z′ nucleotides; or all three of the Z nucleotides all form base pairs with the corresponding Z′ nucleotides.

[0256] When the RNAi agent is represented as formula (IIIc) or (IIId), at least one of the X nucleotides may form a base pair with one of the X′ nucleotides. Alternatively, at least two of the X nucleotides form base pairs with the corresponding X′ nucleotides; or all three of the X nucleotides all form base pairs with the corresponding X′ nucleotides.

[0257] In one embodiment, the modification on the Y nucleotide is different than the modification on the Y′ nucleotide, the modification on the Z nucleotide is different than the modification on the Z′ nucleotide, and/or the modification on the X nucleotide is different than the modification on the X′ nucleotide.

[0258] In one embodiment, when the RNAi agent is represented by formula (IIId), the N.sub.a modifications are 2′-O-methyl or 2′-fluoro modifications. In another embodiment, when the RNAi agent is represented by formula (IIId), the N.sub.a modifications are 2′-O-methyl or 2′-fluoro modifications and n.sub.p′>0 and at least one n.sub.p′ is linked to a neighboring nucleotide a via phosphorothioate linkage. In yet another embodiment, when the RNAi agent is represented by formula (IIId), the N.sub.a modifications are 2′-O-methyl or 2′-fluoro modifications, n.sub.p′>0 and at least one n.sub.p′ is linked to a neighboring nucleotide via phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a monovalent, a bivalent or a trivalent branched linker. In another embodiment, when the RNAi agent is represented by formula (IIId), the N.sub.a modifications are 2′-O-methyl or 2′-fluoro modifications, n.sub.p′>0 and at least one n.sub.p′ is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a monovalent, a bivalent or a trivalent branched linker.

[0259] In one embodiment, when the RNAi agent is represented by formula (IIIa), the N.sub.a modifications are 2′-O-methyl or 2′-fluoro modifications, n.sub.p′>0 and at least one n.sub.p′ is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a monovalent, a bivalent or a trivalent branched linker.

[0260] In one embodiment, the RNAi agent is a multimer containing at least two duplexes represented by formula (III), (IIIa), (IIIb), (IIIc), (IIId), and (IIIe), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

[0261] In one embodiment, the RNAi agent is a multimer containing three, four, five, six or more duplexes represented by formula (III), (IIIa), (IIIb), (IIIc), (IIId), and (IIIe), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

[0262] In one embodiment, two RNAi agents represented by formula (III), (IIIa), (IIIb), (IIIc), (IIId), and (IIIe) are linked to each other at the 5′ end, and one or both of the 3′ ends and are optionally conjugated to a ligand. Each of the agents can target the same gene or two different genes; or each of the agents can target same gene at two different target sites.

[0263] Various publications describe multimeric RNAi agents that can be used in the methods of the invention. Such publications include WO2007/091269, U.S. Pat. No. 7,858,769, WO2010/141511, WO2007/117686, WO2009/014887 and WO2011/031520 the entire contents of each of which are hereby incorporated herein by reference.

[0264] The RNAi agent that contains conjugations of one or more carbohydrate moieties to a RNAi agent can optimize one or more properties of the RNAi agent. In many cases, the carbohydrate moiety will be attached to a modified subunit of the RNAi agent. For example, the ribose sugar of one or more ribonucleotide subunits of a dsRNA agent can be replaced with another moiety, e.g., a non-carbohydrate (preferably cyclic) carrier to which is attached a carbohydrate ligand. A ribonucleotide subunit in which the ribose sugar of the subunit has been so replaced is referred to herein as a ribose replacement modification subunit (RRMS). A cyclic carrier may be a carbocyclic ring system, i.e., all ring atoms are carbon atoms, or a heterocyclic ring system, i.e., one or more ring atoms may be a heteroatom, e.g., nitrogen, oxygen, sulfur. The cyclic carrier may be a monocyclic ring system, or may contain two or more rings, e.g. fused rings. The cyclic carrier may be a fully saturated ring system, or it may contain one or more double bonds.

[0265] The ligand may be attached to the polynucleotide via a carrier. The carriers include (i) at least one "backbone attachment point," preferably two "backbone attachment points" and (ii) at least one "tethering attachment point." A "backbone attachment point" as used herein refers to a functional group, e.g. a hydroxyl group, or generally, a bond available for, and that is suitable for incorporation of the carrier into the backbone, e.g., the phosphate, or modified phosphate, e.g., sulfur containing, backbone, of a ribonucleic acid. A "tethering attachment point" (TAP) in some embodiments refers to a constituent ring atom of the cyclic carrier, e.g., a carbon atom or a heteroatom (distinct from an atom which provides a backbone attachment point), that connects a selected moiety. The moiety can be, e.g., a carbohydrate, e.g. monosaccharide, disaccharide, trisaccharide, tetrasaccharide, oligosaccharide and polysaccharide. Optionally, the selected moiety is connected by an intervening tether to the cyclic carrier. Thus, the cyclic carrier will often include a functional group, e.g., an amino group, or generally, provide a bond, that is suitable for incorporation or tethering of another chemical entity, e.g., a ligand to the constituent ring.

[0266] The RNAi agents may be conjugated to a ligand via a carrier, wherein the carrier can be cyclic group or acyclic group; preferably, the cyclic group is selected from pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolane, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuryl and and decalin; preferably, the acyclic group is selected from serinol backbone or diethanolamine backbone.

[0267] In certain specific embodiments, the RNAi agent for use in the methods of the invention is an agent selected from the group of agents listed in any one of Tables 3A, 3B, 4A, 4B, 6, and 7. These agents may further comprise a ligand.

IV. iRNAs Conjugated to Ligands

[0268] Another modification of the RNA of an iRNA of the invention involves chemically linking to the RNA one or more ligands, moieties or conjugates that enhance the activity, cellular distribution or cellular uptake of the iRNA. Such moieties include but are not limited to lipid moieties such as a cholesterol moiety (Letsinger et al., *Proc. Natl. Acid. Sci. USA*, 1989, 86: 6553-6556), cholic acid (Manoharan et al., *Biorg. Med. Chem. Let.*, 1994, 4:1053-1060), a thioether, e.g., beryl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660:306-309; Manoharan et al., *Biorg. Med. Chem. Let.*, 1993, 3:2765-2770), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20:533-538), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., *EMBO J*, 1991, 10:1111-1118; Kabanov et al., *FEBS Lett.*, 1990, 259:327-330; Svinarchuk et al., *Biochimie*, 1993, 75:49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651-3654; Shea et al., *Nucl. Acids Res.*, 1990, 18:3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides* & *Nucleotides*, 1995, 14:969-973), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651-3654), a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264:229-237), or an octadecylamine or hexylamino-carbonyloxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277:923-937).

[0269] In one embodiment, a ligand alters the distribution, targeting or lifetime of an iRNA agent into which it is incorporated. In preferred embodiments a ligand provides an enhanced affinity for a selected target, e.g., molecule, cell or cell type, compartment, e.g., a cellular or organ compartment, tissue, organ or region of the body, as, e.g., compared to a species absent such a ligand. Preferred ligands will not take part in duplex pairing in a duplexed nucleic acid.

[0270] Ligands can include a naturally occurring substance, such as a protein (e.g., human serum albumin (HSA), low-density lipoprotein (LDL), or globulin); carbohydrate (e.g., a dextran, pullulan, chitin, chitosan, inulin, cyclodextrin, N-acetylgalactosamine, or hyaluronic acid); or a lipid. The ligand can also be a recombinant or synthetic molecule, such as a synthetic polymer, e.g., a synthetic polyamino acid. Examples of polyamino acids include polyamino acid is a polylysine (PLL), poly L-aspartic acid, poly L-glutamic acid, styrene-maleic acid anhydride copolymer, poly(L-lactide-co-glycolied) copolymer, divinyl ether-maleic anhydride copolymer, N-(2-hydroxypropyl)methacrylamide copolymer (HMPA), polyethylene glycol (PEG), polyvinyl alcohol (PVA), polyurethane, poly(2-ethylacryllic acid), N-isopropylacrylamide polymers, or polyphosphazine. Example of polyamines include: polyethylenimine, polylysine (PLL), spermine, spermidine, polyamine, pseudopeptide-polyamine, peptidomimetic polyamine, dendrimer polyamine, arginine, amidine, protamine, cationic lipid, cationic porphyrin, quaternary salt of a polyamine, or an alpha helical peptide.

[0271] Ligands can also include targeting groups, e.g., a cell or tissue targeting agent, e.g., a lectin, glycoprotein, lipid or protein, e.g., an antibody, that binds to a specified cell type such as a kidney cell. A targeting group can be a thyrotropin, melanotropin, lectin, glycoprotein, surfactant protein A, Mucin carbohydrate, multivalent lactose, monovalent or multivalent galactose, N-acetyl-galactosamine, N-acetyl-gulucoseamine multivalent mannose, multivalent fucose, glycosylated polyaminoacids, transferrin, bisphosphonate, polyglutamate, polyaspartate, a lipid, cholesterol, a steroid, bile acid, folate, vitamin B12, vitamin A, biotin, or an RGD peptide or RGD peptide mimetic. In certain embodiments, ligands include monovalent or multivalent galactose. In certain embodiments, ligands include cholesterol.

[0272] Other examples of ligands include dyes, intercalating agents (e.g. acridines), cross-linkers (e.g. psoralene, mitomycin C), porphyrins (TPPC4, texaphyrin, Sapphyrin), polycyclic aromatic hydrocarbons (e.g., phenazine, dihydrophenazine), artificial endonucleases (e.g. EDTA), lipophilic molecules, e.g., cholesterol, cholic acid, adamantane acetic acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-Bis-O(hexadecyl)glycerol, geranyloxyhexyl group, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid, O3-(oleoyl)lithocholic acid, O3-(oleoyl)cholenic acid, dimethoxytrityl, or phenoxazine) and peptide conjugates (e.g., antennapedia peptide, Tat peptide), alkylating agents, phosphate, amino, mercapto, PEG (e.g., PEG-40K), MPEG, [MPEG].sub.2, polyamino, alkyl, substituted alkyl, radiolabeled markers, enzymes, haptens (e.g. biotin), transport/absorption facilitators (e.g., aspirin, vitamin E, folic acid), synthetic ribonucleases (e.g., imidazole, bisimidazole, histamine, imidazole clusters, acridine-imidazole conjugates, Eu^{3+} complexes of tetraazamacrocycles), dinitrophenyl, HRP, or AP.

[0273] Ligands can be proteins, e.g., glycoproteins, or peptides, e.g., molecules having a specific affinity for a co-ligand, or antibodies e.g., an antibody, that binds to a specified cell type such as a hepatic cell. Ligands can also include hormones and hormone receptors. They can also include non-peptidic species, such as lipids, lectins, carbohydrates, vitamins, cofactors, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-gulucosamine multivalent mannose, or multivalent fucose. The ligand can be, for example, a lipopolysaccharide, an activator of p38 MAP kinase, or an activator of NF-κB.

[0274] The ligand can be a substance, e.g., a drug, which can increase the uptake of the iRNA agent into the cell, for example, by disrupting the cell's cytoskeleton, e.g., by disrupting the cell's microtubules, microfilaments, and/or intermediate filaments. The drug can be, for example, taxon, vincristine, vinblastine, cytochalasin, nocodazole, japlakinolide, latrunculin A, phalloidin, swinholide A, indanocine, or myoservin.

[0275] In some embodiments, a ligand attached to an iRNA as described herein acts as a pharmacokinetic modulator (PK modulator). PK modulators include lipophiles, bile acids, steroids, phospholipid analogues, peptides, protein binding agents, PEG, vitamins etc. Exemplary PK modulators include, but are not limited to, cholesterol, fatty acids, cholic acid, lithocholic acid, dialkylglycerides, diacylglyceride, phospholipids, sphingolipids, naproxen, ibuprofen, vitamin E, biotin etc. Oligonucleotides that comprise a number of phosphorothioate linkages are also known to bind to serum protein, thus short oligonucleotides, e.g., oligonucleotides of about 5 bases, 10 bases, 15 bases or 20 bases, comprising multiple of phosphorothioate linkages in the backbone are also amenable to the present invention as ligands (e.g. as PK modulating ligands). In addition, aptamers that bind serum components (e.g. serum proteins) are also suitable for use as PK modulating ligands in the embodiments described herein.

[0276] Ligand-conjugated oligonucleotides of the invention may be synthesized by the use of an oligonucleotide that bears a pendant reactive functionality, such as that derived from the attachment of a linking molecule onto the oligonucleotide (described below). This reactive oligonucleotide may be reacted directly with commercially-available ligands, ligands that are synthesized bearing any of a variety of protecting groups, or ligands that have a linking moiety attached thereto.

[0277] The oligonucleotides used in the conjugates of the present invention may be conveniently and routinely made through the well-known technique of solid-phase synthesis. Equipment for such synthesis is sold by several vendors including, for example, Applied Biosystems (Foster City, Calif.). Any other means for such synthesis known in the art may additionally or alternatively be employed. It is also known to use similar techniques to prepare other oligonucleotides, such as the phosphorothioates and alkylated derivatives.

[0278] In the ligand-conjugated oligonucleotides and ligand-molecule bearing sequence-specific linked nucleosides of the present invention, the oligonucleotides and oligonucleosides may be assembled on a suitable DNA synthesizer utilizing standard nucleotide or nucleoside precursors, or nucleotide or nucleoside conjugate precursors that already bear the linking moiety, ligand-nucleotide or nucleoside-conjugate precursors that already bear the ligand molecule, or non-

nucleoside ligand-bearing building blocks.
[0279] When using nucleotide-conjugate precursors that already bear a linking moiety, the synthesis of the sequence-specific linked nucleosides is typically completed, and the ligand molecule is then reacted with the linking moiety to form the ligand-conjugated oligonucleotide. In some embodiments, the oligonucleotides or linked nucleosides of the present invention are synthesized by an automated synthesizer using phosphoramidites derived from ligand-nucleoside conjugates in addition to the standard phosphoramidites and non-standard phosphoramidites that are commercially available and routinely used in oligonucleotide synthesis.

A. Lipid Conjugates

[0280] In one embodiment, the ligand or conjugate is a lipid or lipid-based molecule. Such a lipid or lipid-based molecule preferably binds a serum protein, e.g., human serum albumin (HSA). An HSA binding ligand allows for distribution of the conjugate to a target tissue, e.g., a non-kidney target tissue of the body. For example, the target tissue can be the liver, including parenchymal cells of the liver. Other molecules that can bind HSA can also be used as ligands. For example, naproxen or aspirin can be used. A lipid or lipid-based ligand can (a) increase resistance to degradation of the conjugate, (b) increase targeting or transport into a target cell or cell membrane, and/or (c) can be used to adjust binding to a serum protein, e.g., HSA.
[0281] A lipid based ligand can be used to inhibit, e.g., control the binding of the conjugate to a target tissue. For example, a lipid or lipid-based ligand that binds to HSA more strongly will be less likely to be targeted to the kidney and therefore less likely to be cleared from the body. A lipid or lipid-based ligand that binds to HSA less strongly can be used to target the conjugate to the kidney.
[0282] In a preferred embodiment, the lipid based ligand binds HSA. Preferably, it binds HSA with a sufficient affinity such that the conjugate will be preferably distributed to a non-kidney tissue. However, it is preferred that the affinity not be so strong that the HSA-ligand binding cannot be reversed.
[0283] In another preferred embodiment, the lipid based ligand binds HSA weakly or not at all, such that the conjugate will be preferably distributed to the kidney. Other moieties that target to kidney cells can also be used in place of or in addition to the lipid based ligand.
[0284] In another aspect, the ligand is a moiety, e.g., a vitamin, which is taken up by a target cell, e.g., a proliferating cell. These are particularly useful for treating disorders characterized by unwanted cell proliferation, e.g., of the malignant or non-malignant type, e.g., cancer cells. Exemplary vitamins include vitamin A, E, and K. Other exemplary vitamins include are B vitamin, e.g., folic acid, B12, riboflavin, biotin, pyridoxal or other vitamins or nutrients taken up by target cells such as liver cells. Also included are HSA and low density lipoprotein (LDL).

B. Cell Permeation Agents

[0285] In another aspect, the ligand is a cell-permeation agent, preferably a helical cell-permeation agent. Preferably, the agent is amphipathic. An exemplary agent is a peptide such as tat or antennopedia. If the agent is a peptide, it can be modified, including a peptidylmimetic, invertomers, non-peptide or pseudo-peptide linkages, and use of D-amino acids. The helical agent is preferably an alpha-helical agent, which preferably has a lipophilic and a lipophobic phase.
[0286] The ligand can be a peptide or peptidomimetic. A peptidomimetic (also referred to herein as an oligopeptidomimetic) is a molecule capable of folding into a defined three-dimensional structure similar to a natural peptide. The attachment of peptide and peptidomimetics to iRNA agents can affect pharmacokinetic distribution of the iRNA, such as by enhancing cellular recognition and absorption. The peptide or peptidomimetic moiety can be about 5-50 amino acids long, e.g., about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 amino acids long.
[0287] A peptide or peptidomimetic can be, for example, a cell permeation peptide, cationic peptide, amphipathic peptide, or hydrophobic peptide (e.g., consisting primarily of Tyr, Trp or Phe). The peptide moiety can be a dendrimer peptide, constrained peptide or crosslinked peptide. In another alternative, the peptide moiety can include a hydrophobic membrane translocation

sequence (MTS). An exemplary hydrophobic MTS-containing peptide is RFGF having the amino acid sequence AAVALLPAVLLALLAP (SEQ ID NO: 9). An RFGF analogue (e.g., amino acid sequence AALLPVLLAAP (SEQ ID NO: 10) containing a hydrophobic MTS can also be a targeting moiety. The peptide moiety can be a "delivery" peptide, which can carry large polar molecules including peptides, oligonucleotides, and protein across cell membranes. For example, sequences from the HIV Tat protein (GRKKRRQRRRPPQ (SEQ ID NO: 11) and the *Drosophila* Antennapedia protein (RQIKIWFQNRRMKWKK (SEQ ID NO: 12) have been found to be capable of functioning as delivery peptides. A peptide or peptidomimetic can be encoded by a random sequence of DNA, such as a peptide identified from a phage-display library, or one-bead-one-compound (OBOC) combinatorial library (Lam et al., *Nature,* 354:82-84, 1991). Examples of a peptide or peptidomimetic tethered to a dsRNA agent via an incorporated monomer unit for cell targeting purposes is an arginine-glycine-aspartic acid (RGD)-peptide, or RGD mimic. A peptide moiety can range in length from about 5 amino acids to about 40 amino acids. The peptide moieties can have a structural modification, such as to increase stability or direct conformational properties. Any of the structural modifications described below can be utilized.

[0288] An RGD peptide for use in the compositions and methods of the invention may be linear or cyclic, and may be modified, e.g., glycosylated or methylated, to facilitate targeting to a specific tissue(s). RGD-containing peptides and peptidiomimemtics may include D-amino acids, as well as synthetic RGD mimics. In addition to RGD, one can use other moieties that target the integrin ligand. Preferred conjugates of this ligand target PECAM-1 or VEGF.

[0289] A "cell permeation peptide" is capable of permeating a cell, e.g., a microbial cell, such as a bacterial or fungal cell, or a mammalian cell, such as a human cell. A microbial cell-permeating peptide can be, for example, a α-helical linear peptide (e.g., LL-37 or Ceropin P1), a disulfide bond-containing peptide (e.g., α-defensin, β-defensin or bactenecin), or a peptide containing only one or two dominating amino acids (e.g., PR-39 or indolicidin). A cell permeation peptide can also include a nuclear localization signal (NLS). For example, a cell permeation peptide can be a bipartite amphipathic peptide, such as MPG, which is derived from the fusion peptide domain of HIV-1 gp41 and the NLS of SV40 large T antigen (Simeoni et al., Nucl. Acids Res. 31:2717-2724, 2003).

C. Carbohydrate Conjugates

[0290] In some embodiments of the compositions and methods of the invention, an iRNA oligonucleotide further comprises a carbohydrate. The carbohydrate conjugated iRNA are advantageous for the in vivo delivery of nucleic acids, as well as compositions suitable for in vivo therapeutic use, as described herein. As used herein, "carbohydrate" refers to a compound which is either a carbohydrate per se made up of one or more monosaccharide units having at least 6 carbon atoms (which can be linear, branched or cyclic) with an oxygen, nitrogen or sulfur atom bonded to each carbon atom; or a compound having as a part thereof a carbohydrate moiety made up of one or more monosaccharide units each having at least six carbon atoms (which can be linear, branched or cyclic), with an oxygen, nitrogen or sulfur atom bonded to each carbon atom. Representative carbohydrates include the sugars (mono-, di-, tri- and oligosaccharides containing from about 4, 5, 6, 7, 8, or 9 monosaccharide units), and polysaccharides such as starches, glycogen, cellulose and polysaccharide gums. Specific monosaccharides include C5 and above (e.g., C5, C6, C7, or C8) sugars; di- and trisaccharides include sugars having two or three monosaccharide units (e.g., C5, C6, C7, or C8).

[0291] In one embodiment, a carbohydrate conjugate for use in the compositions and methods of the invention is a monosaccharide. In another embodiment, a carbohydrate conjugate for use in the compositions and methods of the invention is selected from the group consisting of:

##STR00003## ##STR00004## ##STR00005## ##STR00006##

[0292] In one embodiment, the monosaccharide is an N-acetylgalactosamine, such as

##STR00007##

[0293] Another representative carbohydrate conjugate for use in the embodiments described herein includes, but is not limited to,
##STR00008## [0294] when one of X or Y is an oligonucleotide, the other is a hydrogen.
[0295] In certain embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a monovalent linker. In some embodiments, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a bivalent linker. In yet other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a trivalent linker.
[0296] In one embodiment, the double stranded RNAi agents of the invention comprise one GalNAc or GalNAc derivative attached to the iRNA agent. In another embodiment, the double stranded RNAi agents of the invention comprise a plurality (e.g., 2, 3, 4, 5, or 6) GalNAc or GalNAc derivatives, each independently attached to a plurality of nucleotides of the double stranded RNAi agent through a plurality of monovalent linkers.
[0297] In some embodiments, for example, when the two strands of an iRNA agent of the invention are part of one larger molecule connected by an uninterrupted chain of nucleotides between the 3′-end of one strand and the 5′-end of the respective other strand forming a hairpin loop comprising, a plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker. The hairpin loop may also be formed by an extended overhang in one strand of the duplex.
[0298] In some embodiments, the carbohydrate conjugate further comprises one or more additional ligands as described above, such as, but not limited to, a PK modulator and/or a cell permeation peptide.
[0299] Additional carbohydrate conjugates suitable for use in the present invention include those described in PCT Publication Nos. WO 2014/179620 and WO 2014/179627, the entire contents of each of which are incorporated herein by reference.
D. Linkers
[0300] In some embodiments, the conjugate or ligand described herein can be attached to an iRNA oligonucleotide with various linkers that can be cleavable or non-cleavable.
[0301] The term “linker” or “linking group” means an organic moiety that connects two parts of a compound, e.g., covalently attaches two parts of a compound. Linkers typically comprise a direct bond or an atom such as oxygen or sulfur, a unit such as NR8, C(O), C(O)NH, SO, SO.sub.2, SO.sub.2NH or a chain of atoms, such as, but not limited to, substituted or unsubstituted alkyl, substituted or unsubstituted alkenyl, substituted or unsubstituted alkynyl, arylalkyl, arylalkenyl, arylalkynyl, heteroarylalkyl, heteroarylalkenyl, heteroarylalkynyl, heterocyclylalkyl, heterocyclylalkenyl, heterocyclylalkynyl, aryl, heteroaryl, heterocyclyl, cycloalkyl, cycloalkenyl, alkylarylalkyl, alkylarylalkenyl, alkylarylalkynyl, alkenylarylalkyl, alkenylarylalkenyl, alkenylarylalkynyl, alkynylarylalkyl, alkynylarylalkenyl, alkynylarylalkynyl, alkylheteroarylalkyl, alkylheteroarylalkenyl, alkylheteroarylalkynyl, alkenylheteroarylalkyl, alkenylheteroarylalkenyl, alkenylheteroarylalkynyl, alkynylheteroarylalkyl, alkynylheteroarylalkenyl, alkynylheteroarylalkynyl, alkylheterocyclylalkyl, alkylheterocyclylalkenyl, alkylhererocyclylalkynyl, alkenylheterocyclylalkyl, alkenylheterocyclylalkenyl, alkenylheterocyclylalkynyl, alkynylheterocyclylalkyl, alkynylheterocyclylalkenyl, alkynylheterocyclylalkynyl, alkylaryl, alkenylaryl, alkynylaryl, alkylheteroaryl, alkenylheteroaryl, alkynylhereroaryl, which one or more methylenes can be interrupted or terminated by O, S, S(O), SO.sub.2, N(R8), C(O), substituted or unsubstituted aryl, substituted or unsubstituted heteroaryl, substituted or unsubstituted heterocyclic; where R8 is hydrogen, acyl, aliphatic or substituted aliphatic. In one embodiment, the linker is between about 1-24 atoms, 2-24, 3-24, 4-24, 5-24, 6-24, 6-18, 7-18, 8-18 atoms, 7-17, 8-17, 6-16, 7-16, or 8-16 atoms.
[0302] A cleavable linking group is one which is sufficiently stable outside the cell, but which upon entry into a target cell is cleaved to release the two parts the linker is holding together. In a

preferred embodiment, the cleavable linking group is cleaved at least about 10 times, 20, times, 30 times, 40 times, 50 times, 60 times, 70 times, 80 times, 90 times or more, or at least about 100 times faster in a target cell or under a first reference condition (which can, e.g., be selected to mimic or represent intracellular conditions) than in the blood of a subject, or under a second reference condition (which can, e.g., be selected to mimic or represent conditions found in the blood or serum).

[0303] Cleavable linking groups are susceptible to cleavage agents, e.g., pH, redox potential or the presence of degradative molecules. Generally, cleavage agents are more prevalent or found at higher levels or activities inside cells than in serum or blood. Examples of such degradative agents include: redox agents which are selected for particular substrates or which have no substrate specificity, including, e.g., oxidative or reductive enzymes or reductive agents such as mercaptans, present in cells, that can degrade a redox cleavable linking group by reduction; esterases; endosomes or agents that can create an acidic environment, e.g., those that result in a pH of five or lower; enzymes that can hydrolyze or degrade an acid cleavable linking group by acting as a general acid, peptidases (which can be substrate specific), and phosphatases.

[0304] A cleavable linkage group, such as a disulfide bond can be susceptible to pH. The pH of human serum is 7.4, while the average intracellular pH is slightly lower, ranging from about 7.1-7.3. Endosomes have a more acidic pH, in the range of 5.5-6.0, and lysosomes have an even more acidic pH at around 5.0. Some linkers will have a cleavable linking group that is cleaved at a preferred pH, thereby releasing a cationic lipid from the ligand inside the cell, or into the desired compartment of the cell.

[0305] A linker can include a cleavable linking group that is cleavable by a particular enzyme. The type of cleavable linking group incorporated into a linker can depend on the cell to be targeted. For example, a liver-targeting ligand can be linked to a cationic lipid through a linker that includes an ester group. Liver cells are rich in esterases, and therefore the linker will be cleaved more efficiently in liver cells than in cell types that are not esterase-rich. Other cell-types rich in esterases include cells of the lung, renal cortex, and testis.

[0306] Linkers that contain peptide bonds can be used when targeting cell types rich in peptidases, such as liver cells and synoviocytes.

[0307] In general, the suitability of a candidate cleavable linking group can be evaluated by testing the ability of a degradative agent (or condition) to cleave the candidate linking group. It will also be desirable to also test the candidate cleavable linking group for the ability to resist cleavage in the blood or when in contact with other non-target tissue. Thus, one can determine the relative susceptibility to cleavage between a first and a second condition, where the first is selected to be indicative of cleavage in a target cell and the second is selected to be indicative of cleavage in other tissues or biological fluids, e.g., blood or serum. The evaluations can be carried out in cell free systems, in cells, in cell culture, in organ or tissue culture, or in whole animals. It can be useful to make initial evaluations in cell-free or culture conditions and to confirm by further evaluations in whole animals. In preferred embodiments, useful candidate compounds are cleaved at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or about 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood or serum (or under in vitro conditions selected to mimic extracellular conditions).

i. Redox Cleavable Linking Groups

[0308] In one embodiment, a cleavable linking group is a redox cleavable linking group that is cleaved upon reduction or oxidation. An example of reductively cleavable linking group is a disulphide linking group (—S—S—). To determine if a candidate cleavable linking group is a suitable “reductively cleavable linking group,” or for example is suitable for use with a particular iRNA moiety and particular targeting agent one can look to methods described herein. For example, a candidate can be evaluated by incubation with dithiothreitol (DTT), or other reducing agent using reagents know in the art, which mimic the rate of cleavage which would be observed in a cell, e.g., a target cell. The candidates can also be evaluated under conditions which are selected to mimic blood or serum conditions. In one, candidate compounds are cleaved by at most about 10% in the blood. In other embodiments, useful candidate compounds are degraded at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or about 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood (or under in vitro conditions selected to mimic extracellular conditions). The rate of cleavage of candidate compounds can be determined using standard enzyme kinetics assays under conditions chosen to mimic intracellular media and compared to conditions chosen to mimic extracellular media.

ii. Phosphate-Based Cleavable Linking Groups

[0309] In another embodiment, a cleavable linker comprises a phosphate-based cleavable linking group. A phosphate-based cleavable linking group is cleaved by agents that degrade or hydrolyze the phosphate group. An example of an agent that cleaves phosphate groups in cells are enzymes such as phosphatases in cells. Examples of phosphate-based linking groups are —O—P(O)(ORk)-O—, —O—P(S)(ORk)-O—, —O—P(S)(SRk)-O—, —S—P(O)(ORk)-O—, —O—P(O)(ORk)-S—, —S—P(O)(ORk)-S—, —O—P(S)(ORk)-S—, —S—P(S)(ORk)-O—, —O—P(O)(Rk)-O—, —O—P(S)(Rk)-O—, —S—P(O)(Rk)-O—, —S—P(S)(Rk)-O—, —S—P(O)(Rk)-S—, —O—P(S)(Rk)-S—. Preferred embodiments are —O—P(O)(OH)—O—, —O—P(S)(OH)—O—, —O—P(S)(SH)—O—, —S—P(O)(OH)—O—, —O—P(O)(OH)—S—, —S—P(O)(OH)—S—, —O—P(S)(OH)—S—, —S—P(S)(OH)—O—, —O—P(O)(H)—O—, —O—P(S)(H)—O—, —S—P(O)(H)—O, —S—P(S)(H)—O—, —S—P(O)(H)—S—, —O—P(S)(H)—S—. A preferred embodiment is —O—P(O)(OH)—O—. These candidates can be evaluated using methods analogous to those described above.

iii. Acid Cleavable Linking Groups

[0310] In another embodiment, a cleavable linker comprises an acid cleavable linking group. An acid cleavable linking group is a linking group that is cleaved under acidic conditions. In preferred embodiments acid cleavable linking groups are cleaved in an acidic environment with a pH of about 6.5 or lower (e.g., about 6.0, 5.75, 5.5, 5.25, 5.0, or lower), or by agents such as enzymes that can act as a general acid. In a cell, specific low pH organelles, such as endosomes and lysosomes can provide a cleaving environment for acid cleavable linking groups. Examples of acid cleavable linking groups include but are not limited to hydrazones, esters, and esters of amino acids. Acid cleavable groups can have the general formula —C═NN—, C(O)O, or —OC(O). A preferred embodiment is when the carbon attached to the oxygen of the ester (the alkoxy group) is an aryl group, substituted alkyl group, or tertiary alkyl group such as dimethyl pentyl or t-butyl. These candidates can be evaluated using methods analogous to those described above.

iv. Ester-Based Linking Groups

[0311] In another embodiment, a cleavable linker comprises an ester-based cleavable linking group. An ester-based cleavable linking group is cleaved by enzymes such as esterases and amidases in cells. Examples of ester-based cleavable linking groups include but are not limited to esters of alkylene, alkenylene and alkynylene groups. Ester cleavable linking groups have the general formula —C(O)O—, or —OC(O)—. These candidates can be evaluated using methods analogous to those described above.

v. Peptide-Based Cleaving Groups

[0312] In yet another embodiment, a cleavable linker comprises a peptide-based cleavable linking group. A peptide-based cleavable linking group is cleaved by enzymes such as peptidases and proteases in cells. Peptide-based cleavable linking groups are peptide bonds formed between amino acids to yield oligopeptides (e.g., dipeptides, tripeptides etc.) and polypeptides. Peptide-based cleavable groups do not include the amide group (—C(O)NH—). The amide group can be formed between any alkylene, alkenylene or alkynelene. A peptide bond is a special type of amide bond formed between amino acids to yield peptides and proteins. The peptide based cleavage group is generally limited to the peptide bond (i.e., the amide bond) formed between amino acids yielding peptides and proteins and does not include the entire amide functional group. Peptide-based cleavable linking groups have the general formula —NHCHRAC(O)NHCHRBC(O)—, where RA and RB are the R groups of the two adjacent amino acids. These candidates can be evaluated using methods analogous to those described above.

[0313] In one embodiment, an iRNA of the invention is conjugated to a carbohydrate through a linker. Non-limiting examples of iRNA carbohydrate conjugates with linkers of the compositions and methods of the invention include, but are not limited to,

##STR00009## ##STR00010## ##STR00011##

[0314] when one of X or Y is an oligonucleotide, the other is a hydrogen.

[0315] In certain embodiments of the compositions and methods of the invention, a ligand is one or more “GalNAc” (N-acetylgalactosamine) derivatives attached through a bivalent or trivalent branched linker.

[0316] In one embodiment, a dsRNA of the invention is conjugated to a bivalent or trivalent branched linker selected from the group of structures shown in any of formula (XXXII)—(XXXV):

##STR00012##

wherein: [0317] q.sup.2A, q.sup.2B, q.sup.3A, q.sup.3B, q.sup.4A, q.sup.4B, q.sup.5A, q.sup.5B and q.sup.5C represent independently for each occurrence 0-20 and wherein the repeating unit can be the same or different; [0318] P.sup.2A, P.sup.2B, P.sup.3A, P.sup.3B, P.sup.4A, P.sup.4B, P.sup.5A, P.sup.5B, P.sup.5C, T.sup.2A, T.sup.2B, T.sup.3A, T.sup.3B, T.sup.4A, T.sup.4B, T.sup.5A, T.sup.5B, T.sup.5C are each independently for each occurrence absent, CO, NH, O, S, OC(O), NHC(O), CH.sub.2, CH.sub.2NH or CH.sub.2O; [0319] Q.sup.2A, Q.sup.2B, Q.sup.3A, Q.sup.3B, Q.sup.4A, Q.sup.4B, Q.sup.5A, Q.sup.5B, Q.sup.5C are independently for each occurrence absent, alkylene, substituted alkylene wherein one or more methylenes can be interrupted or terminated by one or more of O, S, S(O), SO.sub.2, N(R.sup.N), C(R′)═C(R″), C≡C or C(O); [0320] R.sup.2A, R.sup.2B, R.sup.3A, R.sup.3B, R.sup.4A, R.sup.4B, R.sup.5A, R.sup.5B, R.sup.5C are each independently for each occurrence absent, NH, O, S, CH.sub.2, C(O)O, C(O)NH, NHCH(R.sup.a)C(O), —C(O)—CH(R.sup.a)—NH—, CO, CH═N—O,

##STR00013##

or heterocyclyl; [0321] L.sup.2A, L.sup.2B, L.sup.3A, L.sup.3B, L.sup.4A, L.sup.4B, L.sup.5A, L.sup.5B and L.sup.5C represent the ligand; i.e. each independently for each occurrence a monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, oligosaccharide, or polysaccharide; and R.sup.a is H or amino acid side chain. Trivalent conjugating GalNAc derivatives are particularly useful for use with RNAi agents for inhibiting the expression of a target gene, such as those of formula (XXXV):

##STR00014## [0322] wherein L.sup.5A, L.sup.5B and L.sup.5C represent a monosaccharide, such as GalNAc derivative.

[0323] Examples of suitable bivalent and trivalent branched linker groups conjugating GalNAc derivatives include, but are not limited to, the structures recited above as formulas II, VII, XI, X, and XIII.

[0324] Representative U.S. patents that teach the preparation of RNA conjugates include, but are not limited to, U.S. Pat. Nos. 4,828,979; 4,948,882; 5,218,105; 5,525,465; 5,541,313; 5,545,730; 5,552,538; 5,578,717, 5,580,731; 5,591,584; 5,109,124; 5,118,802; 5,138,045; 5,414,077; 5,486,603; 5,512,439; 5,578,718; 5,608,046; 4,587,044; 4,605,735; 4,667,025; 4,762,779; 4,789,737; 4,824,941; 4,835,263; 4,876,335; 4,904,582; 4,958,013; 5,082,830; 5,112,963; 5,214,136; 5,082,830; 5,112,963; 5,214,136; 5,245,022; 5,254,469; 5,258,506; 5,262,536; 5,272,250; 5,292,873; 5,317,098; 5,371,241, 5,391,723; 5,416,203, 5,451,463; 5,510,475; 5,512,667; 5,514,785; 5,565,552; 5,567,810; 5,574,142; 5,585,481; 5,587,371; 5,595,726; 5,597,696; 5,599,923; 5,599,928 and 5,688,941; 6,294,664; 6,320,017; 6,576,752; 6,783,931; 6,900,297; 7,037,646; 8,106,022, the entire contents of each of which are hereby incorporated herein by reference.

[0325] It is not necessary for all positions in a given compound to be uniformly modified, and in fact more than one of the aforementioned modifications can be incorporated in a single compound or even at a single nucleoside within an iRNA. The present invention also includes iRNA compounds that are chimeric compounds.

[0326] “Chimeric” iRNA compounds or “chimeras,” in the context of this invention, are iRNA compounds, preferably dsRNAs, which contain two or more chemically distinct regions, each made up of at least one monomer unit, i.e., a nucleotide in the case of a dsRNA compound. These iRNAs typically contain at least one region wherein the RNA is modified so as to confer upon the iRNA increased resistance to nuclease degradation, increased cellular uptake, and/or increased binding affinity for the target nucleic acid. An additional region of the iRNA can serve as a substrate for enzymes capable of cleaving RNA:DNA or RNA:RNA hybrids. By way of example, RNase H is a cellular endonuclease which cleaves the RNA strand of an RNA:DNA duplex. Activation of RNase H, therefore, results in cleavage of the RNA target, thereby greatly enhancing the efficiency of iRNA inhibition of gene expression. Consequently, comparable results can often be obtained with shorter iRNAs when chimeric dsRNAs are used, compared to phosphorothioate deoxy dsRNAs hybridizing to the same target region. Cleavage of the RNA target can be routinely detected by gel electrophoresis and, if necessary, associated nucleic acid hybridization techniques known in the art.

[0327] In certain instances, the RNA of an iRNA can be modified by a non-ligand group. A number of non-ligand molecules have been conjugated to iRNAs in order to enhance the activity, cellular distribution or cellular uptake of the iRNA, and procedures for performing such conjugations are available in the scientific literature. Such non-ligand moieties have included lipid moieties, such as cholesterol (Kubo, T. et al., *Biochem. Biophys. Res. Comm.*, 2007, 365(1):54-61; Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86:6553), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4:1053), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660:306; Manoharan et al., *Bioorg. Med. Chem. Let.*, 1993, 3:2765), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20:533), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., *EMBO J.*, 1991, 10:111; Kabanov et al., *FEBS Lett.*, 1990, 259:327; Svinarchuk et al., *Biochimie*, 1993, 75:49), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethylammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651; Shea et al., *Nucl. Acids Res.*, 1990, 18:3777), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14:969), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651), a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264:229), or an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277:923). Representative United States patents that teach the preparation of such RNA conjugates have been listed above. Typical conjugation protocols involve the synthesis of an RNAs bearing an aminolinker at one or more positions of the sequence. The amino group is then reacted with the molecule being conjugated using appropriate coupling or activating reagents. The conjugation reaction can be performed either with the RNA still bound to the solid support or following cleavage of the RNA, in solution phase. Purification of the RNA conjugate by HPLC typically affords the pure conjugate.

V. Delivery of an iRNA of the Invention

[0328] The delivery of an iRNA of the invention to a cell e.g., a cell within a subject, such as a human subject (e.g., a subject in need thereof, such as a subject having a APCS-associated disease as described herein) can be achieved in a number of different ways. For example, delivery may be performed by contacting a cell with an iRNA of the invention either in vitro or in vivo. In vivo delivery may also be performed directly by administering a composition comprising an iRNA, e.g., a dsRNA, to a subject. Alternatively, in vivo delivery may be performed indirectly by administering one or more vectors that encode and direct the expression of the iRNA. These alternatives are discussed further below.

[0329] In general, any method of delivering a nucleic acid molecule (in vitro or in vivo) can be adapted for use with an iRNA of the invention (see e.g., Akhtar S. and Julian R L. (1992) *Trends Cell. Biol.* 2(5):139-144 and WO94/02595, which are incorporated herein by reference in their entireties). For in vivo delivery, factors to consider in order to deliver an iRNA molecule include, for example, biological stability of the delivered molecule, prevention of non-specific effects, and accumulation of the delivered molecule in the target tissue. The non-specific effects of an iRNA can be minimized by local administration, for example, by direct injection or implantation into a tissue or topically administering the preparation. Local administration to a treatment site maximizes local concentration of the agent, limits the exposure of the agent to systemic tissues that can otherwise be harmed by the agent or that can degrade the agent, and permits a lower total dose of the iRNA molecule to be administered. Several studies have shown successful knockdown of gene products when an iRNA is administered locally. For example, intraocular delivery of a VEGF dsRNA by intravitreal injection in cynomolgus monkeys (Tolentino, M J., et al (2004) *Retina* 24:132-138) and subretinal injections in mice (Reich, S J., et al (2003) *Mol. Vis.* 9:210-216) were both shown to prevent neovascularization in an experimental model of age-related macular degeneration. In addition, direct intratumoral injection of a dsRNA in mice reduces tumor volume (Pille, J., et al (2005) *Mol. Ther.* 11:267-274) and can prolong survival of tumor-bearing mice (Kim, W J., et al (2006) *Mol. Ther.* 14:343-350; Li, S., et al (2007) *Mol. Ther.* 15:515-523). RNA interference has also shown success with local delivery to the CNS by direct injection (Dorn, G., et al. (2004) *Nucleic Acids* 32:e49; Tan, P H., et al (2005) *Gene Ther.* 12:59-66; Makimura, H., et al (2002) *BMC Neurosci.* 3:18; Shishkina, G T., et al (2004) *Neuroscience* 129:521-528; Thakker, E R., et al (2004) *Proc. Natl. Acad. Sci. U.S.A.* 101:17270-17275; Akaneya, Y., et al (2005) *J. Neurophysiol.* 93:594-602) and to the lungs by intranasal administration (Howard, K A., et al (2006) *Mol. Ther.* 14:476-484; Zhang, X., et al (2004) *J. Biol. Chem.* 279:10677-10684; Bitko, V., et al (2005) *Nat. Med.* 11:50-55). For administering an iRNA systemically for the treatment of a disease, the RNA can be modified or alternatively delivered using a drug delivery system; both methods act to prevent the rapid degradation of the dsRNA by endo- and exo-nucleases in vivo. Modification of the RNA or the pharmaceutical carrier can also permit targeting of the iRNA composition to the target tissue and avoid undesirable off-target effects. iRNA molecules can be modified by chemical conjugation to lipophilic groups such as cholesterol to enhance cellular uptake and prevent degradation. For example, an iRNA directed against ApoB conjugated to a lipophilic cholesterol moiety was injected systemically into mice and resulted in knockdown of apoB mRNA in both the liver and jejunum (Soutschek, J., et al (2004) *Nature* 432:173-178). Conjugation of an iRNA to an aptamer has been shown to inhibit tumor growth and mediate tumor regression in a mouse model of prostate cancer (McNamara, J O., et al (2006) *Nat. Biotechnol.* 24:1005-1015). In an alternative embodiment, the iRNA can be delivered using drug delivery systems such as a nanoparticle, a dendrimer, a polymer, liposomes, or a cationic delivery system. Positively charged cationic delivery systems facilitate binding of an iRNA molecule (negatively charged) and also enhance interactions at the negatively charged cell membrane to permit efficient uptake of an iRNA by the cell. Cationic lipids, dendrimers, or polymers can either be bound to an iRNA, or induced to form a vesicle or micelle (see e.g., Kim S H., et al (2008) *Journal of Controlled Release* 129(2):107-116) that encases an iRNA. The formation of vesicles or micelles further prevents degradation of the iRNA when administered systemically. Methods for making and administering cationic-iRNA complexes are well within the abilities of one skilled in the art (see e.g., Sorensen, D R., et al (2003) *J. Mol. Biol* 327:761-766; Verma, U N., et al (2003) *Clin. Cancer Res.* 9:1291-1300; Arnold, A S et al (2007) *J. Hypertens.* 25:197-205, which are incorporated herein by reference in their entirety). Some non-limiting examples of drug delivery systems useful for systemic delivery of iRNAs include DOTAP (Sorensen, D R., et al (2003), supra; Verma, U N., et al (2003), supra), Oligofectamine, “solid nucleic acid lipid particles” (Zimmermann, T S., et al (2006) *Nature* 441:111-114), cardiolipin (Chien, P Y., et al (2005) *Cancer Gene Ther.* 12:321-328;

Pal, A., et al (2005) *Int J. Oncol.* 26:1087-1091), polyethyleneimine (Bonnet M E., et al (2008) *Pharm. Res*. August 16 Epub ahead of print; *Aigner, A.* (2006) *J. Biomed. Biotechnol.* 71659), Arg-Gly-Asp (RGD) peptides (Liu, S. (2006) *Mol. Pharm.* 3:472-487), and polyamidoamines (Tomalia, D A., et al (2007) *Biochem. Soc. Trans.* 35:61-67; Yoo, H., et al (1999) *Pharm. Res.* 16:1799-1804). In some embodiments, an iRNA forms a complex with cyclodextrin for systemic administration. Methods for administration and pharmaceutical compositions of iRNAs and cyclodextrins can be found in U.S. Pat. No. 7,427,605, which is herein incorporated by reference in its entirety.

A. Vector Encoded iRNAs of the Invention

[0330] iRNA targeting a APCS gene can be expressed from transcription units inserted into DNA or RNA vectors (see, e.g., Couture, A, et al., *TIG*. (1996), 12:5-10; Skillern, A., et al., International PCT Publication No. WO 00/22113, Conrad, International PCT Publication No. WO 00/22114, and Conrad, U.S. Pat. No. 6,054,299). Expression can be transient (on the order of hours to weeks) or sustained (weeks to months or longer), depending upon the specific construct used and the target tissue or cell type. These transgenes can be introduced as a linear construct, a circular plasmid, or a viral vector, which can be an integrating or non-integrating vector. The transgene can also be constructed to permit it to be inherited as an extrachromosomal plasmid (Gassmann, et al., *Proc. Natl. Acad. Sci. USA* (1995) 92:1292).

[0331] The individual strand or strands of an iRNA can be transcribed from a promoter on an expression vector. Where two separate strands are to be expressed to generate, for example, a dsRNA, two separate expression vectors can be co-introduced (e.g., by transfection or infection) into a target cell. Alternatively each individual strand of a dsRNA can be transcribed by promoters both of which are located on the same expression plasmid. In one embodiment, a dsRNA is expressed as inverted repeat polynucleotides joined by a linker polynucleotide sequence such that the dsRNA has a stem and loop structure.

[0332] iRNA expression vectors are generally DNA plasmids or viral vectors. Expression vectors compatible with eukaryotic cells, preferably those compatible with vertebrate cells, can be used to produce recombinant constructs for the expression of an iRNA as described herein. Eukaryotic cell expression vectors are well known in the art and are available from a number of commercial sources. Typically, such vectors are provided containing convenient restriction sites for insertion of the desired nucleic acid segment. Delivery of iRNA expressing vectors can be systemic, such as by intravenous or intramuscular administration, by administration to target cells ex-planted from the patient followed by reintroduction into the patient, or by any other means that allows for introduction into a desired target cell.

[0333] iRNA expression plasmids can be transfected into target cells as a complex with cationic lipid carriers (e.g., Oligofectamine) or non-cationic lipid-based carriers (e.g., Transit-TKO™) Multiple lipid transfections for iRNA-mediated knockdowns targeting different regions of a target RNA over a period of a week or more are also contemplated by the invention. Successful introduction of vectors into host cells can be monitored using various known methods. For example, transient transfection can be signaled with a reporter, such as a fluorescent marker, such as Green Fluorescent Protein (GFP). Stable transfection of cells ex vivo can be ensured using markers that provide the transfected cell with resistance to specific environmental factors (e.g., antibiotics and drugs), such as hygromycin B resistance.

[0334] Viral vector systems which can be utilized with the methods and compositions described herein include, but are not limited to, (a) adenovirus vectors; (b) retrovirus vectors, including but not limited to lentiviral vectors, moloney murine leukemia virus, etc.; (c) adeno-associated virus vectors; (d) herpes simplex virus vectors; (e) SV 40 vectors; (f) polyoma virus vectors; (g) papilloma virus vectors; (h) picornavirus vectors; (i) pox virus vectors such as an orthopox, e.g., vaccinia virus vectors or avipox, e.g. canary pox or fowl pox; and (j) a helper-dependent or gutless adenovirus. Replication-defective viruses can also be advantageous. Different vectors will or will not become incorporated into the cells' genome. The constructs can include viral sequences for transfection, if desired. Alternatively, the construct can be incorporated into vectors capable of episomal replication, e.g. EPV and EBV vectors. Constructs for the recombinant expression of an iRNA will generally require regulatory elements, e.g., promoters, enhancers, etc., to ensure the expression of the iRNA in target cells. Other aspects to consider for vectors and constructs are further described below.

[0335] Vectors useful for the delivery of an iRNA will include regulatory elements (promoter, enhancer, etc.) sufficient for expression of the iRNA in the desired target cell or tissue. The regulatory elements can be chosen to provide either constitutive or regulated/inducible expression.

[0336] Expression of the iRNA can be precisely regulated, for example, by using an inducible regulatory sequence that is sensitive to certain physiological regulators, e.g., circulating glucose levels, or hormones (Docherty et al., 1994, *FASEB J.* 8:20-24). Such inducible expression systems, suitable for the control of dsRNA expression in cells or in mammals include, for example, regulation by ecdysone, by estrogen, progesterone, tetracycline, chemical inducers of dimerization, and isopropyl-beta-D1-thiogalactopyranoside (IPTG). A person skilled in the art would be able to choose the appropriate regulatory/promoter sequence based on the intended use of the iRNA transgene.

[0337] Viral vectors that contain nucleic acid sequences encoding an iRNA can be used. For example, a retroviral vector can be used (see Miller et al., *Meth. Enzymol.* 217:581-599 (1993)). These retroviral vectors contain the components necessary for the correct packaging of the viral genome and integration into the host cell DNA. The nucleic acid sequences encoding an iRNA are cloned into one or more vectors, which facilitate delivery of the nucleic acid into a patient. More detail about retroviral vectors can be found, for example, in Boesen et al., *Biotherapy* 6:291-302 (1994), which describes the use of a retroviral vector to deliver the mdr1 gene to hematopoietic stem cells in order to make the stem cells more resistant to chemotherapy. Other references illustrating the use of retroviral vectors in gene therapy are: Clowes et al., *J. Clin. Invest.* 93:644-651 (1994); Kiem et al., *Blood* 83:1467-1473 (1994); Salmons and Gunzberg, *Human Gene Therapy* 4:129-141 (1993); and Grossman and Wilson, *Curr. Opin. in Genetics and Devel.* 3:110-114 (1993). Lentiviral vectors contemplated for use include, for example, the HIV based vectors described in U.S. Pat. Nos. 6,143,520; 5,665,557; and 5,981,276, which are herein incorporated by reference.

[0338] Adenoviruses are also contemplated for use in delivery of iRNAs of the invention. Adenoviruses are especially attractive vehicles, e.g., for delivering genes to respiratory epithelia. Adenoviruses naturally infect respiratory epithelia where they cause a mild disease. Other targets for adenovirus-based delivery systems are liver, the central nervous system, endothelial cells, and muscle. Adenoviruses have the advantage of being capable of infecting non-dividing cells. Kozarsky and Wilson, *Current Opinion in Genetics and Development* 3:499-503 (1993) present a review of adenovirus-based gene therapy. Bout et al., *Human Gene Therapy* 5:3-10 (1994) demonstrated the use of adenovirus vectors to transfer genes to the respiratory epithelia of rhesus monkeys. Other instances of the use of adenoviruses in gene therapy can be found in Rosenfeld et al., *Science* 252:431-434 (1991); Rosenfeld et al., *Cell* 68:143-155 (1992); Mastrangeli et al., *J. Clin. Invest.* 91:225-234 (1993); PCT Publication WO94/12649; and Wang, et al., *Gene Therapy* 2:775-783 (1995). A suitable AV vector for expressing an iRNA featured in the invention, a method for constructing the recombinant AV vector, and a method for delivering the vector into target cells, are described in Xia H et al. (2002), *Nat. Biotech.* 20: 1006-1010.

[0339] Adeno-associated virus (AAV) vectors may also be used to delivery an iRNA of the invention (Walsh et al., *Proc. Soc. Exp. Biol. Med.* 204:289-300 (1993); U.S. Pat. No. 5,436,146). In one embodiment, the iRNA can be expressed as two separate, complementary single-stranded RNA molecules from a recombinant AAV vector having, for example, either the U6 or H1 RNA promoters, or the cytomegalovirus (CMV) promoter. Suitable AAV vectors for expressing the dsRNA featured in the invention, methods for constructing the recombinant AV vector, and methods for delivering the vectors into target cells are described in Samulski R et al. (1987), *J. Virol.* 61: 3096-3101; Fisher K J et al. (1996), *J. Virol*, 70: 520-532; Samulski R et al. (1989), *J. Virol.* 63: 3822-3826; U.S. Pat. Nos. 5,252,479; 5,139,941; International Patent Application No. WO 94/13788; and International Patent Application No. WO 93/24641, the entire disclosures of which are herein incorporated by reference.

[0340] Another viral vector suitable for delivery of an iRNA of the invention is a pox virus such as a vaccinia virus, for example an attenuated vaccinia such as Modified Virus Ankara (MVA) or NYVAC, an avipox such as fowl pox or canary pox.

[0341] The tropism of viral vectors can be modified by pseudotyping the vectors with envelope proteins or other surface antigens from other viruses, or by substituting different viral capsid proteins, as appropriate. For example, lentiviral vectors can be pseudotyped with surface proteins from vesicular stomatitis virus (VSV), rabies, Ebola, Mokola, and the like. AAV vectors can be made to target different cells by engineering the vectors to express different capsid protein serotypes; see, e.g., Rabinowitz J E et al. (2002), *J Virol* 76:791-801, the entire disclosure of which is herein incorporated by reference.

[0342] The pharmaceutical preparation of a vector can include the vector in an acceptable diluent, or can include a slow release matrix in which the gene delivery vehicle is imbedded. Alternatively, where the complete gene delivery vector can be produced intact from recombinant cells, e.g., retroviral vectors, the pharmaceutical preparation can include one or more cells which produce the gene delivery system.

VI. Pharmaceutical Compositions of the Invention

[0343] The present invention also includes pharmaceutical compositions and formulations which include the iRNAs of the invention. In one embodiment, provided herein are pharmaceutical compositions containing an iRNA, as described herein, and a pharmaceutically acceptable carrier. The pharmaceutical compositions containing the iRNA are useful for treating a disease or disorder associated with the expression or activity of an APCS gene, e.g. an APCS-associated disease as descried herein. Such pharmaceutical compositions are formulated based on the mode of delivery. One example is compositions that are formulated for systemic administration via parenteral delivery, e.g., by subcutaneous (SC), intramuscular (IM), or intravenous (IV) delivery. Another example is compositions that are formulated for direct delivery into the brain parenchyma, e.g., by infusion into the brain, such as by continuous pump infusion. The pharmaceutical compositions of the invention may be administered in dosages sufficient to inhibit expression of the target gene.

[0344] The pharmaceutical compositions of the present invention can be administered in a number of ways depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration can be topical (e.g., by a transdermal patch), pulmonary, e.g., by inhalation or insufflation of powders or aerosols, including by nebulizer; intratracheal, intranasal, epidermal and transdermal, oral or parenteral. Parenteral administration includes intravenous, intraarterial, subcutaneous, intraperitoneal or intramuscular injection or infusion; subdermal, e.g., via an implanted device; or intracranial, e.g., by intraparenchymal, intrathecal or intraventricular, administration.

[0345] The iRNA can be delivered in a manner to target a particular tissue, such as the liver (e.g., the hepatocytes of the liver).

[0346] In some embodiments, the pharmaceutical compositions of the invention are suitable for intramuscular administration to a subject. In other embodiments, the pharmaceutical compositions of the invention are suitable for intravenous administration to a subject. In some embodiments of the invention, the pharmaceutical compositions of the invention are suitable for subcutaneous administration to a subject, e.g., using a 29 g or 30 g needle.

[0347] The pharmaceutical compositions of the invention may include an RNAi agent of the invention in an unbuffered solution, such as saline or water, or in a buffer solution, such as a buffer solution comprising acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof.

[0348] In one embodiment, the pharmaceutical compositions of the invention, e.g., such as the compositions suitable for subcutaneous administration, comprise an RNAi agent of the invention in phosphate buffered saline (PBS). Suitable concentrations of PBS include, for example, 1 mM, 1.5 mM, 2 mM, 2.5 mM, 3 mM, 3.5 mM, 4 mM, 4.5 mM, 5 mM, 6.5 mM, 7 mM, 7.5 mM, 9 mM, 8.5 mM, 9 mM, 9.5 mM, or about 10 mM PBS. In one embodiment of the invention, a pharmaceutical composition of the invention comprises an RNAi agent of the invention dissolved in a solution of about 5 mM PBS (e.g., 0.64 mM NaH.sub.2PO.sub.4, 4.36 mM Na.sub.2HPO.sub.4, 85 mM NaCl). Values intermediate to the above recited ranges and values are also intended to be part of this invention. In addition, ranges of values using a combination of any of the above recited values as upper and/or lower limits are intended to be included.

[0349] The pH of the pharmaceutical compositions of the invention may be between about 5.0 to about 8.0, about 5.5 to about 8.0, about 6.0 to about 8.0, about 6.5 to about 8.0, about 7.0 to about 8.0, about 5.0 to about 7.5, about 5.5 to about 7.5, about 6.0 to about 7.5, about 6.5 to about 7.5, about 5.0 to about 7.2, about 5.25 to about 7.2, about 5.5 to about 7.2, about 5.75 to about 7.2, about 6.0 to about 7.2, about 6.5 to about 7.2, or about 6.8 to about 7.2. Ranges and values intermediate to the above recited ranges and values are also intended to be part of this invention.

[0350] The osmolality of the pharmaceutical compositions of the invention may be suitable for subcutaneous administration, such as no more than about 400 mOsm/kg, e.g., between 50 and 400 mOsm/kg, between 75 and 400 mOsm/kg, between 100 and 400 mOsm/kg, between 125 and 400 mOsm/kg, between 150 and 400 mOsm/kg, between 175 and 400 mOsm/kg, between 200 and 400 mOsm/kg, between 250 and 400 mOsm/kg, between 300 and 400 mOsm/kg, between 50 and 375 mOsm/kg, between 75 and 375 mOsm/kg, between 100 and 375 mOsm/kg, between 125 and 375 mOsm/kg, between 150 and 375 mOsm/kg, between 175 and 375 mOsm/kg, between 200 and 375 mOsm/kg, between 250 and 375 mOsm/kg, between 300 and 375 mOsm/kg, between 50 and 350 mOsm/kg, between 75 and 350 mOsm/kg, between 100 and 350 mOsm/kg, between 125 and 350 mOsm/kg, between 150 and 350 mOsm/kg, between 175 and 350 mOsm/kg, between 200 and 350 mOsm/kg, between 250 and 350 mOsm/kg, between 50 and 325 mOsm/kg, between 75 and 325 mOsm/kg, between 100 and 325 mOsm/kg, between 125 and 325 mOsm/kg, between 150 and 325 mOsm/kg, between 175 and 325 mOsm/kg, between 200 and 325 mOsm/kg, between 250 and 325 mOsm/kg, between 300 and 325 mOsm/kg, between 300 and 350 mOsm/kg, between 50 and 300 mOsm/kg, between 75 and 300 mOsm/kg, between 100 and 300 mOsm/kg, between 125 and 300 mOsm/kg, between 150 and 300 mOsm/kg, between 175 and 300 mOsm/kg, between 200 and 300 mOsm/kg, between 250 and 300, between 50 and 250 mOsm/kg, between 75 and 250 mOsm/kg, between 100 and 250 mOsm/kg, between 125 and 250 mOsm/kg, between 150 and 250 mOsm/kg, between 175 and 350 mOsm/kg, between 200 and 250 mOsm/kg, e.g., about 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 295, 300, 305, 310, 320, 325, 330, 335, 340, 345, 350, 355, 360, 365, 370, 375, 380, 385, 390, 395, or about 400 mOsm/kg. Ranges and values intermediate to the above recited ranges and values are also intended to be part of this invention.

[0351] The pharmaceutical compositions of the invention comprising the RNAi agents of the invention, may be present in a vial that contains about 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, or about 2.0 mL of the pharmaceutical composition. The concentration of the RNAi agents in the pharmaceutical compositions of the invention may be about 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 130, 125, 130, 135, 140, 145, 150, 175, 180, 185, 190, 195, 200, 205, 210, 215, 230, 225, 230, 235, 240, 245, 250, 275, 280, 285, 290, 295, 300, 305, 310, 315, 330, 325, 330, 335, 340, 345, 350, 375, 380, 385, 390, 395, 400, 405, 410, 415, 430, 425, 430, 435, 440, 445, 450, 475, 480, 485, 490, 495, or about 500 mg/mL. In one embodiment, the concentration of the RNAi agents in the pharmaceutical compositions of the invention is about 100 mg/mL. Values intermediate to the above recited ranges and values are also intended to be part of this invention.

[0352] The pharmaceutical compositions of the invention may comprise a dsRNA agent of the invention in a free acid form. In other embodiments of the invention, the pharmaceutical compositions of the invention may comprise a dsRNA agent of the invention in a salt form, such as a sodium salt form. In certain embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for substantially all of the phosphodiester and/or phosphorothiotate groups present in the agent. Agents in which substantially all of the phosphodiester and/or phosphorothioate linkages have a sodium counterion include not more than 5, 4, 3, 2, or 1 phosphodiester and/or phosphorothioate linkages without a sodium counterion. In some embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for all of the phosphodiester and/or phosphorothiotate groups present in the agent.

[0353] Pharmaceutical compositions and formulations for topical administration can include transdermal patches, ointments, lotions, creams, gels, drops, suppositories, sprays, liquids and powders. Conventional pharmaceutical carriers, aqueous, powder, or oily bases, thickeners and the like can be necessary or desirable. Coated condoms, gloves and the like can also be useful. Suitable topical formulations include those in which the iRNAs featured in the invention are in admixture with a topical delivery agent such as lipids, liposomes, fatty acids, fatty acid esters, steroids, chelating agents and surfactants. Suitable lipids and liposomes include neutral (e.g., dioleoylphosphatidyl DOPE ethanolamine, dimyristoylphosphatidyl choline DMPC, distearolyphosphatidyl choline) negative (e.g., dimyristoylphosphatidyl glycerol DMPG), and cationic (e.g., dioleoyltetramethylaminopropyl DOTAP and dioleoylphosphatidyl ethanolamine DOTMA). iRNAs featured in the invention can be encapsulated within liposomes or can form complexes thereto, in particular to cationic liposomes. Alternatively, iRNAs can be complexed to lipids, in particular to cationic lipids. Suitable fatty acids and esters include but are not limited to arachidonic acid, oleic acid, eicosanoic acid, lauric acid, caprylic acid, capric acid, myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprate, tricaprate, monoolein, dilaurin, glyceryl 1-monocaprate, 1-dodecylazacycloheptan-2-one, an acylcarnitine, an acylcholine, or a C.sub.1-20 alkyl ester (e.g., isopropylmyristate IPM), monoglyceride or diglyceride or pharmaceutically acceptable salt thereof. Topical formulations are described in detail in U.S. Pat. No. 6,747,014, which is incorporated herein by reference.

A. iRNA Formulations Comprising Membranous Molecular Assemblies

[0354] An iRNA for use in the compositions and methods of the invention can be formulated for delivery in a membranous molecular assembly, e.g., a liposome or a micelle. As used herein, the term “liposome” refers to a vesicle composed of amphiphilic lipids arranged in at least one bilayer, e.g., one bilayer or a plurality of bilayers. Liposomes include unilamellar and multilamellar vesicles that have a membrane formed from a lipophilic material and an aqueous interior. The aqueous portion contains the iRNA composition. The lipophilic material isolates the aqueous interior from an aqueous exterior, which typically does not include the iRNA composition, although in some examples, it may. Liposomes are useful for the transfer and delivery of active ingredients to the site of action. Because the liposomal membrane is structurally similar to biological membranes, when liposomes are applied to a tissue, the liposomal bilayer fuses with bilayer of the cellular membranes. As the merging of the liposome and cell progresses, the internal aqueous contents that include the iRNA are delivered into the cell where the iRNA can specifically bind to a target RNA and can mediate RNAi. In some cases the liposomes are also specifically targeted, e.g., to direct the iRNA to particular cell types.

[0355] A liposome containing an RNAi agent can be prepared by a variety of methods. In one example, the lipid component of a liposome is dissolved in a detergent so that micelles are formed with the lipid component. For example, the lipid component can be an amphipathic cationic lipid or lipid conjugate. The detergent can have a high critical micelle concentration and may be nonionic.

Exemplary detergents include cholate, CHAPS, octylglucoside, deoxycholate, and lauroyl sarcosine. The RNAi agent preparation is then added to the micelles that include the lipid component. The cationic groups on the lipid interact with the RNAi agent and condense around the RNAi agent to form a liposome. After condensation, the detergent is removed, e.g., by dialysis, to yield a liposomal preparation of RNAi agent.

[0356] If necessary a carrier compound that assists in condensation can be added during the condensation reaction, e.g., by controlled addition. For example, the carrier compound can be a polymer other than a nucleic acid (e.g., spermine or spermidine). pH can also adjusted to favor condensation.

[0357] Methods for producing stable polynucleotide delivery vehicles, which incorporate a polynucleotide/cationic lipid complex as structural components of the delivery vehicle, are further described in, e.g., WO 96/37194, the entire contents of which are incorporated herein by reference. Liposome formation can also include one or more aspects of exemplary methods described in Felgner, P. L. et al., *Proc. Natl. Acad. Sci., USA* 8:7413-7417, 1987; U.S. Pat. Nos. 4,897,355; 5,171,678; Bangham, et al. *M. Mol. Biol.* 23:238, 1965; Olson, et al. *Biochim. Biophys. Acta* 557:9, 1979; Szoka, et al. *Proc. Natl. Acad. Sci.* 75: 4194, 1978; Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984; Kim, et al. *Biochim. Biophys. Acta* 728:339, 1983; and Fukunaga, et al. *Endocrinol.* 115:757, 1984. Commonly used techniques for preparing lipid aggregates of appropriate size for use as delivery vehicles include sonication and freeze-thaw plus extrusion (see, e.g., Mayer, et al. *Biochim. Biophys. Acta* 858:161, 1986). Microfluidization can be used when consistently small (50 to 200 nm) and relatively uniform aggregates are desired (Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984). These methods are readily adapted to packaging RNAi agent preparations into liposomes.

[0358] Liposomes fall into two broad classes. Cationic liposomes are positively charged liposomes which interact with the negatively charged nucleic acid molecules to form a stable complex. The positively charged nucleic acid/liposome complex binds to the negatively charged cell surface and is internalized in an endosome. Due to the acidic pH within the endosome, the liposomes are ruptured, releasing their contents into the cell cytoplasm (Wang et al., *Biochem. Biophys. Res. Commun.*, 1987, 147, 980-985).

[0359] Liposomes which are pH-sensitive or negatively-charged, entrap nucleic acids rather than complex with it. Since both the nucleic acid and the lipid are similarly charged, repulsion rather than complex formation occurs. Nevertheless, some nucleic acid is entrapped within the aqueous interior of these liposomes. pH-sensitive liposomes have been used to deliver nucleic acids encoding the thymidine kinase gene to cell monolayers in culture. Expression of the exogenous gene was detected in the target cells (Zhou et al., *Journal of Controlled Release,* 1992, 19, 269-274).

[0360] One major type of liposomal composition includes phospholipids other than naturally-derived phosphatidylcholine. Neutral liposome compositions, for example, can be formed from dimyristoyl phosphatidylcholine (DMPC) or dipalmitoyl phosphatidylcholine (DPPC). Anionic liposome compositions generally are formed from dimyristoyl phosphatidylglycerol, while anionic fusogenic liposomes are formed primarily from dioleoyl phosphatidylethanolamine (DOPE). Another type of liposomal composition is formed from phosphatidylcholine (PC) such as, for example, soybean PC, and egg PC. Another type is formed from mixtures of phospholipid and/or phosphatidylcholine and/or cholesterol.

[0361] Examples of other methods to introduce liposomes into cells in vitro and in vivo include U.S. Pat. Nos. 5,283,185; 5,171,678; WO 94/00569; WO 93/24640; WO 91/16024; Felgner, *J. Biol. Chem.* 269:2550, 1994*; Nabel, Proc. Natl. Acad. Sci.* 90:11307, 1993; Nabel, *Human Gene Ther.* 3:649, 1992*; Gershon, Biochem.* 32:7143, 1993; and Strauss *EMBO J.* 11:417, 1992.

[0362] Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol.

Non-ionic liposomal formulations comprising Novasome™ I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and Novasome™ II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver cyclosporin-A into the dermis of mouse skin. Results indicated that such non-ionic liposomal systems were effective in facilitating the deposition of cyclosporine A into different layers of the skin (Hu et al. *S.T.P.Pharma. Sci.,* 1994, 4(6) 466).

[0363] Liposomes also include “sterically stabilized” liposomes, a term which, as used herein, refers to liposomes comprising one or more specialized lipids that, when incorporated into liposomes, result in enhanced circulation lifetimes relative to liposomes lacking such specialized lipids. Examples of sterically stabilized liposomes are those in which part of the vesicle-forming lipid portion of the liposome (A) comprises one or more glycolipids, such as monosialoganglioside G.sub.M1, or (B) is derivatized with one or more hydrophilic polymers, such as a polyethylene glycol (PEG) moiety. While not wishing to be bound by any particular theory, it is thought in the art that, at least for sterically stabilized liposomes containing gangliosides, sphingomyelin, or PEG-derivatized lipids, the enhanced circulation half-life of these sterically stabilized liposomes derives from a reduced uptake into cells of the reticuloendothelial system (RES) (Allen et al., *FEBS Letters*, 1987, 223, 42; Wu et al., *Cancer Research,* 1993, 53, 3765).

[0364] Various liposomes comprising one or more glycolipids are known in the art. Papahadjopoulos et al. (*Ann. N.Y. Acad. Sci.,* 1987, 507, 64) reported the ability of monosialoganglioside G.sub.M1, galactocerebroside sulfate and phosphatidylinositol to improve blood half-lives of liposomes. These findings were expounded upon by Gabizon et al. (*Proc. Natl. Acad. Sci. U.S.A.,* 1988, 85, 6949). U.S. Pat. No. 4,837,028 and WO 88/04924, both to Allen et al., disclose liposomes comprising (1) sphingomyelin and (2) the ganglioside G.sub.M1 or a galactocerebroside sulfate ester. U.S. Pat. No. 5,543,152 (Webb et al.) discloses liposomes comprising sphingomyelin. Liposomes comprising 1,2-sn-dimyristoylphosphatidylcholine are disclosed in WO 97/13499 (Lim et al).

[0365] In one embodiment, cationic liposomes are used. Cationic liposomes possess the advantage of being able to fuse to the cell membrane. Non-cationic liposomes, although not able to fuse as efficiently with the plasma membrane, are taken up by macrophages in vivo and can be used to deliver RNAi agents to macrophages.

[0366] Further advantages of liposomes include: liposomes obtained from natural phospholipids are biocompatible and biodegradable; liposomes can incorporate a wide range of water and lipid soluble drugs; liposomes can protect encapsulated RNAi agents in their internal compartments from metabolism and degradation (Rosoff, in “Pharmaceutical Dosage Forms,” Lieberman, Rieger and Banker (Eds.), 1988, volume 1, p. 245). Important considerations in the preparation of liposome formulations are the lipid surface charge, vesicle size and the aqueous volume of the liposomes.

[0367] A positively charged synthetic cationic lipid, N-[1-(2,3-dioleyloxy)propyl]-N,N,N-trimethylammonium chloride (DOTMA) can be used to form small liposomes that interact spontaneously with nucleic acid to form lipid-nucleic acid complexes which are capable of fusing with the negatively charged lipids of the cell membranes of tissue culture cells, resulting in delivery of RNAi agent (see, e.g., Felgner, P. L. et al., *Proc. Natl. Acad. Sci., USA* 8:7413-7417, 1987 and U.S. Pat. No. 4,897,355 for a description of DOTMA and its use with DNA).

[0368] A DOTMA analogue, 1,2-bis(oleoyloxy)-3-(trimethylammonia)propane (DOTAP) can be used in combination with a phospholipid to form DNA-complexing vesicles. Lipofectin™ Bethesda Research Laboratories, Gaithersburg, Md.) is an effective agent for the delivery of highly anionic nucleic acids into living tissue culture cells that comprise positively charged DOTMA liposomes which interact spontaneously with negatively charged polynucleotides to form complexes. When enough positively charged liposomes are used, the net charge on the resulting complexes is also positive. Positively charged complexes prepared in this way spontaneously attach to negatively charged cell surfaces, fuse with the plasma membrane, and efficiently deliver functional nucleic acids into, for example, tissue culture cells. Another commercially available cationic lipid, 1,2-bis(oleoyloxy)-3,3-(trimethylammonia)propane ("DOTAP") (Boehringer Mannheim, Indianapolis, Indiana) differs from DOTMA in that the oleoyl moieties are linked by ester, rather than ether linkages.

[0369] Other reported cationic lipid compounds include those that have been conjugated to a variety of moieties including, for example, carboxyspermine which has been conjugated to one of two types of lipids and includes compounds such as 5-carboxyspermylglycine dioctaoleoylamide ("DOGS") (Transfectam™, Promega, Madison, Wisconsin) and dipalmitoylphosphatidylethanolamine 5-carboxyspermyl-amide ("DPPES") (see, e.g., U.S. Pat. No. 5,171,678).

[0370] Another cationic lipid conjugate includes derivatization of the lipid with cholesterol ("DC-Chol") which has been formulated into liposomes in combination with DOPE (See, Gao, X. and Huang, L., *Biochim. Biophys. Res. Commun.* 179:280, 1991). Lipopolylysine, made by conjugating polylysine to DOPE, has been reported to be effective for transfection in the presence of serum (Zhou, X. et al., *Biochim. Biophys. Acta* 1065:8, 1991). For certain cell lines, these liposomes containing conjugated cationic lipids, are said to exhibit lower toxicity and provide more efficient transfection than the DOTMA-containing compositions. Other commercially available cationic lipid products include DMRIE and DMRIE-HP (Vical, La Jolla, California) and Lipofectamine (DOSPA) (Life Technology, Inc., Gaithersburg, Maryland). Other cationic lipids suitable for the delivery of oligonucleotides are described in WO 98/39359 and WO 96/37194.

[0371] Liposomal formulations are particularly suited for topical administration, liposomes present several advantages over other formulations. Such advantages include reduced side effects related to high systemic absorption of the administered drug, increased accumulation of the administered drug at the desired target, and the ability to administer RNAi agent into the skin. In some implementations, liposomes are used for delivering RNAi agent to epidermal cells and also to enhance the penetration of RNAi agent into dermal tissues, e.g., into skin. For example, the liposomes can be applied topically. Topical delivery of drugs formulated as liposomes to the skin has been documented (see, e.g., Weiner et al., *Journal of Drug Targeting,* 1992, vol. 2,405-410 and du Plessis et al., *Antiviral Research,* 18, 1992, 259-265; Mannino, R. J. and Fould-Fogerite, S., Biotechniques 6:682-690, 1988; Itani, T. et al. *Gene* 56:267-276. 1987; Nicolau, C. et al. *Meth. Enz.* 149:157-176, 1987; Straubinger, *R. M. and Papahadjopoulos, D. Meth. Enz.* 101:512-527, 1983; Wang, C. Y. and Huang, *L., Proc. Natl. Acad. Sci. USA* 84:7851-7855, 1987).

[0372] Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol. Non-ionic liposomal formulations comprising Novasome I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and Novasome II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver a drug into the dermis of mouse skin. Such formulations with RNAi agent are useful for treating a dermatological disorder.

[0373] Liposomes that include iRNA can be made highly deformable. Such deformability can enable the liposomes to penetrate through pore that are smaller than the average radius of the liposome. For example, transfersomes are a type of deformable liposomes. Transferosomes can be made by adding surface edge activators, usually surfactants, to a standard liposomal composition. Transfersomes that include RNAi agent can be delivered, for example, subcutaneously by infection in order to deliver RNAi agent to keratinocytes in the skin. In order to cross intact mammalian skin, lipid vesicles must pass through a series of fine pores, each with a diameter less than 50 nm, under the influence of a suitable transdermal gradient. In addition, due to the lipid properties, these transferosomes can be self-optimizing (adaptive to the shape of pores, e.g., in the skin), self-repairing, and can frequently reach their targets without fragmenting, and often self-loading.

[0374] Other formulations amenable to the present invention are described in PCT Publication No. WO 2008/042973.
[0375] Transfersomes are yet another type of liposomes, and are highly deformable lipid aggregates which are attractive candidates for drug delivery vehicles. Transfersomes can be described as lipid droplets which are so highly deformable that they are easily able to penetrate through pores which are smaller than the droplet. Transfersomes are adaptable to the environment in which they are used, e.g., they are self-optimizing (adaptive to the shape of pores in the skin), self-repairing, frequently reach their targets without fragmenting, and often self-loading. To make transfersomes it is possible to add surface edge-activators, usually surfactants, to a standard liposomal composition. Transfersomes have been used to deliver serum albumin to the skin. The transfersome-mediated delivery of serum albumin has been shown to be as effective as subcutaneous injection of a solution containing serum albumin.
[0376] Surfactants find wide application in formulations such as emulsions (including microemulsions) and liposomes. The most common way of classifying and ranking the properties of the many different types of surfactants, both natural and synthetic, is by the use of the hydrophile/lipophile balance (HLB). The nature of the hydrophilic group (also known as the “head”) provides the most useful means for categorizing the different surfactants used in formulations (Rieger, in “Pharmaceutical Dosage Forms”, Marcel Dekker, Inc., New York, N.Y., 1988, p. 285).
[0377] If the surfactant molecule is not ionized, it is classified as a nonionic surfactant. Nonionic surfactants find wide application in pharmaceutical and cosmetic products and are usable over a wide range of pH values. In general their HLB values range from 2 to about 18 depending on their structure. Nonionic surfactants include nonionic esters such as ethylene glycol esters, propylene glycol esters, glyceryl esters, polyglyceryl esters, sorbitan esters, sucrose esters, and ethoxylated esters. Nonionic alkanolamides and ethers such as fatty alcohol ethoxylates, propoxylated alcohols, and ethoxylated/propoxylated block polymers are also included in this class. The polyoxyethylene surfactants are the most popular members of the nonionic surfactant class.
[0378] If the surfactant molecule carries a negative charge when it is dissolved or dispersed in water, the surfactant is classified as anionic. Anionic surfactants include carboxylates such as soaps, acyl lactylates, acyl amides of amino acids, esters of sulfuric acid such as alkyl sulfates and ethoxylated alkyl sulfates, sulfonates such as alkyl benzene sulfonates, acyl isethionates, acyl taurates and sulfosuccinates, and phosphates. The most important members of the anionic surfactant class are the alkyl sulfates and the soaps.
[0379] If the surfactant molecule carries a positive charge when it is dissolved or dispersed in water, the surfactant is classified as cationic. Cationic surfactants include quaternary ammonium salts and ethoxylated amines. The quaternary ammonium salts are the most used members of this class.
[0380] If the surfactant molecule has the ability to carry either a positive or negative charge, the surfactant is classified as amphoteric. Amphoteric surfactants include acrylic acid derivatives, substituted alkylamides, N-alkylbetaines and phosphatides.
[0381] The use of surfactants in drug products, formulations and in emulsions has been reviewed (Rieger, in “Pharmaceutical Dosage Forms”, Marcel Dekker, Inc., New York, N.Y., 1988, p. 285).
[0382] The iRNA for use in the methods of the invention can also be provided as micellar formulations. “Micelles” are defined herein as a particular type of molecular assembly in which amphipathic molecules are arranged in a spherical structure such that all the hydrophobic portions of the molecules are directed inward, leaving the hydrophilic portions in contact with the surrounding aqueous phase. The converse arrangement exists if the environment is hydrophobic.
[0383] A mixed micellar formulation suitable for delivery through transdermal membranes may be prepared by mixing an aqueous solution of the siRNA composition, an alkali metal C.sub.8 to C.sub.22 alkyl sulphate, and a micelle forming compounds. Exemplary micelle forming

compounds include lecithin, hyaluronic acid, pharmaceutically acceptable salts of hyaluronic acid, glycolic acid, lactic acid, chamomile extract, cucumber extract, oleic acid, linoleic acid, linolenic acid, monoolein, monooleates, monolaurates, borage oil, evening of primrose oil, menthol, trihydroxy oxo cholanyl glycine and pharmaceutically acceptable salts thereof, glycerin, polyglycerin, lysine, polylysine, triolein, polyoxyethylene ethers and analogues thereof, polidocanol alkyl ethers and analogues thereof, chenodeoxycholate, deoxycholate, and mixtures thereof. The micelle forming compounds may be added at the same time or after addition of the alkali metal alkyl sulphate. Mixed micelles will form with substantially any kind of mixing of the ingredients but vigorous mixing in order to provide smaller size micelles.

[0384] In one method a first micellar composition is prepared which contains the siRNA composition and at least the alkali metal alkyl sulphate. The first micellar composition is then mixed with at least three micelle forming compounds to form a mixed micellar composition. In another method, the micellar composition is prepared by mixing the siRNA composition, the alkali metal alkyl sulphate and at least one of the micelle forming compounds, followed by addition of the remaining micelle forming compounds, with vigorous mixing.

[0385] Phenol and/or m-cresol may be added to the mixed micellar composition to stabilize the formulation and protect against bacterial growth. Alternatively, phenol and/or m-cresol may be added with the micelle forming ingredients. An isotonic agent such as glycerin may also be added after formation of the mixed micellar composition.

[0386] For delivery of the micellar formulation as a spray, the formulation can be put into an aerosol dispenser and the dispenser is charged with a propellant. The propellant, which is under pressure, is in liquid form in the dispenser. The ratios of the ingredients are adjusted so that the aqueous and propellant phases become one, i.e., there is one phase. If there are two phases, it is necessary to shake the dispenser prior to dispensing a portion of the contents, e.g., through a metered valve. The dispensed dose of pharmaceutical agent is propelled from the metered valve in a fine spray.

[0387] Propellants may include hydrogen-containing chlorofluorocarbons, hydrogen-containing fluorocarbons, dimethyl ether and diethyl ether. In certain embodiments, HFA 134a (1,1,1,2 tetrafluoroethane) may be used.

[0388] The specific concentrations of the essential ingredients can be determined by relatively straightforward experimentation. For absorption through the oral cavities, it is often desirable to increase, e.g., at least double or triple, the dosage for through injection or administration through the gastrointestinal tract.

B. Lipid Particles

[0389] iRNAs, e.g., dsRNAs of in the invention may be fully encapsulated in a lipid formulation, e.g., a LNP, or other nucleic acid-lipid particle.

[0390] As used herein, the term “LNP” refers to a stable nucleic acid-lipid particle. LNPs typically contain a cationic lipid, a non-cationic lipid, and a lipid that prevents aggregation of the particle (e.g., a PEG-lipid conjugate). LNPs are extremely useful for systemic applications, as they exhibit extended circulation lifetimes following intravenous (i.v.) injection and accumulate at distal sites (e.g., sites physically separated from the administration site). LNPs include “pSPLP,” which include an encapsulated condensing agent-nucleic acid complex as set forth in PCT Publication No. WO 00/03683. The particles of the present invention typically have a mean diameter of about 50 nm to about 150 nm, more typically about 60 nm to about 130 nm, more typically about 70 nm to about 110 nm, most typically about 70 nm to about 90 nm, and are substantially nontoxic. In addition, the nucleic acids when present in the nucleic acid-lipid particles of the present invention are resistant in aqueous solution to degradation with a nuclease. Nucleic acid-lipid particles and their method of preparation are disclosed in, e.g., U.S. Pat. Nos. 5,976,567; 5,981,501; 6,534,484; 6,586,410; 6,815,432; U.S. Publication No. 2010/0324120 and PCT Publication No. WO 96/40964.

[0391] In one embodiment, the lipid to drug ratio (mass/mass ratio) (e.g., lipid to dsRNA ratio) will be in the range of from about 1:1 to about 50:1, from about 1:1 to about 25:1, from about 3:1 to about 15:1, from about 4:1 to about 10:1, from about 5:1 to about 9:1, or about 6:1 to about 9:1. Ranges intermediate to the above recited ranges are also contemplated to be part of the invention.

[0392] The cationic lipid can be, for example, N,N-dioleyl-N,N-dimethylammonium chloride (DODAC), N,N-distearyl-N,N-dimethylammonium bromide (DDAB), N—(I-(2,3-dioleoyloxy)propyl)-N,N,N-trimethylammonium chloride (DOTAP), N—(I-(2,3-dioleyloxy)propyl)-N,N,N-trimethylammonium chloride (DOTMA), N,N-dimethyl-2,3-dioleyloxy)propylamine (DODMA), 1,2-DiLinoleyloxy-N,N-dimethylaminopropane (DLinDMA), 1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLenDMA), 1,2-Dilinoleylcarbamoyloxy-3-dimethylaminopropane (DLin-C-DAP), 1,2-Dilinoleyoxy-3-(dimethylamino)acetoxypropane (DLin-DAC), 1,2-Dilinoleyoxy-3-morpholinopropane (DLin-MA), 1,2-Dilinoleoyl-3-dimethylaminopropane (DLinDAP), 1,2-Dilinoleylthio-3-dimethylaminopropane (DLin-S-DMA), 1-Linoleoyl-2-linoleyloxy-3-dimethylaminopropane (DLin-2-DMAP), 1,2-Dilinoleyloxy-3-trimethylaminopropane chloride salt (DLin-TMA.Cl), 1,2-Dilinoleoyl-3-trimethylaminopropane chloride salt (DLin-TAP.Cl), 1,2-Dilinoleyloxy-3-(N-methylpiperazino)propane (DLin-MPZ), or 3-(N,N-Dilinoleylamino)-1,2-propanediol (DLinAP), 3-(N,N-Dioleylamino)-1,2-propanedio (DOAP), 1,2-Dilinoleyloxo-3-(2-N,N-dimethylamino)ethoxypropane (DLin-EG-DMA), 1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLinDMA), 2,2-Dilinoleyl-4-dimethylaminomethyl-[1,3]-dioxolane (DLin-K-DMA) or analogs thereof, (3aR,5s,6aS)—N,N-dimethyl-2,2-di((9Z,12Z)-octadeca-9,12-dienyl)tetrahydro-3aH-cyclopenta[d][1,3]dioxol-5-amine (ALN100), (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yl 4-(dimethylamino)butanoate (MC3), 1,1′-(2-(4-(2-((2-(bis(2-hydroxydodecyl)amino)ethyl)(2-hydroxydodecyl)amino)ethyl)piperazin-1-yl)ethylazanediyl)didodecan-2-ol (Tech G1), or a mixture thereof. The cationic lipid can comprise from about 20 mol % to about 50 mol % or about 40 mol % of the total lipid present in the particle.

[0393] In another embodiment, the compound 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane can be used to prepare lipid-siRNA nanoparticles. Synthesis of 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane is described in U.S. provisional patent application No. 61/107,998 filed on Oct. 23, 2008, which is herein incorporated by reference.

[0394] In one embodiment, the lipid-siRNA particle includes 40% 2, 2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane: 10% DSPC: 40% Cholesterol: 10% PEG-C-DOMG (mole percent) with a particle size of 63.0±20 nm and a 0.027 siRNA/Lipid Ratio.

[0395] The ionizable/non-cationic lipid can be an anionic lipid or a neutral lipid including, but not limited to, distearoylphosphatidylcholine (DSPC), dioleoylphosphatidylcholine (DOPC), dipalmitoylphosphatidylcholine (DPPC), dioleoylphosphatidylglycerol (DOPG), dipalmitoylphosphatidylglycerol (DPPG), dioleoyl-phosphatidylethanolamine (DOPE), palmitoyloleoylphosphatidylcholine (POPC), palmitoyloleoylphosphatidylethanolamine (POPE), dioleoyl-phosphatidylethanolamine 4-(N-maleimidomethyl)-cyclohexane-1-carboxylate (DOPE-mal), dipalmitoyl phosphatidyl ethanolamine (DPPE), dimyristoylphosphoethanolamine (DMPE), distearoyl-phosphatidyl-ethanolamine (DSPE), 16-O-monomethyl PE, 16-O-dimethyl PE, 18-1-trans PE, 1-stearoyl-2-oleoyl-phosphatidyethanolamine (SOPE), cholesterol, or a mixture thereof. The non-cationic lipid can be from about 5 mol % to about 90 mol %, about 10 mol %, or about 58 mol % if cholesterol is included, of the total lipid present in the particle.

[0396] The conjugated lipid that inhibits aggregation of particles can be, for example, a polyethyleneglycol (PEG)-lipid including, without limitation, a PEG-diacylglycerol (DAG), a PEG-dialkyloxypropyl (DAA), a PEG-phospholipid, a PEG-ceramide (Cer), or a mixture thereof. The PEG-DAA conjugate can be, for example, a PEG-dilauryloxypropyl (Ci.sub.2), a PEG-dimyristyloxypropyl (Ci.sub.4), a PEG-dipalmityloxypropyl (Ci.sub.6), or a PEG-distearyloxypropyl (C].sub.8). The conjugated lipid that prevents aggregation of particles can be from 0 mol % to about 20 mol % or about 2 mol % of the total lipid present in the particle.

[0397] In some embodiments, the nucleic acid-lipid particle further includes cholesterol at, e.g., about 10 mol % to about 60 mol % or about 48 mol % of the total lipid present in the particle.

[0398] In one embodiment, the lipidoid ND98·4HCl (MW 1487) (see U.S. patent application Ser. No. 12/056,230, filed Mar. 26, 2008, which is incorporated herein by reference), Cholesterol (Sigma-Aldrich), and PEG-Ceramide C16 (Avanti Polar Lipids) can be used to prepare lipid-dsRNA nanoparticles (i.e., LNP01 particles). Stock solutions of each in ethanol can be prepared as follows: ND98, 133 mg/ml; Cholesterol, 25 mg/ml, PEG-Ceramide C16, 100 mg/ml. The ND98, Cholesterol, and PEG-Ceramide C16 stock solutions can then be combined in a, e.g., 42:48:10 molar ratio. The combined lipid solution can be mixed with aqueous dsRNA (e.g., in sodium acetate pH 5) such that the final ethanol concentration is about 35-45% and the final sodium acetate concentration is about 100-300 mM. Lipid-dsRNA nanoparticles typically form spontaneously upon mixing. Depending on the desired particle size distribution, the resultant nanoparticle mixture can be extruded through a polycarbonate membrane (e.g., 100 nm cut-off) using, for example, a thermobarrel extruder, such as Lipex Extruder (Northern Lipids, Inc). In some cases, the extrusion step can be omitted. Ethanol removal and simultaneous buffer exchange can be accomplished by, for example, dialysis or tangential flow filtration. Buffer can be exchanged with, for example, phosphate buffered saline (PBS) at about pH 7, e.g., about pH 6.9, about pH 7.0, about pH 7.1, about pH 7.2, about pH 7.3, or about pH 7.4.

##STR00015##

[0399] LNP01 formulations are described, e.g., in International Application Publication No. WO 2008/042973, which is hereby incorporated by reference.

[0400] Additional exemplary lipid-dsRNA formulations are described in Table 1.

TABLE-US-00001 TABLE 1 cationic lipid/non-cationic lipid/cholesterol/PEG conjugate Ionizable/Cationic Lipid Lipid:siRNA ratio SNALP-1 1,2-Dilinolenyloxy-N,N-dimethylaminopropane DLinDMA/DPPC/Cholesterol/PEG-cDMA (DLinDMA) (57.1/7.1/34.4/1.4) lipid:siRNA~7:l 2-XTC 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-diox text missing or illegible when filed XTC/DPPC/Cholesterol/PEG-cDMA (XTC) 57.1/7.1/34.4/1.4 lipid:siRNA~7:1 LNP05 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-diox text missing or illegible when filed XTC/DSPC/Cholesterol/PEG-DMG (XTC) 57.5/7.5/31.5/3.5 lipid:siRNA~6:1 LNP06 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-diox text missing or illegible when filed XTC/DSPC/Cholesterol/PEG-DMG (XTC) 57.5/7.5/31.5/3.5 lipid:siRNA~11:1 LNP07 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-diox text missing or illegible when filed XTC/DSPC/Cholesterol/PEG-DMG (XTC) 60/7.5/31/1.5, lipid:siRNA~6:1 LNP08 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-diox text missing or illegible when filed XTC/DSPC/Cholesterol/PEG-DMG (XTC) 60/7.5/31/1.5, lipid:siRNA~11:1 LNP09 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-diox text missing or illegible when filed XTC/DSPC/Cholesterol/PEG-DMG (XTC) 50/10/38.5/1.5 Lipid:siRNA 10:1 LNP10 (3aR,5s,6aS)-N,N-dimethyl-2,2-di((9Z,12Z)-oc text missing or illegible when filed ALN100/DSPC/Cholesterol/PEG-DMG 9,12-dienyl)tetrahydro-3aH-cyclopenta[d][1,3]d text missing or illegible when filed 50/10/38.5/1.5 amine (ALN100) Lipid:siRNA 10:1 LNP11 (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetra text missing or illegible when filed MC-3/DSPC/Cholesterol/PEG-DMG 4-(dimethylamino)butanoate (MC3) 50/10/38.5/1.5 Lipid:siRNA 10:1 LNP12 1,1′-(2-(4-(2-((2-(bis(2-hydroxydodecyl)amino) Tech G1/DSPC/Cholesterol/PEG-DMG hydroxydodecyl)amino)ethyl)piperazin-1- 50/10/38.5/1.5 yl)ethylazanediyl)didodecan-2-ol (Tech G1) Lipid:siRNA 10:1 LNP13 XTC XTC/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 33:1 LNP14 MC3 MC3/DSPC/Chol/PEG-DMG 40/15/40/5 Lipid:siRNA: 11:1 LNP15 MC3 MC3/DSPC/Chol/PEG-DSG/GalNAc-PEG-DSG 50/10/35/4.5/0.5 Lipid:siRNA: 11:1 LNP16 MC3 MC3/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 7:1 LNP17 MC3 MC3/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 10:1 LNP18 MC3 MC3/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 12:1 LNP19 MC3

MC3/DSPC/Chol/PEG-DMG 50/10/35/5 Lipid:siRNA: 8:1 LNP20 MC3 MC3/DSPC/Chol/PEG-DPG 50/10/38.5/1.5 Lipid:siRNA: 10:1 LNP21 C12-200 C12-200/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 7:1 LNP22 XTC XTC/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 10:1 DSPC: distearoylphosphatidylcholine DPPC: dipalmitoylphosphatidylcholine PEG-DMG: PEG-didimyristoyl glycerol (C14-PEG, or PEG-C14) (PEG with avg mol wt of 2000); PEG-DSG: PEG-distyryl glycerol (C18-PEG, or PEG-C18) (PEG with avg mol wt of 2000); PEG-cDMA: PEG-carbamoyl-1,2-dimyristyloxypropylamine (PEG with avg mol wt of 2000); SNALP (1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLinDMA)) comprising formulations are described in International Publication No. WO2009/127060, filed Apr. 15, 2009, which is hereby incorporated by reference. XTC comprising formulations are described in PCT Publication No. WO 2010/088537, the entire contents of which are hereby incorporated herein by reference. MC3 comprising formulations are described, e.g., in U.S. Publication No. 2010/0324120, filed Jun. 10, 2010, the entire contents of which are hereby incorporated by reference. ALNY-100 comprising formulations are described in PCT Publication No. WO 2010/054406, the entire contents of which are hereby incorporated herein by reference. C12-200 comprising formulations are described in PCT Publication No. WO 2010/129709, the entire contents of which are hereby incorporated herein by reference. text missing or illegible when filed indicates data missing or illegible when filed

[0401] Compositions and formulations for oral administration include powders or granules, microparticulates, nanoparticulates, suspensions or solutions in water or non-aqueous media, capsules, gel capsules, sachets, tablets or minitablets. Thickeners, flavoring agents, diluents, emulsifiers, dispersing aids or binders can be desirable. In some embodiments, oral formulations are those in which dsRNAs featured in the invention are administered in conjunction with one or more penetration enhancer surfactants and chelators. Suitable surfactants include fatty acids and/or esters or salts thereof, bile acids and/or salts thereof. Suitable bile acids/salts include chenodeoxycholic acid (CDCA) and ursodeoxychenodeoxycholic acid (UDCA), cholic acid, dehydrocholic acid, deoxycholic acid, glucholic acid, glycholic acid, glycodeoxycholic acid, taurocholic acid, taurodeoxycholic acid, sodium tauro-24,25-dihydro-fusidate and sodium glycodihydrofusidate. Suitable fatty acids include arachidonic acid, undecanoic acid, oleic acid, lauric acid, caprylic acid, capric acid, myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprate, tricaprate, monoolein, dilaurin, glyceryl 1-monocaprate, 1-dodecylazacycloheptan-2-one, an acylcarnitine, an acylcholine, or a monoglyceride, a diglyceride or a pharmaceutically acceptable salt thereof (e.g., sodium). In some embodiments, combinations of penetration enhancers are used, for example, fatty acids/salts in combination with bile acids/salts. One exemplary combination is the sodium salt of lauric acid, capric acid and UDCA. Further penetration enhancers include polyoxyethylene-9-lauryl ether, polyoxyethylene-20-cetyl ether. RNAi agents featured in the invention can be delivered orally, in granular form including sprayed dried particles, or complexed to form micro or nanoparticles. RNAi complexing agents include poly-amino acids; polyimines; polyacrylates; polyalkylacrylates, polyoxethanes, polyalkylcyanoacrylates; cationized gelatins, albumins, starches, acrylates, polyethyleneglycols (PEG) and starches; polyalkylcyanoacrylates; DEAE-derivatized polyimines, pollulans, celluloses and starches. Suitable complexing agents include chitosan, N-trimethylchitosan, poly-L-lysine, polyhistidine, polyornithine, polyspermines, protamine, polyvinylpyridine, polythiodiethylaminomethylethylene P(TDAE), polyaminostyrene (e.g., p-amino), poly(methylcyanoacrylate), poly(ethylcyanoacrylate), poly(butylcyanoacrylate), poly(isobutylcyanoacrylate), poly(isohexylcynaoacrylate), DEAE-methacrylate, DEAE-hexylacrylate, DEAE-acrylamide, DEAE-albumin and DEAE-dextran, polymethylacrylate, polyhexylacrylate, poly(D,L-lactic acid), poly(DL-lactic-co-glycolic acid (PLGA), alginate, and polyethyleneglycol (PEG). Oral formulations for dsRNAs and their preparation are described in detail in U.S. Pat. No. 6,887,906, US Publn. No. 20030027780, and U.S. Pat. No. 6,747,014, each of which is incorporated herein by reference.

[0402] Compositions and formulations for parenteral, intraparenchymal (into the brain), intrathecal, intraventricular or intrahepatic administration can include sterile aqueous solutions which can also contain buffers, diluents and other suitable additives such as, but not limited to, penetration enhancers, carrier compounds and other pharmaceutically acceptable carriers or excipients.
[0403] Pharmaceutical compositions of the present invention include, but are not limited to, solutions, emulsions, and liposome-containing formulations. These compositions can be generated from a variety of components that include, but are not limited to, preformed liquids, self-emulsifying solids and self-emulsifying semisolids. Particularly preferred are formulations that target the liver when treating hepatic disorders such as hepatic carcinoma.
[0404] The pharmaceutical formulations of the present invention, which can conveniently be presented in unit dosage form, can be prepared according to conventional techniques well known in the pharmaceutical industry. Such techniques include the step of bringing into association the active ingredients with the pharmaceutical carrier(s) or excipient(s). In general, the formulations are prepared by uniformly and intimately bringing into association the active ingredients with liquid carriers or finely divided solid carriers or both, and then, if necessary, shaping the product.
[0405] The compositions of the present invention can be formulated into any of many possible dosage forms such as, but not limited to, tablets, capsules, gel capsules, liquid syrups, soft gels, suppositories, and enemas. The compositions of the present invention can also be formulated as suspensions in aqueous, non-aqueous or mixed media. Aqueous suspensions can further contain substances which increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol and/or dextran. The suspension can also contain stabilizers.
C. Additional Formulations
i. Emulsions
[0406] The compositions of the present invention can be prepared and formulated as emulsions. Emulsions are typically heterogeneous systems of one liquid dispersed in another in the form of droplets usually exceeding 0.1 μm in diameter (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199; Rosoff, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., Volume 1, p. 245; Block in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 2, p. 335; Higuchi et al., in Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pa., 1985, p. 301). Emulsions are often biphasic systems comprising two immiscible liquid phases intimately mixed and dispersed with each other. In general, emulsions can be of either the water-in-oil (w/o) or the oil-in-water (o/w) variety. When an aqueous phase is finely divided into and dispersed as minute droplets into a bulk oily phase, the resulting composition is called a water-in-oil (w/o) emulsion. Alternatively, when an oily phase is finely divided into and dispersed as minute droplets into a bulk aqueous phase, the resulting composition is called an oil-in-water (o/w) emulsion. Emulsions can contain additional components in addition to the dispersed phases, and the active drug which can be present as a solution in either the aqueous phase, oily phase or itself as a separate phase. Pharmaceutical excipients such as emulsifiers, stabilizers, dyes, and anti-oxidants can also be present in emulsions as needed. Pharmaceutical emulsions can also be multiple emulsions that are comprised of more than two phases such as, for example, in the case of oil-in-water-in-oil (o/w/o) and water-in-oil-in-water (w/o/w) emulsions. Such complex formulations often provide certain advantages that simple binary emulsions do not. Multiple emulsions in which individual oil droplets of an o/w emulsion enclose small water droplets constitute a w/o/w emulsion. Likewise, a system of oil droplets enclosed in globules of water stabilized in an oily continuous phase provides an o/w/o emulsion.
[0407] Emulsions are characterized by little or no thermodynamic stability. Often, the dispersed or

discontinuous phase of the emulsion is well dispersed into the external or continuous phase and maintained in this form through the means of emulsifiers or the viscosity of the formulation. Either of the phases of the emulsion can be a semisolid or a solid, as is the case of emulsion-style ointment bases and creams. Other means of stabilizing emulsions entail the use of emulsifiers that can be incorporated into either phase of the emulsion. Emulsifiers can broadly be classified into four categories: synthetic surfactants, naturally occurring emulsifiers, absorption bases, and finely dispersed solids (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

[0408] Synthetic surfactants, also known as surface active agents, have found wide applicability in the formulation of emulsions and have been reviewed in the literature (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rieger, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), Marcel Dekker, Inc., New York, N.Y., 1988, volume 1, p. 199). Surfactants are typically amphiphilic and comprise a hydrophilic and a hydrophobic portion. The ratio of the hydrophilic to the hydrophobic nature of the surfactant has been termed the hydrophile/lipophile balance (HLB) and is a valuable tool in categorizing and selecting surfactants in the preparation of formulations. Surfactants can be classified into different classes based on the nature of the hydrophilic group: nonionic, anionic, cationic and amphoteric (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY Rieger, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285).

[0409] Naturally occurring emulsifiers used in emulsion formulations include lanolin, beeswax, phosphatides, lecithin and acacia. Absorption bases possess hydrophilic properties such that they can soak up water to form w/o emulsions yet retain their semisolid consistencies, such as anhydrous lanolin and hydrophilic petrolatum. Finely divided solids have also been used as good emulsifiers especially in combination with surfactants and in viscous preparations. These include polar inorganic solids, such as heavy metal hydroxides, nonswelling clays such as bentonite, attapulgite, hectorite, kaolin, montmorillonite, colloidal aluminum silicate and colloidal magnesium aluminum silicate, pigments and nonpolar solids such as carbon or glyceryl tristearate.

[0410] A large variety of non-emulsifying materials are also included in emulsion formulations and contribute to the properties of emulsions. These include fats, oils, waxes, fatty acids, fatty alcohols, fatty esters, humectants, hydrophilic colloids, preservatives and antioxidants (Block, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 335; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

[0411] Hydrophilic colloids or hydrocolloids include naturally occurring gums and synthetic polymers such as polysaccharides (for example, acacia, agar, alginic acid, carrageenan, guar gum, karaya gum, and tragacanth), cellulose derivatives (for example, carboxymethylcellulose and carboxypropylcellulose), and synthetic polymers (for example, carbomers, cellulose ethers, and carboxyvinyl polymers). These disperse or swell in water to form colloidal solutions that stabilize emulsions by forming strong interfacial films around the dispersed-phase droplets and by increasing the viscosity of the external phase.

[0412] Since emulsions often contain a number of ingredients such as carbohydrates, proteins, sterols and phosphatides that can readily support the growth of microbes, these formulations often incorporate preservatives. Commonly used preservatives included in emulsion formulations include methyl paraben, propyl paraben, quaternary ammonium salts, benzalkonium chloride, esters of phydroxybenzoic acid, and boric acid. Antioxidants are also commonly added to emulsion formulations to prevent deterioration of the formulation. Antioxidants used can be free radical scavengers such as tocopherols, alkyl gallates, butylated hydroxyanisole, butylated hydroxytoluene, or reducing agents such as ascorbic acid and sodium metabisulfite, and antioxidant synergists such as citric acid, tartaric acid, and lecithin.

[0413] The application of emulsion formulations via dermatological, oral and parenteral routes and methods for their manufacture have been reviewed in the literature (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199). Emulsion formulations for oral delivery have been very widely used because of ease of formulation, as well as efficacy from an absorption and bioavailability standpoint (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199). Mineral-oil base laxatives, oil-soluble vitamins and high fat nutritive preparations are among the materials that have commonly been administered orally as o/w emulsions.

ii. Microemulsions

[0414] In one embodiment of the present invention, the compositions of iRNAs and nucleic acids are formulated as microemulsions. A microemulsion can be defined as a system of water, oil and amphiphile which is a single optically isotropic and thermodynamically stable liquid solution (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245). Typically microemulsions are systems that are prepared by first dispersing an oil in an aqueous surfactant solution and then adding a sufficient amount of a fourth component, generally an intermediate chain-length alcohol to form a transparent system. Therefore, microemulsions have also been described as thermodynamically stable, isotropically clear dispersions of two immiscible liquids that are stabilized by interfacial films of surface-active molecules (Leung and Shah, in: Controlled Release of Drugs: Polymers and Aggregate Systems, Rosoff, M., Ed., 1989, VCH Publishers, New York, pages 185-215). Microemulsions commonly are prepared via a combination of three to five components that include oil, water, surfactant, cosurfactant and electrolyte. Whether the microemulsion is of the water-in-oil (w/o) or an oil-in-water (o/w) type is dependent on the properties of the oil and surfactant used and on the structure and geometric packing of the polar heads and hydrocarbon tails of the surfactant molecules (Schott, in Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pa., 1985, p. 271).

[0415] The phenomenological approach utilizing phase diagrams has been extensively studied and has yielded a comprehensive knowledge, to one skilled in the art, of how to formulate microemulsions (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245; Block, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 335). Compared to conventional emulsions, microemulsions offer the advantage of solubilizing water-insoluble drugs in a formulation of thermodynamically stable droplets that are formed spontaneously.

[0416] Surfactants used in the preparation of microemulsions include, but are not limited to, ionic surfactants, non-ionic surfactants, Brij 96, polyoxyethylene oleyl ethers, polyglycerol fatty acid esters, tetraglycerol monolaurate (ML310), tetraglycerol monooleate (MO310), hexaglycerol monooleate (PO310), hexaglycerol pentaoleate (PO500), decaglycerol monocaprate (MCA750), decaglycerol monooleate (MO750), decaglycerol sequioleate (SO750), decaglycerol decaoleate (DAO750), alone or in combination with cosurfactants. The cosurfactant, usually a short-chain alcohol such as ethanol, 1-propanol, and 1-butanol, serves to increase the interfacial fluidity by penetrating into the surfactant film and consequently creating a disordered film because of the void space generated among surfactant molecules. Microemulsions can, however, be prepared without the use of cosurfactants and alcohol-free self-emulsifying microemulsion systems are known in the art. The aqueous phase can typically be, but is not limited to, water, an aqueous solution of the drug, glycerol, PEG300, PEG400, polyglycerols, propylene glycols, and derivatives of ethylene glycol. The oil phase can include, but is not limited to, materials such as Captex 300, Captex 355, Capmul MCM, fatty acid esters, medium chain (C8-C12) mono, di, and tri-glycerides, polyoxyethylated glyceryl fatty acid esters, fatty alcohols, polyglycolized glycerides, saturated polyglycolized C8-C10 glycerides, vegetable oils and silicone oil.

[0417] Microemulsions are particularly of interest from the standpoint of drug solubilization and the enhanced absorption of drugs. Lipid based microemulsions (both o/w and w/o) have been proposed to enhance the oral bioavailability of drugs, including peptides (see e.g., U.S. Pat. Nos. 6,191,105; 7,063,860; 7,070,802; 7,157,099; Constantinides et al., *Pharmaceutical Research*, 1994, 11, 1385-1390; Ritschel, *Meth. Find. Exp. Clin. Pharmacol.*, 1993, 13, 205). Microemulsions afford advantages of improved drug solubilization, protection of drug from enzymatic hydrolysis, possible enhancement of drug absorption due to surfactant-induced alterations in membrane fluidity and permeability, ease of preparation, ease of oral administration over solid dosage forms, improved clinical potency, and decreased toxicity (see e.g., U.S. Pat. Nos. 6,191,105; 7,063,860; 7,070,802; 7,157,099; Constantinides et al., *Pharmaceutical Research*, 1994, 11, 1385; Ho et al., *J. Pharm. Sci.*, 1996, 85, 138-143). Often microemulsions can form spontaneously when their components are brought together at ambient temperature. This can be particularly advantageous when formulating thermolabile drugs, peptides or iRNAs. Microemulsions have also been effective in the transdermal delivery of active components in both cosmetic and pharmaceutical applications. It is expected that the microemulsion compositions and formulations of the present invention will facilitate the increased systemic absorption of iRNAs and nucleic acids from the gastrointestinal tract, as well as improve the local cellular uptake of iRNAs and nucleic acids.

[0418] Microemulsions of the present invention can also contain additional components and additives such as sorbitan monostearate (Grill 3), Labrasol, and penetration enhancers to improve the properties of the formulation and to enhance the absorption of the iRNAs and nucleic acids of the present invention. Penetration enhancers used in the microemulsions of the present invention can be classified as belonging to one of five broad categories—surfactants, fatty acids, bile salts, chelating agents, and non-chelating non-surfactants (Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92). Each of these classes has been discussed above.

iii. Microparticles

[0419] An RNAi agent of the invention may be incorporated into a particle, e.g., a microparticle. Microparticles can be produced by spray-drying, but may also be produced by other methods including lyophilization, evaporation, fluid bed drying, vacuum drying, or a combination of these techniques.

iv. Penetration Enhancers

[0420] In one embodiment, the present invention employs various penetration enhancers to effect the efficient delivery of nucleic acids, particularly iRNAs, to the skin of animals. Most drugs are present in solution in both ionized and nonionized forms. However, usually only lipid soluble or lipophilic drugs readily cross cell membranes. It has been discovered that even non-lipophilic drugs can cross cell membranes if the membrane to be crossed is treated with a penetration enhancer. In addition to aiding the diffusion of non-lipophilic drugs across cell membranes, penetration enhancers also enhance the permeability of lipophilic drugs.

[0421] Penetration enhancers can be classified as belonging to one of five broad categories, i.e., surfactants, fatty acids, bile salts, chelating agents, and non-chelating non-surfactants (see e.g., Malmsten, M. Surfactants and polymers in drug delivery, Informa Health Care, New York, NY, 2002; Lee et al., Critical Reviews in Therapeutic Drug Carrier Systems, 1991, p. 92). Each of the above mentioned classes of penetration enhancers are described below in greater detail. Such compounds are well known in the art.

v. Carriers

[0422] Certain compositions of the present invention also incorporate carrier compounds in the formulation. As used herein, “carrier compound” or “carrier” can refer to a nucleic acid, or analog thereof, which is inert (i.e., does not possess biological activity per se) but is recognized as a nucleic acid by in vivo processes that reduce the bioavailability of a nucleic acid having biological activity by, for example, degrading the biologically active nucleic acid or promoting its removal from circulation. The coadministration of a nucleic acid and a carrier compound, typically with an excess of the latter substance, can result in a substantial reduction of the amount of nucleic acid recovered in the liver, kidney or other extracirculatory reservoirs, presumably due to competition between the carrier compound and the nucleic acid for a common receptor. For example, the recovery of a partially phosphorothioate dsRNA in hepatic tissue can be reduced when it is coadministered with polyinosinic acid, dextran sulfate, polycytidic acid or 4-acetamido-4′isothiocyano-stilbene-2,2′-disulfonic acid (Miyao et al., DsRNA Res. Dev., 1995, 5, 115-121; Takakura et al., DsRNA & Nucl. Acid Drug Dev., 1996, 6, 177-183.

vi. Excipients

[0423] In contrast to a carrier compound, a “pharmaceutical carrier” or “excipient” is a pharmaceutically acceptable solvent, suspending agent or any other pharmacologically inert vehicle for delivering one or more nucleic acids to an animal. The excipient can be liquid or solid and is selected, with the planned manner of administration in mind, so as to provide for the desired bulk, consistency, etc., when combined with a nucleic acid and the other components of a given pharmaceutical composition. Typical pharmaceutical carriers include, but are not limited to, binding agents (e.g., pregelatinized maize starch, polyvinylpyrrolidone or hydroxypropyl methylcellulose, etc.); fillers (e.g., lactose and other sugars, microcrystalline cellulose, pectin, gelatin, calcium sulfate, ethyl cellulose, polyacrylates or calcium hydrogen phosphate, etc.); lubricants (e.g., magnesium stearate, talc, silica, colloidal silicon dioxide, stearic acid, metallic stearates, hydrogenated vegetable oils, corn starch, polyethylene glycols, sodium benzoate, sodium acetate, etc.); disintegrants (e.g., starch, sodium starch glycolate, etc.); and wetting agents (e.g., sodium lauryl sulphate, etc).

[0424] Pharmaceutically acceptable organic or inorganic excipients suitable for non-parenteral administration which do not deleteriously react with nucleic acids can also be used to formulate the compositions of the present invention. Suitable pharmaceutically acceptable carriers include, but are not limited to, water, salt solutions, alcohols, polyethylene glycols, gelatin, lactose, amylose, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulose, polyvinylpyrrolidone and the like.

[0425] Formulations for topical administration of nucleic acids can include sterile and non-sterile aqueous solutions, non-aqueous solutions in common solvents such as alcohols, or solutions of the nucleic acids in liquid or solid oil bases. The solutions can also contain buffers, diluents and other suitable additives. Pharmaceutically acceptable organic or inorganic excipients suitable for non-parenteral administration which do not deleteriously react with nucleic acids can be used.

[0426] Suitable pharmaceutically acceptable excipients include, but are not limited to, water, salt solutions, alcohol, polyethylene glycols, gelatin, lactose, amylose, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulose, polyvinylpyrrolidone and the like.

vii. Other Components

[0427] The compositions of the present invention can additionally contain other adjunct components conventionally found in pharmaceutical compositions, at their art-established usage levels. Thus, for example, the compositions can contain additional, compatible, pharmaceutically-active materials such as, for example, antipruritics, astringents, local anesthetics or anti-inflammatory agents, or can contain additional materials useful in physically formulating various dosage forms of the compositions of the present invention, such as dyes, flavoring agents, preservatives, antioxidants, opacifiers, thickening agents and stabilizers. However, such materials, when added, should not unduly interfere with the biological activities of the components of the compositions of the present invention. The formulations can be sterilized and, if desired, mixed with auxiliary agents, e.g., lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure, buffers, colorings, flavorings and/or aromatic substances and the like which do not deleteriously interact with the nucleic acid(s) of the formulation.
[0428] Aqueous suspensions can contain substances which increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol and/or dextran. The suspension can also contain stabilizers.
[0429] In some embodiments, pharmaceutical compositions featured in the invention include (a) one or more iRNA agents of the invention and (b) one or more agents which function by a non-RNAi mechanism and which are useful in treating a APCS-associated disease or disorder, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.
[0430] Toxicity and therapeutic efficacy of iRNA agents of the invention can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD50 (the dose lethal to 50% of the population) and the ED50 (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD.sub.50/ED.sub.50. Compounds that exhibit high therapeutic indices are preferred.
[0431] The data obtained from cell culture assays and animal studies can be used in formulating a range of dosage for use in humans. The dosage of compositions featured herein in the invention lies generally within a range of circulating concentrations that include the ED.sub.50 with little or no toxicity. The dosage can vary within this range depending upon the dosage form employed and the route of administration utilized. For any compound used in the methods featured in the invention, the therapeutically effective dose can be estimated initially from cell culture assays. A dose can be formulated in animal models to achieve a circulating plasma concentration range of the compound or, when appropriate, of the polypeptide product of a target sequence (e.g., achieving a decreased concentration of the polypeptide) that includes the IC.sub.50 (i.e., the concentration of the test compound which achieves a half-maximal inhibition of symptoms) as determined in cell culture. Such information can be used to more accurately determine useful doses in humans. Levels in plasma can be measured, for example, by high performance liquid chromatography.
[0432] In addition to their administration, as discussed above, the iRNAs featured in the invention can be administered in combination with other known agents effective in treatment of pathological processes mediated by APCS expression. In any event, the administering physician can adjust the amount and timing of iRNA administration on the basis of results observed using standard measures of efficacy known in the art or described herein.
VII. Methods for Inhibiting APCS Expression
[0433] The present invention provides methods of inhibiting expression of an APCS gene as described herein. In one aspect, the present invention provides methods of inhibiting expression of APCS in a cell. The methods include contacting a cell with an RNAi agent, e.g., a double stranded RNAi agent, in an amount effective to inhibit expression of the APCS in the cell, thereby inhibiting expression of the APCS in the cell.
[0434] Contacting of a cell with an RNAi agent, e.g., a double stranded RNAi agent, may be done in vitro or in vivo. Contacting a cell in vivo with the RNAi agent includes contacting a cell or

group of cells within a subject, e.g., a human subject, with the RNAi agent. Combinations of in vitro and in vivo methods of contacting a cell or a group of cells are also possible. Contacting a cell or a group of cells may be direct or indirect. Thus, for example, the RNAi agent may be put into physical contact with the cell by the individual performing the method, or alternatively, the RNAi agent may be put into a situation that will permit or cause it to subsequently come into contact with the cell. Furthermore, contacting a cell or a group of cells may be accomplished via a targeting ligand, including any ligand described herein or known in the art. In preferred embodiments, the targeting ligand is a carbohydrate moiety, e.g., a GalNAc3 ligand, or any other ligand that directs the RNAi agent to a site of interest, e.g., the liver of a subject.

[0435] Contacting a cell in vitro may be done, for example, by incubating the cell with the RNAi agent. Contacting a cell in vivo may be done, for example, by injecting the RNAi agent into or near the tissue where the cell is located, or by injecting the RNAi agent into another area, e.g., the bloodstream or the subcutaneous space, such that the agent will subsequently reach the tissue where the cell to be contacted is located. For example, the RNAi agent may contain and/or be coupled to a ligand, e.g., GalNAc3, that directs the RNAi agent to a site of interest, e.g., the liver. Combinations of in vitro and in vivo methods of contacting are also possible. For example, a cell may also be contacted in vitro with an RNAi agent and subsequently transplanted into a subject.

[0436] The term “inhibiting,” as used herein, is used interchangeably with “reducing,” “silencing,” “downregulating”, “suppressing”, and other similar terms, and includes any level of inhibition. Preferably inhibiting includes a statistically significant or clinically significant inhibition.

[0437] The phrase “inhibiting expression of a APCS gene” is intended to refer to inhibition of expression of any APCS gene (such as, e.g., a mouse APCS gene, a rat APCS gene, a monkey APCS gene, or a human APCS gene) as well as variants or mutants of an APCS gene.

[0438] The phrase “inhibiting expression of a APCS gene” is intended to refer to inhibition of expression of any APCS gene (such as, e.g., a mouse APCS gene, a rat APCS gene, a monkey APCS gene, or a human APCS gene) as well as variants or mutants of an APCS gene. Thus, the APCS gene may be a wild-type APCS gene, a mutant APCS gene, or a transgenic APCS gene in the context of a genetically manipulated cell, group of cells, or an organism.

[0439] “Inhibiting expression of an APCS gene” includes any level of inhibition of an APCS gene, e.g., at least partial suppression of the expression of an APCS gene. The expression of the APCS gene may be assessed based on the level, or the change in the level, of any variable associated with APCS gene expression, e.g., APCS mRNA level, APCS protein level (SAP), or the severity of an APCS-associated disease. This level may be assessed in an individual cell or in a group of cells, including, for example, a sample derived from a subject.

[0440] In some embodiments of the methods of the invention, expression of an APCS gene, is inhibited by at least about 5%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%%, or to below the level of detection of the assay. In some embodiments, the inhibition of expression of an APCS gene results in normalization of the level of the APCS gene, such that the difference between the level before treatment and a normal control level is reduced by at least 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95%. In some embodiments, the inhibition is a clinically relevant inhibition.

[0441] Inhibition of the expression of the target gene, e.g., an APCS gene may be manifested by a reduction of the amount of mRNA expressed by a first cell or group of cells (such cells may be present, for example, in a sample derived from a subject) in which a target gene is transcribed and which has or have been treated (e.g., by contacting the cell or cells with an RNAi agent of the invention, or by administering an RNAi agent of the invention to a subject in which the cells are or were present) such that the expression of a target gene is inhibited, as compared to a second cell or group of cells substantially identical to the first cell or group of cells but which has not or have not been so treated (control cell(s)). In preferred embodiments, the inhibition is assessed by the rtPCR method, with in vitro assays being performed in an appropriately matched cell line with the duplex at a 10 nM concentration, and expressing the level of mRNA in treated cells as a percentage of the level of mRNA in control cells, using the following formula:

[00001] $$\frac{(\text{mRNAincontrolcells}) - (\text{mRNAintreatedcells})}{(\text{mRNAincontrolcells})} \text{.Math. } 100\%$$

[0442] Alternatively, inhibition of the expression of an APCS gene may be assessed in terms of a reduction of a parameter that is functionally linked to APCS gene expression, e.g., APCS protein expression. APCS gene silencing may be determined in any cell expressing an APCS gene, either constitutively or by genomic engineering, and by any assay known in the art.

[0443] Inhibition of the expression of an APCS gene may be manifested by a reduction in the level of the protein encoded by the APCS gene (SAP) that is produced by a cell or group of cells (e.g., the level of protein produced in a sample derived from a subject). As explained above for the assessment of mRNA suppression, the inhibition of protein expression levels in a treated cell or group of cells may similarly be expressed as a percentage of the level of protein in a control cell or group of cells.

[0444] A control cell or group of cells that may be used to assess the inhibition of the expression of a target gene includes a cell or group of cells that has not yet been contacted with an RNAi agent of the invention. For example, the control cell or group of cells may be derived from an individual subject (e.g., a human or animal subject) prior to treatment of the subject with an RNAi agent. In alternative embodiments, the level may be compared to an appropriate control sample, e.g., a known population control sample.

[0445] The level of APCS mRNA that is expressed by a cell or group of cells may be determined using any method known in the art for assessing mRNA expression. In one embodiment, the level of expression of APCS in a sample is determined by detecting a transcribed polynucleotide, or portion thereof, e.g., mRNA of the APCS gene. RNA may be extracted from cells using RNA extraction techniques including, for example, using acid phenol/guanidine isothiocyanate extraction (RNAzol B; Biogenesis), RNeasy RNA preparation kits (Qiagen) or PAXgene (PreAnalytix, Switzerland). Typical assay formats utilizing ribonucleic acid hybridization include nuclear run-on assays, RT-PCR, RNase protection assays (Melton et al., *Nuc. Acids Res.* 12:7035), Northern blotting, in situ hybridization, and microarray analysis. Circulating APCS mRNA may be detected using methods the described in PCT Publication No. WO 2012/177906, the entire contents of which are hereby incorporated herein by reference.

[0446] In one embodiment, the level of expression of APCS is determined using a nucleic acid probe. The term “probe”, as used herein, refers to any molecule that is capable of selectively binding to a specific APCS. Probes can be synthesized by one of skill in the art, or derived from appropriate biological preparations. Probes may be specifically designed to be labeled. Examples of molecules that can be utilized as probes include, but are not limited to, RNA, DNA, proteins, antibodies, and organic molecules.

[0447] Isolated mRNA can be used in hybridization or amplification assays that include, but are not limited to, Southern or Northern analyses, polymerase chain reaction (PCR) analyses and probe arrays. One method for the determination of mRNA levels involves contacting the isolated mRNA with a nucleic acid molecule (probe) that can hybridize, e.g., specifically hybridize, to APCS mRNA. In one embodiment, the mRNA is immobilized on a solid surface and contacted with a probe, for example by running the isolated mRNA on an agarose gel and transferring the mRNA from the gel to a membrane, such as nitrocellulose. In an alternative embodiment, the probe(s) are immobilized on a solid surface and the mRNA is contacted with the probe(s), for example, in an Affymetrix gene chip array. A skilled artisan can readily adapt known mRNA detection methods for use in determining the level of APCS mRNA.

[0448] An alternative method for determining the level of expression of APCS in a sample involves the process of nucleic acid amplification and/or reverse transcriptase (to prepare cDNA) of for example mRNA in the sample, e.g., by RT-PCR (the experimental embodiment set forth in Mullis, 1987, U.S. Pat. No. 4,683,202), ligase chain reaction (Barany (1991) *Proc. Natl. Acad. Sci. USA* 88:189-193), self sustained sequence replication (Guatelli et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:1874-1878), transcriptional amplification system (Kwoh et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:1173-1177), Q-Beta Replicase (Lizardi et al. (1988) *Bio/Technology* 6:1197), rolling circle replication (Lizardi et al., U.S. Pat. No. 5,854,033) or any other nucleic acid amplification method, followed by the detection of the amplified molecules using techniques well known to those of skill in the art. These detection schemes are especially useful for the detection of nucleic acid molecules if such molecules are present in very low numbers. In particular aspects of the invention, the level of expression of APCS is determined by quantitative fluorogenic RT-PCR (i.e., the TaqMan™ System).

[0449] The expression levels of APCS mRNA may be monitored using a membrane blot (such as used in hybridization analysis such as Northern, Southern, dot, and the like), or microwells, sample tubes, gels, beads or fibers (or any solid support comprising bound nucleic acids). See U.S. Pat. Nos. 5,770,722, 5,874,219, 5,744,305, 5,677,195 and 5,445,934, which are incorporated herein by reference. The determination of APCS expression level may also comprise using nucleic acid probes in solution.

[0450] In preferred embodiments, the level of APCS mRNA expression is assessed using branched DNA (bDNA) assays or real time PCR (qPCR).

[0451] The level of APCS protein (SAP) expression may be determined using any method known in the art for the measurement of protein levels. Such methods include, for example, electrophoresis, capillary electrophoresis, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), hyperdiffusion chromatography, fluid or gel precipitin reactions, absorption spectroscopy, a colorimetric assays, spectrophotometric assays, flow cytometry, immunodiffusion (single or double), immunoelectrophoresis, Western blotting, radioimmunoassay (RIA), enzyme-linked immunosorbent assays (ELISAs), immunofluorescent assays, electrochemiluminescence assays, and the like.

[0452] In some embodiments, the efficacy of the methods of the invention can be monitored by detecting or monitoring a reduction in a symptom of an APCS-associated disease. Symptoms may be assessed in vitro or in vivo using any method known in the art. For example, efficacy of the methods of the invention in treating an APCS-associated disease, such as amyloidosis, may be assessed by measuring levels of SAP in plasma collected from a subject suffering from an APCS-associated disease; measuring an amyloid load by using, e.g., whole-body .sup.123I-SAP scintigraphy; measuring the expansion of the extracellular volume by systemic amyloid deposits by using, e.g., equilibrium magnetic resonance imaging (MRI); or measuring liver stiffness if the amyloidosis has hepatic involvement (see Richards et al., *N. Engl. J. Med.* 373(12), 1106-1114, 2015). In another example, efficacy of the methods of the invention in treating an APCS-associated disease, such as a cardiovascular disease, e.g., coronary atherosclerotic heart disease, may be assessed by measuring the size of atherosclerotic lesions.

[0453] The term “sample” as used herein refers to a collection of similar fluids, cells, or tissues isolated from a subject, as well as fluids, cells, or tissues present within a subject. Examples of biological fluids include blood, serum and serosal fluids, plasma, lymph, urine, cerebrospinal fluid, saliva, ocular fluids, and the like. Tissue samples may include samples from tissues, organs or localized regions. For example, samples may be derived from particular organs, parts of organs, or fluids or cells within those organs. In certain embodiments, samples may be derived from the liver (e.g., whole liver or certain segments of liver or certain types of cells in the liver, such as, e.g., hepatocytes), the retina or parts or parts of the retina (e.g., retinal pigment epithelium), the central

nervous system or parts of the central nervous system (e.g., ventricles or choroid plexus), or the pancreas or certain cells or parts of the pancreas. In preferred embodiments, a “sample derived from a subject” refers to blood drawn from the subject or plasma derived therefrom. In further embodiments, a “sample derived from a subject” refers to liver tissue (or subcomponents thereof) or retinal tissue (or subcomponents thereof) derived from the subject.
[0454] In some embodiments of the methods of the invention, the RNAi agent is administered to a subject such that the RNAi agent is delivered to a specific site within the subject. The inhibition of expression of APCS may be assessed using measurements of the level or change in the level of APCS mRNA and/or APCS protein in a sample derived from fluid or tissue from the specific site within the subject. In preferred embodiments, the site is the liver. The site may also be a subsection or subgroup of cells from any one of the aforementioned sites. The site may also include cells that express a particular type of receptor.
VIII. Methods for Treating or Preventing an APCS-Associated Disease or Disorder
[0455] The present invention provides therapeutic and prophylactic methods which include administering to a subject having an APCS-associated disease, as described herein, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease, an iRNA agent, pharmaceutical composition comprising an iRNA agent, or vector comprising an iRNA of the invention.
[0456] In one aspect, the present invention provides methods of treating a subject having a disorder that would benefit from reduction in APCS expression, e.g., an APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease. The treatment methods (and uses) of the invention include administering to the subject, e.g., a human, a therapeutically effective amount of an iRNA agent targeting an APCS gene or a pharmaceutical composition comprising an iRNA agent targeting an APCS gene, thereby treating the subject having a disorder that would benefit from reduction in APCS expression.
[0457] In one aspect, the invention provides methods of preventing at least one symptom in a subject having a disorder that would benefit from reduction in APCS expression, e.g., an amyloid-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease. The methods include administering to the subject a therapeutically effective amount of the iRNA agent, e.g., dsRNA, or vector of the invention, thereby preventing at least one symptom in the subject having a disorder that would benefit from reduction in APCS expression. For example, the invention provides methods for preventing formation of, or reducing the size of amyloid deposits or atherosclerotic lesions in a subject suffering from a disorder that would benefit from reduction in APCS expression, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.
[0458] In another aspect, the present invention provides uses of a therapeutically effective amount of an iRNA agent of the invention for treating a subject, e.g., a subject that would benefit from a reduction and/or inhibition of APCS expression.
[0459] In yet another aspect, the present invention provides uses of an iRNA agent, e.g., a dsRNA, of the invention targeting an APCS gene or a pharmaceutical composition comprising an iRNA agent targeting an APCS gene in the manufacture of a medicament for treating a subject, e.g., a subject that would benefit from a reduction and/or inhibition of APCS expression, such as a subject having a disorder that would benefit from reduction in APCS expression, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.
[0460] In another aspect, the invention provides uses of an iRNA, e.g., a dsRNA, of the invention for preventing at least one symptom in a subject suffering from a disorder that would benefit from a reduction and/or inhibition of APCS expression, such as an APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.
[0461] In a further aspect, the present invention provides uses of an iRNA agent of the invention in the manufacture of a medicament for preventing at least one symptom in a subject suffering from a disorder that would benefit from a reduction and/or inhibition of APCS expression, such as an

APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.

[0462] In one embodiment, an iRNA agent targeting APCS is administered to a subject having an APCS-associated disease such that the expression of a APCS gene, e.g., in a cell, tissue, blood or other tissue or fluid of the subject is reduced by at least about 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 62%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or at least about 99% or more, or to a level below the level of detection of the assay, when the dsRNA agent is administered to the subject.

[0463] The methods and uses of the invention include administering a composition described herein such that expression of the target APCS gene is decreased for an extended duration, e.g., at least one month, preferably at least three months.

[0464] Administration of the dsRNA according to the methods and uses of the invention may result in a reduction of the severity, signs, symptoms, and/or markers of such diseases or disorders in a patient with an APCS-associated disease. By “reduction” in this context is meant a statistically or clinically significant decrease in such level. The reduction can be, for example, at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or about 100%.

[0465] Efficacy of treatment or prevention of disease can be assessed, for example by measuring disease progression, disease remission, symptom severity, reduction in pain, quality of life, dose of a medication required to sustain a treatment effect, level of a disease marker or any other measurable parameter appropriate for a given disease being treated or targeted for prevention. It is well within the ability of one skilled in the art to monitor efficacy of treatment or prevention by measuring any one of such parameters, or any combination of parameters. Comparison of the later readings with the initial readings, or historically relevant population controls, provide a physician an indication of whether the treatment is effective. It is well within the ability of one skilled in the art to monitor efficacy of treatment or prevention by measuring any one of such parameters, or any combination of parameters. In connection with the administration of an iRNA targeting an APCS or pharmaceutical composition thereof, “effective against” an APCS-associated disease indicates that administration in a clinically appropriate manner results in a beneficial effect for at least a statistically significant fraction of patients, such as improvement of symptoms, a cure, a reduction in disease, extension of life, improvement in quality of life, or other effect generally recognized as positive by medical doctors familiar with treating an APCS-associated disease and the related causes.

[0466] A treatment or preventive effect is evident when there is a statistically significant improvement in one or more parameters of disease status, or by a failure to worsen or to develop symptoms where they would otherwise be anticipated. As an example, a favorable change of at least 10% in a measurable parameter of disease, and preferably at least 20%, 30%, 40%, 50% or more can be indicative of effective treatment. Efficacy for a given iRNA drug or formulation of that drug can also be judged using an experimental animal model for the given disease as known in the art. When using an experimental animal model, efficacy of treatment is evidenced when a statistically significant reduction in a marker or symptom is observed.

[0467] Subjects can be administered a therapeutic amount of iRNA, such as about 0.01 mg/kg to about 200 mg/kg. Values and ranges intermediate to the recited values are also intended to be part of this invention. Typically, a suitable dose of an iRNA of the invention will be in the range of about 0.1 mg/kg to about 5.0 mg/kg, preferably about 0.3 mg/kg and about 3.0 mg/kg.

[0468] The iRNA can be administered by intravenous infusion over a period of time, on a regular basis, e.g., once per month, once every other month, once per quarter.

[0469] In certain embodiments, one or more loading doses is administered.

[0470] Administration of the iRNA can reduce the presence of an APCS protein (SAP), e.g., in a cell, tissue, blood, serum, plasma, urine or other compartment of the patient by at least about 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or at least about 99% or more, or below the level of detection of the assay method used.

[0471] Before administration of a full dose of the iRNA, patients can be administered a smaller dose, such as a 5% infusion, and monitored for adverse effects, such as an allergic reaction. In another example, the patient can be monitored for unwanted immunostimulatory effects, such as increased cytokine (e.g., TNF-alpha or INF-alpha) levels.

[0472] Owing to the inhibitory effects on APCS gene expression, a composition according to the invention or a pharmaceutical composition prepared therefrom can enhance the quality of life.

[0473] An iRNA of the invention may be administered in “naked” form, where the modified or unmodified iRNA agent is directly suspended in aqueous or suitable buffer solvent, as a “free iRNA.” A free iRNA is administered in the absence of a pharmaceutical composition. The free iRNA may be in a suitable buffer solution. The buffer solution may comprise acetate, citrate, prolamine, carbonate, or phosphate, or any combination thereof. In one embodiment, the buffer solution is phosphate buffered saline (PBS). The pH and osmolarity of the buffer solution containing the iRNA can be adjusted such that it is suitable for administering to a subject.

[0474] Alternatively, an iRNA of the invention may be administered as a pharmaceutical composition, such as a dsRNA liposomal formulation.

[0475] Subjects that would benefit from a reduction and/or inhibition of APCS gene expression are those having an APCS-associated disease or disorder as described herein.

[0476] Treatment of a subject that would benefit from a reduction and/or inhibition of APCS gene expression includes therapeutic and prophylactic treatment.

[0477] Subjects that would benefit from a reduction and/or inhibition of APCS gene expression are those having an APCS-associated disease or disorder as described herein. In one embodiment, an APCS-associated disease is an amyloid-associated disease, such as amyloidosis, e.g., such as primary (systemic AL) amyloidosis, secondary (systemic AA) amyloidosis, dialysis-related amyloidosis (DRA), familial (hereditary FA) amyloidosis, senile systemic amyloidosis (SSA) or organ-specific amyloidosis.

[0478] In another embodiment, the amyloid-associated disease is Alzheimer's disease.

[0479] In another embodiment, the amyloid associated disease is diabetes mellitus type 2; Parkinson's disease; transmissible spongiform encephalopathy (such as bovine spongiform encephalopathy); fatal familial insomnia; Huntington's disease; medullary carcinoma of the thyroid; cardiac arrhythmias; isolated atrial amyloidosis; rheumatoid arthritis; aortic medial amyloid; prolactinoma; familial amyloid polyneuropathy; lattice corneal dystrophy; cerebral amyloid angiopathy; cerebral amyloid angiopathy (Icelandic type); sporadic inclusion body myositis

[0480] In another embodiment, an APCS-associated disease is a cardiovascular disease, e.g., coronary atherosclerotic heart disease.

[0481] The invention further provides methods and uses of an iRNA agent or a pharmaceutical composition thereof for treating a subject that would benefit from reduction and/or inhibition of APCS gene expression, e.g., a subject having an APCS-associated disease, in combination with other pharmaceuticals and/or other therapeutic methods, e.g., with known pharmaceuticals and/or known therapeutic methods, such as, for example, those which are currently employed for treating these disorders.

[0482] Accordingly, in some aspects of the invention, the methods which include either a single iRNA agent of the invention, further include administering to the subject one or more additional therapeutic agents.

[0483] The iRNA agent and an additional therapeutic agent and/or treatment may be administered at the same time and/or in the same combination, e.g., parenterally, or the additional therapeutic agent can be administered as part of a separate composition or at separate times and/or by another method known in the art or described herein.

[0484] Additional therapeutics and therapeutic methods suitable for treating a subject that would benefit from reductone in APCS expression, e.g., a subject having an APCS-associated disease, may include (R-1-[6-[R-2-carboxy-pyrrolidin-1-yl]-6-oxo-hexanoyl]pyrrolidine-2-carboxylic acid) (CPHPC), a proline-derived small molecule that is bound by SAP in the circulation to form stable complexes of pairs of native pentameric SAP molecules cross-linked by the drug. These complexes are immediately cleared by the liver, leading to almost complete depletion of plasma SAP for the duration of drug administration (Al-Shawi et al., *Open Biol.* 6: 150202).

[0485] Additional therapeutics and therapeutic methods suitable for treating a subject that would benefit from reduction in APCS expression, e.g., a subject having an APCS-associated disease, may also include an antibody targeting SAP, such as an anti-SAP antibody, e.g., a monoclonal IgG1 anti-SAP antibody. It is believed that an anti-SAP antibody may activate macrophage destruction of the SAP-containing amyloid deposits in tissues (Al-Shawi et al., *Open Biol.* 6: 150202).

[0486] Additional therapeutics and therapeutic methods suitable for treating a subject that would benefit from reduction in APCS expression, e.g., a subject having an APCS-associated disease, may also include one or more chemotherapeutic agents, e.g., bortezomib, dexamethasone, melphalan, or combinations thereof.

[0487] Additional therapeutics and therapeutic methods suitable for treating a subject that would benefit from reduction in APCS expression, e.g., a subject having an APCS-associated disease, may also include one or more agents useful for treating cardiovascular disease, e.g., statins, such as atorvastatin (Lipitor), fluvastatin (Lescol), lovastatin, pitavastatin (Livalo), pravastatin (Pravachol), rosuvastatin (Crestor) and simvastatin (Zocor); anti-platelet medications, such as aspirin; beta blockers, such as acebutolol (Sectral), atenolol (Tenormin), bisoprolol (Zebeta), metoprolol (Lopressor, Toprol-XL), Nadolol (Corgard), nebivolol (Bystolic) and propranolol (Inderal LA, InnoPran XL); angiotensin-converting enzyme (ACE) inhibitors, such as benazepril (Lotensin, Lotensin Hct), captopril (Capoten), enalapril (Vasotec), fosinopril (Monopril), Lisinopril (Prinivil, Zestril), moexipril (Univasc), perindopril (Aceon) and quinapril (Accupril); calcium channel blockers, such as amlodipine (Norvasc), diltiazem (Cardizem, Tiazac), felodipine, isradipine, nicardipine, nifedipine (Adalat CC, Afeditab CR, Procardia), nisoldipine (Sular) and verapamil (Calan, Verelan); diuretics, or combinations thereof.

[0488] Additional therapeutics and therapeutic methods suitable for treating a subject that would benefit from reduction in APCS expression, e.g., a subject having an APCS-associated disease, may also include one or more agents useful for treating Alzheimer's disease, e.g., donepezil (Aricept), galantamine (Razadyne), memantine (Namenda), rivastigmine (Exelon), or combinations thereof, e.g., a combination of donepezil and memantine (Namzaric).

[0489] Accordingly, in one aspect, the present invention provides methods of treating a subject having a disorder that would benefit from reduction in APCS expression, e.g., amyloidosis or Alzheimer's disease, which include administering to the subject, e.g., a human, a therapeutically effective amount of an iRNA agent targeting an APCS gene or a pharmaceutical composition comprising an iRNA agent targeting an APCS gene, and an additional therapeutic agent, such as CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, thereby treating the subject having a disorder that would benefit from reduction in APCS expression.

[0490] In another aspect, the invention provides methods of preventing at least one symptom in a subject having a disorder that would benefit from reduction in APCS expression, e.g., an APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease. The methods include administering to the subject a therapeutically effective amount of the iRNA agent, e.g., dsRNA, or vector of the invention, and an additional therapeutic agent, such as CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, thereby preventing at least one symptom in the subject having a disorder that would benefit from reduction in APCS expression.
[0491] In another aspect, the present invention provides uses of a therapeutically effective amount of an iRNA agent of the invention and an additional therapeutic agent, such as CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, for treating a subject, e.g., a subject that would benefit from a reduction and/or inhibition of APCS expression.
[0492] In another aspect, the present invention provides uses of an iRNA agent, e.g., a dsRNA, of the invention targeting a APCS gene or a pharmaceutical composition comprising an iRNA agent targeting a APCS gene in the manufacture of a medicament for use in combination with an additional therapeutic agent, such as CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, for treating a subject, e.g., a subject that would benefit from a reduction and/or inhibition of APCS expression, e.g., an APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.
[0493] In yet another aspect, the invention provides uses of an iRNA agent, e.g., a dsRNA, of the invention, and an additional therapeutic agent, such as CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, for preventing at least one symptom in a subject suffering from a disorder that would benefit from a reduction and/or inhibition of APCS expression, such as an APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.
[0494] In a further aspect, the present invention provides uses of an iRNA agent of the invention in the manufacture of a medicament for use in combination with an additional therapeutic agent, such as CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, for preventing at least one symptom in a subject suffering from a disorder that would benefit from a reduction and/or inhibition of APCS expression, such as an APCS-associated disease, e.g., amyloidosis, Alzheimer's disease or coronary atherosclerotic heart disease.
[0495] In one embodiment, an iRNA agent targeting APCS is administered to a subject having an APCS-associated disease as described herein such that APCS levels, e.g., in a cell, tissue, blood, urine or other tissue or fluid of the subject are reduced by at least about 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 62%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or at least about 99% or more and, subsequently, an additional therapeutic is administered to the subject.
[0496] The additional therapeutic, e.g., CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, may be administered to the subject at the same time as the iRNA agent targeting APCS or at a different time.
[0497] Moreover, the additional therapeutic, e.g., CPHPC and/or an anti-SAP antibody, or antigen-binding fragment thereof, may be administered to the subject in the same formulation as the iRNA agent targeting APCS or in a different formulation as the iRNA agent targeting APCS.
[0498] The methods and uses of the invention include administering a composition described herein such that expression of the target APCS gene is decreased, such as for about 1, 2, 3, 4, 5, 6, 7, 8, 12, 16, 18, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, or about 80 hours. In one embodiment, expression of the target gene is decreased for an extended duration, e.g., at least about

two, three, four, five, six, seven days or more, e.g., about one week, two weeks, three weeks, or about four weeks or longer.

[0499] The present invention also provides methods of using an iRNA agent of the invention and/or a composition containing an iRNA agent of the invention to reduce and/or inhibit APCS expression in a subject.

[0500] In other aspects, use of an iRNA of the invention and/or a composition comprising an iRNA of the invention for the manufacture of a medicament for reducing and/or inhibiting APCS gene expression in a subject are provided.

[0501] Reduction in gene expression can be assessed by any methods known in the art. For example, a reduction in the expression of an APCS gene may be determined by determining the mRNA expression level of APCS using methods routine to one of ordinary skill in the art, e.g., Northern blotting, qRT-PCR, by determining the protein level of APCS using methods routine to one of ordinary skill in the art, such as Western blotting, immunological techniques, flow cytometry methods, ELISA, and/or by determining a biological activity of APCS.

[0502] APCS gene expression may be inhibited in the cell by at least about 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or about 100%, or below the level of detection of the assay method used.

[0503] APCS protein (SAP) production may be inhibited in the cell by at least about 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or about 100%, or below the level of detection of the assay method used.

[0504] The in vivo methods and uses of the invention may include administering to a subject a composition containing an iRNA, where the iRNA includes a nucleotide sequence that is complementary to at least a part of an RNA transcript of the APCS gene of the mammal to be treated. When the organism to be treated is a human, the composition can be administered by any means known in the art including, but not limited to subcutaneous, intravenous, oral, intraperitoneal, or parenteral routes, including intracranial (e.g., intraventricular, intraparenchymal and intrathecal), intramuscular, transdermal, airway (aerosol), nasal, rectal, and topical (including buccal and sublingual) administration. In certain embodiments, the compositions are administered by subcutaneous or intravenous infusion or injection. In one embodiment, the compositions are administered by subcutaneous injection. In some embodiments, the compositions are administered by intravenous infusion.

[0505] In some embodiments, the administration is via a depot injection. A depot injection may release the iRNA in a consistent way over a prolonged time period. Thus, a depot injection may reduce the frequency of dosing needed to obtain a desired effect, e.g., a desired inhibition of APCS, or a therapeutic or prophylactic effect. A depot injection may also provide more consistent serum concentrations. Depot injections may include subcutaneous injections or intramuscular injections. In preferred embodiments, the depot injection is a subcutaneous injection.

[0506] In some embodiments, the administration is via a pump. The pump may be an external pump or a surgically implanted pump. In certain embodiments, the pump is a subcutaneously implanted osmotic pump. In other embodiments, the pump is an infusion pump. An infusion pump may be used for intravenous, subcutaneous, arterial, or epidural infusions. In preferred embodiments, the infusion pump is a subcutaneous infusion pump. In other embodiments, the pump is a surgically implanted pump that delivers the iRNA to the subject.

[0507] The mode of administration may be chosen based upon whether local or systemic treatment is desired and based upon the area to be treated. The route and site of administration may be chosen to enhance targeting.

[0508] In one aspect, the present invention also provides methods for inhibiting the expression of an APCS gene in a subject, such as a mammal, e.g., a human. The present invention also provides a composition comprising an iRNA, e.g., a dsRNA, that targets an APCS gene in a cell of a subject, such as a mammal, for use in inhibiting expression of the APCS gene in the subject, such as a mammal. In another aspect, the present invention provides use of an iRNA, e.g., a dsRNA, that targets a APCS gene in a cell of a subject, such as a mammal, in the manufacture of a medicament for inhibiting expression of an APCS gene in the subject, such as a mammal.

[0509] The methods and uses include administering to a subject, such as a mammal, e.g., a human, a composition comprising an iRNA, e.g., a dsRNA, that targets an APCS gene in a cell of the subject, e.g., a mammal, and maintaining the mammal for a time sufficient to obtain degradation of the mRNA transcript of the APCS gene, thereby inhibiting expression of the APCS gene in the mammal.

[0510] In one embodiment, verification of RISC medicated cleavage of target in vivo following administration of iRNA agent is done by performing 5′-RACE or modifications of the protocol as known in the art (Lasham A et al., (2010) *Nucleic Acid Res.*, 38 (3) p-e19) (Zimmermann et al. (2006) *Nature* 441: 111-4).

[0511] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the iRNAs and methods featured in the invention, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

EXAMPLES

Example 1. iRNA Synthesis

Source of Reagents

[0512] Where the source of a reagent is not specifically given herein, such reagent can be obtained from any supplier of reagents for molecular biology at a quality/purity standard for application in molecular biology.

Bioinformatics

[0513] A set of siRNAs targeting the human SAP or APCS, “amyloid P component, serum” (human: NCBI refseqlD NM_001639; NCBI GeneID: 325) as well as toxicology-species APCS orthologs (cynomolgus monkey: XM_005541312; mouse: NM_011318; rat: NM_017170) were designed using custom R and Python scripts. The human NM_001639 REFSEQ mRNA, version 3, has a length of 960 bases. The rationale and method for the set of siRNA designs is as follows: the predicted efficacy for every potential 19mer siRNA from position 10 through the end was determined with a linear model derived the direct measure of mRNA knockdown from more than 20,000 distinct siRNA designs targeting a large number of vertebrate genes. A subset of the APCS siRNAs was designed with perfect or near-perfect matches between human and cynomolgus monkey. A further subset was designed with perfect or near-perfect matches between mouse and rat APCS orthologs. A further subset was designed with perfect matches to human APCS. For each strand of the siRNA, a custom Python script was used in a brute force search to measure the number and positions of mismatches between the siRNA and all potential alignments in the target species transcriptome. Extra weight was given to mismatches in the seed region, defined here as positions 2-9 of the antisense oligonucleotide, as well the cleavage site of the siRNA, defined here as positions 10-11 of the antisense oligonucleotide. The relative weight of the mismatches was 2.8; 1.2: 1 for seed mismatches, cleavage site, and other positions up through antisense position 19. Mismatches in the first position were ignored. A specificity score was calculated for each strand by summing the value of each weighted mismatch. Preference was given to siRNAs whose antisense score in human was >=2.2 and predicted efficacy was >=50% knockdown.

[0514] A detailed list of the unmodified SAP sense and antisense strand sequences is shown in Tables 3A and 3B. A detailed list of the modified SAP sense and antisense strand sequences is shown in Table 4A and 4B.

TABLE-US-00002 TABLE 2 Abbreviations of nucleotide monomers used in nucleic acid sequence representation. It will be understood that these monomers, when present in an oligonucleotide, are mutually linked by 5′-3′-phosphodiester bonds. Abbreviation Nucleotide(s) A Adenosine-3′-phosphate Af 2′-fluoroadenosine-3′-phosphate Afs 2′-fluoroadenosine-3′-phosphorothioate As adenosine-3′-phosphorothioate C cytidine-3′-phosphate Cf 2′-fluorocytidine-3′-phosphate Cfs 2′-fluorocytidine-3′-phosphorothioate Cs cytidine-3′-phosphorothioate G guanosine-3′-phosphate Gf 2′-fluoroguanosine-3′-phosphate Gfs 2′-fluoroguanosine-3′-phosphorothioate Gs guanosine-3′-phosphorothioate T 5′-methyluridine-3′-phosphate Tf 2′-fluoro-5-methyluridine-3′-phosphate Tfs 2′-fluoro-5-methyluridine-3′-phosphorothioate Ts 5-methyluridine-3′-phosphorothioate U Uridine-3′-phosphate Uf 2′-fluorouridine-3′-phosphate Ufs 2′-fluorouridine-3′-phosphorothioate Us uridine-3′-phosphorothioate N any nucleotide (G, A, C, T or U) a 2′-O-methyladenosine-3′-phosphate as 2′-O-methyladenosine-3′-phosphorothioate c 2′-O-methylcytidine-3′-phosphate cs 2′-O-methylcytidine-3′-phosphorothioate g 2′-O-methylguanosine-3′-phosphate gs 2′-O-methylguanosine-3′-phosphorothioate t 2′-O-methyl-5-methyluridine-3′-phosphate ts 2′-O-methyl-5-methyluridine-3′-phosphorothioate u 2′-O-methyluridine-3′-phosphate us 2′-O-methyluridine-3′-phosphorothioate s phosphorothioate linkage dT 2′-deoxythymidine-3′-phosphate dTs 2′-deoxythymidine-3′-phosphorothioate L96 N-[tris(GalNAc-alkyl)-amidodecanoyl)]-4-hydroxyprolinol Hyp-(GalNAc-alkyl)3 5′-VP 5′-vinyl phosphate

In Vitro Screen

Dual-Glo® Luciferase Assay

[0515] Cos7 cells (ATCC, Manassas, VA) were grown to near confluence at 37° C. in an atmosphere of 5% CO_2 in DMEM (ATCC) supplemented with 10% FBS, before being released from the plate by trypsinization. siRNA and psiCHECK2-APCS plasmid transfection was carried out by adding 5 μl of siRNA duplexes and 5 μl of psiCHECK2-APCS plasmid per well along with 5 μl of Opti-MEM plus 0.1 μl of Lipofectamine 2000 per well (Invitrogen, Carlsbad CA. cat #13778-150) and then incubated at room temperature for 15 minutes. The mixture was then added to the cells which were re-suspended in 35 μl of fresh complete media. The transfected cells were incubated at 37° C. in an atmosphere of 5% CO_2.

[0516] Forty-eight hours after the siRNAs and psiCHECK2-APCS plasmid were transfected; Firefly (transfection control) and *Renilla* (fused to ANGPTL4 target sequence) luciferase were measured. First, media was removed from cells. Then Firefly luciferase activity was measured by adding 20 μl of Dual-Glo® Luciferase Reagent equal to the culture medium volume to each well and mix. The mixture was incubated at room temperature for 30 minutes before luminescence (500 nm) was measured on a Spectramax (Molecular Devices) to detect the Firefly luciferase signal. *Renilla* luciferase activity was measured by adding 20 μl of room temperature of Dual-Glo® Stop & Glo® Reagent to each well and the plates were incubated for 10-15 minutes before luminescence was again measured to determine the *Renilla* luciferase signal. The Dual-Glo® Stop & Glo® Reagent, quenches the firefly luciferase signal and sustained luminescence for the *Renilla* luciferase reaction. siRNA activity was determined by normalizing the *Renilla* (HBV) signal to the Firefly (control) signal within each well. The magnitude of siRNA activity was then assessed relative to cells that were transfected with the same vector but were not treated with siRNA or were treated with a non-targeting siRNA. All transfections were done at n=2 or greater. Table 5A shows the results of a single dose screen of the indicated agents at 10 nm or 0.1 nm and Table 5B shows the results of a single dose screen of the indicated agents at 10 nM, 1 nM, or 0.1 nM. Data are expressed as percent of mRNA remaining relative to untreated Cos7 cells.

TABLE-US-00003 TABLE 3A APCS iRNAs - Unmodified Sequences Duplex SEQ ID SEQ ID Range Name Sense Sequence 5′ to 3′ NO: Antisnse Sequence 5′ to 3′ NO: IN NM-001639.3 AD-75708 ACUGCUUCUGCUAUAACAGCA 13 UGCUGUUAUAGCAGAAGCAGUGA 97 61-83 AD-75723 UAUGAACAAGCCGCUGCUUUA 14 UAAAGCAGCGGCUUGUUCAUAUU 98 94-116 AD-75694 CAGUGGGAAGGUGUUUGUAUA 15 UAUACAAACACCUUCCCACUGAG 99 163-185 AD-75692 GUGUUUGUAUUUCCUAGAGAA 16 UUCUCUAGGAAAUACAAACACCU 100 173-195 AD-75664 UCUGUUACUGAUCAUGUAAAA 17 UUUUACAUGAUCAGUAACAGAUU 101 194-216 AD-75664.2 UCUGUUACUGAUCAUGUAAAA 18 UUUUACAUGAUCAGUAACAGAUU 102 194-216 AD-75679 GUUACUGAUCAUGUAAACUUA 19 UAAGUUUACAUGAUCAGUAACAG 103 197-219 AD-75659 UACUGAUCAUGUAAACUUGAA 20 UUCAAGUUUACAUGAUCAGUAAC 104 199-221 AD-75662 AUCAUGUAAACUUGAUCACAA 21 UUGUGAUCAAGUUUACAUGAUCA 105 204-226 AD-75680 UCAUGUAAACUUGAUCACACA 22 UGUGUGAUCAAGUUUACAUGAUC 106 205-227 AD-75687 CUCUACAGAACUUUACCUUGA 23 UCAAGGUAAAGUUCUGUAGAGGC 107 237-259 AD-75657 ACUUUACCUUGUGUUUUCGAA 24 UUCGAAAACACAAGGUAAAGUUC 108 246-268 AD-75699 GUUUUCGAGCCUAUAGUGAUA 25 UAUCACUAUAGGCUCGAAAACAC 109 258-280 AD-75727 GAAACUGAUCAUGUGAAGCUA 26 UAGCUUCACAUGAUCAGUUUCAG 110 271-293 AD-75731 ACCUCUGCAGAAUUUUACACA 27 UGUGUAAAAUUCUGCAGAGGUUU 111 309-331 AD-75728 GCAGAAUUUUACACUGUGUUA 28 UAACACAGUGUAAAAUUCUGCAG 112 315-337 AD-75737 CAGAAUUUUACACUGUGUUUA 29 UAAACACAGUGUAAAAUUCUGCA 113 316-338 AD-75696 UAAUGAGCUACUAGUUUAUAA 30 UUAUAAACUAGUAGCUCAUUAUC 114 325-347 AD-75718 AAUGAGCUACUAGUUUAUAAA 31 UUUAUAAACUAGUAGCUCAUUAU 115 326-348 AD-75676 AUGAGCUACUAGUUUAUAAAA 32 UUUUAUAAACUAGUAGCUCAUUA 116 327-349 AD-75663 UGAGCUACUAGUUUAUAAAGA 33 UCUUUAUAAACUAGUAGCUCAUU 117 328-350 AD-75669 GAGCUACUAGUUUAUAAAGAA 34 UUCUUUAUAAACUAGUAGCUCAU 118 329-351 AD-75666 AGCUACUAGUUUAUAAAGAAA 35 UUUCUUUAUAAACUAGUAGCUCA 119 330-352 AD-75735 UGUUUCCGAACCUACAGUGAA 36 UUCACUGUAGGUUCGGAAACACA 120 331-353 AD-75686 CUACUAGUUUAUAAAGAAAGA 37 UCUUUCUUUAUAAACUAGUAGCU 121 332-354 AD-75736 CCGAACCUACAGUGACCUUUA 38 UAAAGGUCACUGUAGGUUCGGAA 122 336-358 AD-75674 GUUUAUAAAGAAAGAGUUGGA 39 UCCAACUCUUUCUUUAUAAACUA 123 338-360 AD-75717 UAAAGAAAGAGUUGGAGAGUA 40 UACUCUCCAACUCUUUCUUUAUA 124 343-365 AD-75706 GUUGGAGAGUAUAGUCUAUAA 41 UUAUAGACUAUACUCUCCAACUC 125 353-375 AD-75719 AGUAUAGUCUAUACAUUGGAA 42 UUCCAAUGUAUAGACUAUACUCU 126 360-382 AD-75688 UAGUCUAUACAUUGGAAGACA 43 UGUCUUCCAAUGUAUAGACUAUA 127 364-386 AD-75734 AGAGUCUUUUCUCCUACAGUA 44 UACUGUAGGAGAAAAGACUCUGA 128 365-387 AD-75724 UACAUUGGAAGACACAAAGUA 45

UACUUUGUGUCUUCCAAUGUAUA 129 371-393 AD-75711
UUGGAAGACACAAAGUUACAA 46 UUGUAACUUUGUGUCUUCCAAUG 130 375-397
AD-75703 AGACACAAAGUUACAUCCAAA 47 UUUGGAUGUAACUUUGUGUCUUC 131
380-402 AD-75721 ACAAAGUUACAUCCAAAGUUA 48
UAACUUUGGAUGUAACUUUGUGU 132 384-406 AD-75712
AAGUUACAUCCAAAGUUAUCA 49 UGAUAACUUUGGAUGUAACUUUG 133 387-409
AD-75697 UACAUCCAAAGUUAUCGAAAA 50 UUUUCGAUAACUUUGGAUGUAAC 134
391-413 AD-75726 UCCAAAGUUAUCGAAAAGUUA 51
UAACUUUUCGAUAACUUUGGAUG 135 395-417 AD-75730
GAGACAAUGAGCUACUAAUUA 52 UAAUUAGUAGCUCAUUGUCUCUG 136 395-417
AD-75732 AGACAAUGAGCUACUAAUUUA 53 UAAAUUAGUAGCUCAUUGUCUCU 137
396-418 AD-75733 AAUGAGCUACUAAUUUAUAAA 54
UUUAUAAAUUAGUAGCUCAUUGU 138 400-422 AD-75729
GAGCUACUAAUUUAUAAAGAA 55 UUCUUUAUAAAUUAGUAGCUCAU 139 403-425
AD-75685 CUCAUCAGGUAUUGCUGAAUA 56 UAUUCAGCAAUACCUGAUGAGGA 140
451-473 AD-75673 UCAGGUAUUGCUGAAUUUUGA 57
UCAAAAUUCAGCAAUACCUGAUG 141 455-477 AD-75691
GUAUUGCUGAAUUUUGGAUCA 58 UGAUCCAAAAUUCAGCAAUACCU 142 459-481
AD-75670 GAAGCUCAGCCCAAGAUUGUA 59 UACAAUCUUGGGCUGAGCUUCUA 143
527-549 AD-75739 GCAUUGUUGAAUUUUGGGUCA 60
UGACCCAAAAUUCAACAAUGCCA 144 533-555 AD-75738
UUGUUGAAUUUUGGGUCAAUA 61 UAUUGACCCAAAAUUCAACAAUG 145 536-558
AD-75722 AUUCCUAUGGGGGCAAGUUUA 62 UAAACUUGCCCCCAUAGGAAUCC 146
564-586 AD-75714 AAAAUAUCCUGUCUGCCUAUA 63
UAUAGGCAGACAGGAUAUUUUCU 147 651-673 AD-75681
AAAUAUCCUGUCUGCCUAUCA 64 UGAUAGGCAGACAGGAUAUUUUC 148 652-674
AD-75668 AAUAUCCUGUCUGCCUAUCAA 65 UUGAUAGGCAGACAGGAUAUUUU 149
653-675 AD-75693 CAGGCUCUGAACUAUGAAAUA 66
UAUUUCAUAGUUCAGAGCCUGCC 150 707-729 AD-75677
AGGCUCUGAACUAUGAAAUCA 67 UGAUUUCAUAGUUCAGAGCCUGC 151 708-730
AD-75690 GGCUCUGAACUAUGAAAUCAA 68 UUGAUUUCAUAGUUCAGAGCCUG 152
709-731 AD-75716 GCUCUGAACUAUGAAAUCAGA 69
UCUGAUUUCAUAGUUCAGAGCCU 153 710-732 AD-75682
GAGGAUAUGUCAUCAUCAAAA 70 UUUUGAUGAUGACAUAUCCUCUG 154 729-751
AD-75720 UAUGUCAUCAUCAAACCCUUA 71 UAAGGGUUUGAUGAUGACAUAUC 155
734-756 AD-75725 CAUCAUCAAACCCUUGGUGUA 72
UACACCAAGGGUUUGAUGAUGAC 156 739-761 AD-75695
AACGAGAGCACUUGAAAAUGA 73 UCAUUUUCAAGUGCUCUCGUUGA 157 778-800
AD-75665 CGAGAGCACUUGAAAAUGAAA 74 UUUCAUUUUCAAGUGCUCUCGUU 158
780-802 AD-75661 AGCACUUGAAAAUGAAAUGAA 75
UUCAUUUCAUUUUCAAGUGCUCU 159 784-806 AD-75658
GCACUUGAAAAUGAAAUGACA 76 UGUCAUUUCAUUUUCAAGUGCUC 160 785-807
AD-75700 AAAUGAAAUGACUGUCUAAGA 77 UCUUAGACAGUCAUUUCAUUUUC 161
793-815 AD-75698 AAUGAAAUGACUGUCUAAGAA 78
UUCUUAGACAGUCAUUUCAUUUU 162 794-816 AD-75672
UGAAAUGACUGUCUAAGAGAA 79 UUCUCUUAGACAGUCAUUUCAUU 163 796-818
AD-75684 GAAAUGACUGUCUAAGAGAUA 80 UAUCUCUUAGACAGUCAUUUCAU 164
797-819 AD-75667 AAAUGACUGUCUAAGAGAUCA 81
UGAUCUCUUAGACAGUCAUUUCA 165 798-820 AD-75678
CAACUGGAUACUAGAUCUUAA 82 UUAAGAUCUAGUAUCCAGUUGCU 166 827-849

AD-75660 GCAGCUCUUUCUUCUUUGAAA 83 UUUCAAAGAAGAAAGAGCUGCAG 167 852-874 AD-75701 CUCUUUCUUCUUUGAAUUUCA 84 UGAAAUUCAAAGAAGAAAGAGCU 168 856-878 AD-75707 CUUUCUUCUUUGAAUUUCCUA 85 UAGGAAAUUCAAAGAAGAAAGAG 169 858-880 AD-75675 CUUUGAAUUUCCUAUCUGUAA 86 UUACAGAUAGGAAAUUCAAAGAA 170 865-887 AD-75671 UUUGAAUUUCCUAUCUGUAUA 87 UAUACAGAUAGGAAAUUCAAAGA 171 866-888 AD-75683 UGAAUUUCCUAUCUGUAUGUA 88 UACAUACAGAUAGGAAAUUCAAA 172 868-890 AD-75689 AAUUUCCUAUCUGUAUGUCUA 89 UAGACAUACAGAUAGGAAAUUCA 173 870-892 AD-75715 AUCUGUAUGUCUGCCUAAUUA 90 UAAUUAGGCAGACAUACAGAUAG 174 878-900 AD-75705 UGUAUGUCUGCCUAAUUAAAA 91 UUUUAAUUAGGCAGACAUACAGA 175 881-903 AD-75704 GUAUGUCUGCCUAAUUAAAAA 92 UUUUUAAUUAGGCAGACAUACAG 176 882-904 AD-75713 UAUGUCUGCCUAAUUAAAAAA 93 UUUUUUAAUUAGGCAGACAUACA 177 883-905 AD-75702 UGUCUGCCUAAUUAAAAAAAA 94 UUUUUUUUAAUUAGGCAGACAUA 178 885-907 AD-75709 UUGUAUUAUGCUACCUGCAAA 95 UUUGCAGGUAGCAUAAUACAAUA 179 911-933 AD-75710 UUAUGCUACCUGCAAAAAAAA 96 UUUUUUUUGCAGGUAGCAUAAUA 180 916-938

TABLE-US-00004 TABLE 3B APCS iRNAs - Unmodified Sequences Duplex Sense Sequence 5′-3′ SEQ ID Range in Name Antisense Sequence 5′-3′ Range in NM_011318.2 NO NM_011318.2 AD-197584.1 CAAUGAGCUACUAAUUUAUAA NM_011318.2_401-421_s 401-421_s UUAUAAAUUAGUAGCUCAUUGUC NM_011318.2_399-421_as 399-421_as AD-197510.1 AUUUUACACUGUGUUUCCGAA NM_011318.2_322-342_s 322-342_s UUCGGAAACACAGUGUAAAAUUC NM_011318.2_320-342_as 320-342_as AD-197583.1 ACAAUGAGCUACUAAUUUAUA NM_011318.2_400-420_s 400-420_s UAUAAAUUAGUAGCUCAUUGUCU NM_011318.2_398-420_as 398-420_as AD-75728.3 GCAGAAUUUUACACUGUGUUA NM_011318.2_319-338_s 319-338_s UAACACAGUGUAAAAUUCUGCAG NM_011318.2_317-338_as 317-338_as AD-197508.1 GAAUUUUACACUGUGUUUCCA NM_011318.2_320-340_G21A_s 320-340_G21A_s UGGAAACACAGUGUAAAAUUCUG NM_011318.2_318-340_C1U_as 318-340_C1U_as AD-75737.3 CAGAAUUUUACACUGUGUUUA NM_011318.2_320-339_C21A_s 320-339_C21A_s UAAACACAGUGUAAAAUUCUGCA NM_011318.2_318-339_C21A_as 318-339_C21A_as AD-197550.1 CUCAGAGUCUUUUCUCCUACA NM_011318.2_364-384_s 364-384_s UGUAGGAGAAAAGACUCUGAGAG NM_011318.2_362-384_as 362-384_as AD-197582.1 GACAAUGAGCUACUAAUUUAA NM_011318.2_399-419_s 399-419_s UUAAAUUAGUAGCUCAUUGUCUC NM_011318.2_397-419_as 397-419_as AD-75732.3 AGACAAUGAGCUACUAAUUUA NM_011318.2_400-419_s 400-419_s UAAAUUAGUAGCUCAUUGUCUCU NM_011318.2_398-419_as 398-419_as AD-197726.1 AUGGAAAGCCUUGGGUAAAAA NM_011318.2_556-576_s 556-576_s UUUUUACCCAAGGCUUUCCAUUG NM_011318.2_554-576_as 554-576_as AD-197544.1 CCCGCUCUCAGAGUCUUUUCA NM_011318.2_358-378_s 358-378_s UGAAAAGACUCUGAGAGCGGGAA NM_011318.2_356-378_as 356-378_as AD-197551.1 UCAGAGUCUUUUCUCCUACAA NM_011318.2_365-385_G21A_s 365-385_G21A_s UUGUAGGAGAAAAGACUCUGAGA NM_011318.2_363-385_C1U_as 363-385_C1U_as AD-75730.2 GAGACAAUGAGCUACUAAUUA NM_011318.2_399-418_s 399-418_s UAAUUAGUAGCUCAUUGUCUCUG NM_011318.2_397-418_as 397-418_as AD-197410.1 UUCACCAGCCUUCUUUCAGAA NM_011318.2_198-218_s 198-218_s UUCUGAAAGAAGGCUGGUGAAGA NM_011318.2_196-218_as 196-218_as AD-197556.1

UCUUUUCUCCUACAGUGUCAA NM_011318.2_371-391_s 371-391_s UUGACACUGUAGGAGAAAAGACU NM_011318.2_369-391_as 369-391_as AD-197534.1 AGUGACCUUUCCCGCUCUCAA NM_011318.2_348-368_G21A_s 348-368_G21A_s UUGAGAGCGGGAAAGGUCACUGU NM_011318.2_346-368_C1U_as 346-368_C1U_as AD-75734.2 AGAGUCUUUUCUCCUACAGUA NM_011318.2_369-388_G21A_s 369-388_G21A_s UACUGUAGGAGAAAAGACUCUGA NM_011318.2_367-388_G21A_as 367-388_G21A_as AD-197503.1 CUCUGCAGAAUUUUACACUGA NM_011318.2_313-333_s 313-333_s UCAGUGUAAAAUUCUGCAGAGGU NM_011318.2_311-333_as 311-333_as AD-197406.1 UGUCUUCACCAGCCUUCUUUA NM_011318.2_194-214_C21A_s 194-214_C21A_s UAAAGAAGGCUGGUGAAGACAAA NM_011318.2_192-214_G1U_as 192-214_G1U_as AD-197416.1 AGCCUUCUUUCAGAAGCCUUA NM_011318.2_204-224_s 204-224_s UAAGGCUUCUGAAAGAAGGCUGG NM_011318.2_202-224_as 202-224_as AD-197725.1 AAUGGAAAGCCUUGGGUAAAA NM_011318.2_555-575_s 555-575_s UUUUACCCAAGGCUUUCCAUUGA NM_011318.2_553-575_as 553-575_as AD-197546.1 CGCUCUCAGAGUCUUUUCUCA NM_011318.2_360-380_C21A_s 360-380_C21A_s UGAGAAAAGACUCUGAGAGCGGG NM_011318.2_358-380_G1U_as 358-380_G1U_as AD-197724.1 CAAUGGAAAGCCUUGGGUAAA NM_011318.2_554-574_s 554-574_s UUUACCCAAGGCUUUCCAUUGAC NM_011318.2_552-574_as 552-574_as AD-197408.1 UCUUCACCAGCCUUCUUUCAA NM_011318.2_196-216_G21A_s 196-216_G21A_s UUGAAAGAAGGCUGGUGAAGACA NM_011318.2_194-216_C1U_as 194-216_C1U_as AD-197548.1 CUCUCAGAGUCUUUUCUCCUA NM_011318.2_362-382_s 362-382_s UAGGAGAAAAGACUCUGAGAGCG NM_011318.2_360-382_as 360-382_as AD-197565.1 CAGAGACAAUGAGCUACUAAA NM_011318.2_397-416_s 397-416_s UUUAGUAGCUCAUUGUCUCUGCC NM_011318.2_395-416_as 395-416_as AD-197581.1 AGAGACAAUGAGCUACUAAUA NM_011318.2_396-416_s 396-416_s UAUUAGUAGCUCAUUGUCUCUGC NM_011318.2_394-416_as 394-416_as AD-197552.1 CAGAGUCUUUUCUCCUACAGA NM_011318.2_366-386_s 366-386_s UCUGUAGGAGAAAAGACUCUGAG NM_011318.2_364-386_as 364-386_as AD-197549.1 UCUCAGAGUCUUUUCUCCUAA NM_011318.2_363-383_C21A_s 363-383_C21A_s UUAGGAGAAAAGACUCUGAGAGC NM_011318.2_361-383_G1U_as 361-383_G1U_as AD-197529.1 CCUACAGUGACCUUUCCCGCA NM_011318.2_343-363_s 343-363_s UGCGGGAAAGGUCACUGUAGGUU NM_011318.2_341-363_as 341-363_as AD-197417.1 GCCUUCUUUCAGAAGCCUUUA NM_011318.2_205-225_s 205-225_s UAAAGGCUUCUGAAAGAAGGCUG NM_011318.2_203-225_as 203-225_as AD-197543.1 UCCCGCUCUCAGAGUCUUUUA NM_011318.2_357-377_C21A_s 357-377_C21A_s UAAAAGACUCUGAGAGCGGGAAA NM_011318.2_355-377_G1U_as 355-377_G1U_as AD-197555.1 GUCUUUUCUCCUACAGUGUCA NM_011318.2_370-390_s 370-390_s UGACACUGUAGGAGAAAAGACUC NM_011318.2_368-390_as 368-390_as AD-197530.1 CUACAGUGACCUUUCCCGCUA NM_011318.2_344-364_C21A_s 344-364_C21A_s UAGCGGGAAAGGUCACUGUAGGU NM_011318.2_342-364_G1U_as 342-364_G1U_as AD-197554.1 AGUCUUUUCUCCUACAGUGUA NM_011318.2_369-389_C21A_s 369-389_C21A_s UACACUGUAGGAGAAAAGACUCU NM_011318.2_367-389_G1U_as 367-389_G1U_as AD-197532.1 ACAGUGACCUUUCCCGCUCUA NM_011318.2_346-366_C21A_s 346-366_C21A_s UAGAGCGGGAAAGGUCACUGUAG NM_011318.2_344-366_G1U_as 344-366_G1U_as AD-197412.1 CACCAGCCUUCUUUCAGAAGA NM_011318.2_200-220_C21A_s 200-220_C21A_s UCUUCUGAAAGAAGGCUGGUGAA NM_011318.2_198-220_G1U_as 198-220_G1U_as AD-197713.1 GAAUUUUGGGUCAAUGGAAAA NM_011318.2_543-563_s 543-563_s UUUUCCAUUGACCCAAAAUUCAA NM_011318.2_541-563_as 541-563_as AD-197411.1 UCACCAGCCUUCUUUCAGAAA NM_011318.2_199-219_G21A_s 199-219_G21A_s UUUCUGAAAGAAGGCUGGUGAAG NM_011318.2_197-219_C1U_as 197-219_C1U_as AD-

197557.1 CUUUUCUCCUACAGUGUCAAA NM_011318.2_372-392_G21A_s 372-392_G21A_s UUUGACACUGUAGGAGAAAAGAC NM_011318.2_370-392_C1U_as 370-392_C1U_as AD-197528.1 ACCUACAGUGACCUUUCCCGA NM_011318.2_342-362_C21A_s 342-362_C21A_s UCGGGAAAGGUCACUGUAGGUUC NM_011318.2_340-362_G1U_as 340-362_G1U_as AD-197631.1 UCAAAAGUCACAGUCCGUGGA NM_011318.2_459-479_s 459-479_s UCCACGGACUGUGACUUUUGAUU NM_011318.2_457-479_as 457-479_as AD-197541.1 UUUCCCGCUCUCAGAGUCUUA NM_011318.2_355-375_s 355-375_s UAAGACUCUGAGAGCGGGAAAGG NM_011318.2_353-375_as 353-375_as AD-197533.1 CAGUGACCUUUCCCGCUCUCA NM_011318.2_347-367_s 347-367_s UGAGAGCGGGAAAGGUCACUGUA NM_011318.2_345-367_as 345-367_as AD-197632.1 CAAAAGUCACAGUCCGUGGUA NM_011318.2_460-480_s 460-480_s UACCACGGACUGUGACUUUUGAU NM_011318.2_458-480_as 458-480_as AD-197542.1 UUCCCGCUCUCAGAGUCUUUA NM_011318.2_356-376_s 356-376_s UAAAGACUCUGAGAGCGGGAAAG NM_011318.2_354-376_as 354-376_as AD-197629.1 AAUCAAAAGUCACAGUCCGUA NM_011318.2_457-477_G21A_s 457-477_G21A_s UACGGACUGUGACUUUUGAUUGU NM_011318.2_455-477_C1U_as 455-477_C1U_as

TABLE-US-00005 TABLE 4A APCS iRNAs - Modified Sequences Duplex Name Sense Sequence 5′ to 3′ SEQ ID NO Antisense Sequence 5′ to 3′ SEQ ID NO mRNA target sequence SEQ ID NO AD-75708 ascsugcuUfcUfGfCfuauaacagcaL96 181 VPusGfscugUfuAfUfagcaGfaAfgcagusgsa 265 UCACUGCUUCUGCUAUAACAGCC 349 AD-75723 usasugaaCfaAfGfCfcgcugcuuuaL96 182 VPusAfsaagCfaGfCfggcuUfgUfucauasusu 266 AAUAUGAACAAGCCGCUGCUUUG 350 AD-75694 csasguggGfaAfGfGfuguuuguauaL96 183 VPusAfsuacAfaAfCfaccuUfcCfcacugsasg 267 CUCAGUGGGAAGGUGUUUGUAUU 351 AD-75692 gsusguuuGfuAfUfUfuccuagagaaL96 184 VPusUfscucUfaGfGfaaauAfcAfaacacscsu 268 AGGUGUUUGUAUUUCCUAGAGAA 352 AD-75664 uscsuguuAfcUfGfAfucauguaaaaL96 185 VPusUfsuuaCfaUfGfaucaGfuAfacagasusu 269 AAUCUGUUACUGAUCAUGUAAAC 353 AD-75664.2 uscsuguuAfcUfGfAfucauguaaaaL96 186 VPusUfsuuaCfaUfGfaucaGfuAfacagasusu 270 AAUCUGUUACUGAUCAUGUAAAC 354 AD-75679 gsusuacuGfaUfCfAfuguaaacuuaL96 187 VPusAfsaguUfuAfCfaugaUfcAfguaacsasg 271 CUGUUACUGAUCAUGUAAACUUG 355 AD-75659 usascugaUfcAfUfGfuaaacuugaaL96 188 VPusUfscaaGfuUfUfacauGfaUfcaguasasc 272 GUUACUGAUCAUGUAAACUUGAU 356 AD-75662 asuscaugUfaAfAfCfuugaucacaaL96 189 VPusUfsgugAfuCfAfaguuUfaCfaugauscsa 273 UGAUCAUGUAAACUUGAUCACAC 357 AD-75680 uscsauguAfaAfCfUfugaucacacaL96 190 VPusGfsuguGfaUfCfaaguUfuAfcaugasusc 274 GAUCAUGUAAACUUGAUCACACC 358 AD-75687 csuscuacAfgAfAfCfuuuaccuugaL96 191 VPusCfsaagGfuAfAfaguuCfuGfuagagsgsc 275 GCCUCUACAGAACUUUACCUUGU 359 AD-75657 ascsuuuaCfcUfUfGfuguuuucgaaL96 192 VPusUfscgaAfaAfCfacaaGfgUfaaagususc 276 GAACUUUACCUUGUGUUUUCGAG 360 AD-75699 gsusuuucGfaGfCfCfuauagugauaL96 193 VPusAfsucaCfuAfUfaggcUfcGfaaaacsasc 277 GUGUUUUCGAGCCUAUAGUGAUC 361 AD-75727 gsasaacuGfaUfCfAfugugaagcuaL96 194 VPusAfsgcuUfcAfCfaugaUfcAfguuucsasg 278 CUGAAACUGAUCAUGUGAAGCUG 362 AD-75731 ascscucuGfcAfGfAfauuuuacacaL96 195 VPusGfsuguAfaAfAfuucuGfcAfgaggususu 279 AAACCUCUGCAGAAUUUUACACU 363 AD-75728 gscsagaaUfuUfUfAfcacuguguuaL96 196 VPusAfsacaCfaGfUfguaaAfaUfucugcsasg 280 CUGCAGAAUUUUACACUGUGUUU 364 AD-75737 csasgaauUfuUfAfCfacuguguuuaL96 197 VPusAfsaacAfcAfGfuguaAfaAfuucugscsa 281 UGCAGAAUUUUACACUGUGUUUC 365 AD-75696 usasaugaGfcUfAfCfuaguuuauaaL96 198 VPusUfsauaAfaCfUfaguaGfcUfcauuasusc 282 GAUAAUGAGCUACUAGUUUAUAA 366 AD-75718 asasugagCfuAfCfUfaguuuauaaaL96 199 VPusUfsuauAfaAfCfuaguAfgCfucauusasu 283 AUAAUGAGCUACUAGUUUAUAAA 367 AD-75676 asusgagcUfaCfUfAfguuuauaaaaL96 200 VPusUfsuuaUfaAfAfcuagUfaGfcucaususa 284 UAAUGAGCUACUAGUUUAUAAAG 368 AD-75663 usgsagcuAfcUfAfGfuuuauaaagaL96 201 VPusCfsuuuAfuAfAfacuaGfuAfgcucasusu 285 AAUGAGCUACUAGUUUAUAAAGA 369 AD-

75669 gsasgcuaCfuAfGfUfuuauaaagaaL96 202 VPusUfscuuUfaUfAfaacuAfgUfagcucsasu 286 AUGAGCUACUAGUUUAUAAAGAA 370 AD-75666 asgscuacUfaGfUfUfuauaaagaaaL96 203 VPusUfsucuUfuAfUfaaacUfaGfuagcuscsa 287 UGAGCUACUAGUUUAUAAAGAAA 371 AD-75735 usgsuuucCfgAfAfCfcuacagugaaL96 204 VPusUfscacUfgUfAfgguuCfgGfaaacascsa 288 UGUGUUUCCGAACCUACAGUGAC 372 AD-75686 csusacuaGfuUfUfAfuaaagaaagaL96 205 VPusCfsuuuCfuUfUfauaaAfcUfaguagscsu 289 AGCUACUAGUUUAUAAAGAAAGA 373 AD-75736 cscsgaacCfuAfCfAfgugaccuuuaL96 206 VPusAfsaagGfuCfAfcuguAfgGfuucggsasa 290 UUCCGAACCUACAGUGACCUUUC 374 AD-75674 gsusuuauAfaAfGfAfaagaguuggaL96 207 VPusCfscaaCfuCfUfuucuUfuAfuaaacsusa 291 UAGUUUAUAAAGAAAGAGUUGGA 375 AD-75717 usasaagaAfaGfAfGfuuggagaguaL96 208 VPusAfscucUfcCfAfacucUfuUfcuuuasusa 292 UAUAAAGAAAGAGUUGGAGAGUA 376 AD-75706 gsusuggaGfaGfUfAfuagucuauaaL96 209 VPusUfsauaGfaCfUfauacUfcUfccaacsusc 293 GAGUUGGAGAGUAUAGUCUAUAC 377 AD-75719 asgsuauaGfuCfUfAfuacauuggaaL96 210 VPusUfsccaAfuGfUfauagAfcUfauacuscsu 294 AGAGUAUAGUCUAUACAUUGGAA 378 AD-75688 usasgucuAfuAfCfAfuuggaagacaL96 211 VPusGfsucuUfcCfAfauguAfuAfgacuasusa 295 UAUAGUCUAUACAUUGGAAGACA 379 AD-75734 asgsagucUfuUfUfCfuccuacaguaL96 212 VPusAfscugUfaGfGfagaaAfaGfacucusgsa 296 UCAGAGUCUUUUCUCCUACAGUG 380 AD-75724 usascauuGfgAfAfGfacacaaaguaL96 213 VPusAfscuuUfgUfGfucuuCfcAfauguasusa 297 UAUACAUUGGAAGACACAAAGUU 381 AD-75711 ususggaaGfaCfAfCfaaaguuacaaL96 214 VPusUfsguaAfcUfUfugugUfcUfuccaasusg 298 CAUUGGAAGACACAAAGUUACAU 382 AD-75703 asgsacacAfaAfGfUfuacauccaaaL96 215 VPusUfsuggAfuGfUfaacuUfuGfugucususc 299 GAAGACACAAAGUUACAUCCAAA 383 AD-75721 ascsaaagUfuAfCfAfuccaaaguuaL96 216 VPusAfsacuUfuGfGfauguAfaCfuuugusgsu 300 ACACAAAGUUACAUCCAAAGUUA 384 AD-75712 asasguuaCfaUfCfCfaaaguuaucaL96 217 VPusGfsauaAfcUfUfuggaUfgUfaacuususg 301 CAAAGUUACAUCCAAAGUUAUCG 385 AD-75697 usascaucCfaAfAfGfuuaucgaaaaL96 218 VPusUfsuucGfaUfAfacuuUfgGfauguasasc 302 GUUACAUCCAAAGUUAUCGAAAA 386 AD-75726 uscscaaaGfuUfAfUfcgaaaaguuaL96 219 VPusAfsacuUfuUfCfgauaAfcUfuuggasusg 303 CAUCCAAAGUUAUCGAAAAGUUC 387 AD-75730 gsasgacaAfuGfAfGfcuacuaauuaL96 220 VPusAfsauuAfgUfAfgcucAfuUfgucucsusg 304 CAGAGACAAUGAGCUACUAAUUU 388 AD-75732 asgsacaaUfgAfGfCfuacuaauuuaL96 221 VPusAfsaauUfaGfUfagcuCfaUfugucuscsu 305 AGAGACAAUGAGCUACUAAUUUA 389 AD-75733 asasugagCfuAfCfUfaauuuauaaaL96 222 VPusUfsuauAfaAfUfuaguAfgCfucauusgsu 306 ACAAUGAGCUACUAAUUUAUAAA 390 AD-75729 gsasgcuaCfuAfAfUfuuauaaagaaL96 223 VPusUfscuuUfaUfAfaauuAfgUfagcucsasu 307 AUGAGCUACUAAUUUAUAAAGAA 391 AD-75685 csuscaucAfgGfUfAfuugcugaauaL96 224 VPusAfsuucAfgCfAfauacCfuGfaugagsgsa 308 UCCUCAUCAGGUAUUGCUGAAUU 392 AD-75673 uscsagguAfuUfGfCfugaauuuugaL96 225 VPusCfsaaaAfuUfCfagcaAfuAfccugasusg 309 CAUCAGGUAUUGCUGAAUUUUGG 393 AD-75691 gsusauugCfuGfAfAfuuuuggaucaL96 226 VPusGfsaucCfaAfAfauucAfgCfaauacscsu 310 AGGUAUUGCUGAAUUUUGGAUCA 394 AD-75670 gsasagcuCfaGfCfCfcaagauuguaL96 227 VPusAfscaaUfcUfUfgggcUfgAfgcuucsusa 311 UAGAAGCUCAGCCCAAGAUUGUC 395 AD-75739 gscsauugUfuGfAfAfuuuuggucaL96 228 VPusGfsaccCfaAfAfauucAfaCfaaugcscsa 312 UGGCAUUGUUGAAUUUUGGGUCA 396 AD-75738 ususguugAfaUfUfUfugggucaauaL96 229 VPusAfsuugAfcCfCfaaaaUfuCfaacaasusg 313 CAUUGUUGAAUUUUGGGUCAAUG 397 AD-75722 asusuccuAfuGfGfGfggcaaguuuaL96 230 VPusAfsaacUfuGfCfccccAfuAfggaauscsc 314 GGAUUCCUAUGGGGGCAAGUUUG 398 AD-75714 asasaauaUfcCfUfGfucugccuauaL96 231 VPusAfsuagGfcAfGfacagGfaUfauuuuscsu 315 AGAAAAUAUCCUGUCUGCCUAUC 399 AD-75681 asasauauCfcUfGfUfcugccuaucaL96 232 VPusGfsauaGfgCfAfgacaGfgAfuauuususc 316 GAAAAUAUCCUGUCUGCCUAUCA 400 AD-75668 asasuaucCfuGfUfCfugccuaucaaL96 233 VPusUfsgauAfgGfCfagacAfgGfauauususu 317 AAAAUAUCCUGUCUGCCUAUCAG 401 AD-75693 csasggcuCfuGfAfAfcuaugaaauaL96 234 VPusAfsuuuCfaUfAfguucAfgAfgccugscsc 318 GGCAGGCUCUGAACUAUGAAAUC 402 AD-75677 asgsgcucUfgAfAfCfuaugaaaucaL96 235

VPusGfsauuUfcAfUfaguuCfaGfagccusgsc 319 GCAGGCUCUGAACUAUGAAAUCA 403 AD-75690 gsgscucuGfaAfCfUfaugaaaucaaL96 236 VPusUfsgauUfuCfAfuaguUfcAfgagccsusg 320 CAGGCUCUGAACUAUGAAAUCAG 404 AD-75716 gscsucugAfaCfUfAfugaaaucagaL96 237 VPusCfsugaUfuUfCfauagUfuCfagagcscsu 321 AGGCUCUGAACUAUGAAAUCAGA 405 AD-75682 gsasggauAfuGfUfCfaucaucaaaaL96 238 VPusUfsuugAfuGfAfugacAfuAfuccucsusg 322 CAGAGGAUAUGUCAUCAUCAAAC 406 AD-75720 usasugucAfuCfAfUfcaaacccuuaL96 239 VPusAfsaggGfuUfUfgaugAfuGfacauasusc 323 GAUAUGUCAUCAUCAAACCCUUG 407 AD-75725 csasucauCfaAfAfCfccuugguguaL96 240 VPusAfscacCfaAfGfgguuUfgAfugaugsasc 324 GUCAUCAUCAAACCCUUGGUGUG 408 AD-75695 asascgagAfgCfAfCfuugaaaaugaL96 241 VPusCfsauuUfuCfAfagugCfuCfucguusgsa 325 UCAACGAGAGCACUUGAAAAUGA 409 AD-75665 csgsagagCfaCfUfUfgaaaaugaaaL96 242 VPusUfsucaUfuUfUfcaagUfgCfucucgsusu 326 AACGAGAGCACUUGAAAAUGAAA 410 AD-75661 asgscacuUfgAfAfAfaugaaaugaaL96 243 VPusUfscauUfuCfAfuuuuCfaAfgugcuscsu 327 AGAGCACUUGAAAAUGAAAUGAC 411 AD-75658 gscsacuuGfaAfAfAfugaaaugacaL96 244 VPusGfsucaUfuUfCfauuuUfcAfagugcsusc 328 GAGCACUUGAAAAUGAAAUGACU 412 AD-75700 asasaugaAfaUfGfAfcugucuaagaL96 245 VPusCfsuuaGfaCfAfgucaUfuUfcauuususc 329 GAAAAUGAAAUGACUGUCUAAGA 413 AD-75698 asasugaaAfuGfAfCfugucuaagaaL96 246 VPusUfscuuAfgAfCfagucAfuUfucauususu 330 AAAAUGAAAUGACUGUCUAAGAG 414 AD-75672 usgsaaauGfaCfUfGfucuaagagaaL96 247 VPusUfscucUfuAfGfacagUfcAfuuucasusu 331 AAUGAAAUGACUGUCUAAGAGAU 415 AD-75684 gsasaaugAfcUfGfUfcuaagagauaL96 248 VPusAfsucuCfuUfAfgacaGfuCfauuucsasu 332 AUGAAAUGACUGUCUAAGAGAUC 416 AD-75667 asasaugaCfuGfUfCfuaagagaucaL96 249 VPusGfsaucUfcUfUfagacAfgUfcauuuscsa 333 UGAAAUGACUGUCUAAGAGAUCU 417 AD-75678 csasacugGfaUfAfCfuagaucuuaaL96 250 VPusUfsaagAfuCfUfaguaUfcCfaguugscsu 334 AGCAACUGGAUACUAGAUCUUAC 418 AD-75660 gscsagcuCfuUfUfCfuucuuugaaaL96 251 VPusUfsucaAfaGfAfagaaAfgAfgcugcsasg 335 CUGCAGCUCUUUCUUCUUUGAAU 419 AD-75701 csuscuuuCfuUfCfUfuugaauuucaL96 252 VPusGfsaaaUfuCfAfaagaAfgAfaagagscsu 336 AGCUCUUUCUUCUUUGAAUUUCC 420 AD-75707 csusuucuUfcUfUfUfgaauuuccuaL96 253 VPusAfsggaAfaUfUfcaaaGfaAfgaaagsasg 337 CUCUUUCUUCUUUGAAUUUCCUA 421 AD-75675 csusuugaAfuUfUfCfcuaucuguaaL96 254 VPusUfsacaGfaUfAfggaaAfuUfcaaagsasa 338 UUCUUUGAAUUUCCUAUCUGUAU 422 AD-75671 ususugaaUfuUfCfCfuaucuguauaL96 255 VPusAfsuacAfgAfUfaggaAfaUfucaaasgsa 339 UCUUUGAAUUUCCUAUCUGUAUG 423 AD-75683 usgsaauuUfcCfUfAfucuguauguaL96 256 VPusAfscauAfcAfGfauagGfaAfauucasasa 340 UUUGAAUUUCCUAUCUGUAUGUC 424 AD-75689 asasuuucCfuAfUfCfuguaugucuaL96 257 VPusAfsgacAfuAfCfagauAfgGfaaauuscsa 341 UGAAUUUCCUAUCUGUAUGUCUG 425 AD-75715 asuscuguAfuGfUfCfugccuaauuaL96 258 VPusAfsauuAfgGfCfagacAfuAfcagausasg 342 CUAUCUGUAUGUCUGCCUAAUUA 426 AD-75705 usgsuaugUfcUfGfCfcuaauuaaaaL96 259 VPusUfsuuaAfuUfAfggcaGfaCfauacasgsa 343 UCUGUAUGUCUGCCUAAUUAAAA 427 AD-75704 gsusauguCfuGfCfCfuaauuaaaaaL96 260 VPusUfsuuuAfaUfUfaggcAfgAfcauacsasg 344 CUGUAUGUCUGCCUAAUUAAAAA 428 AD-75713 usasugucUfgCfCfUfaauuaaaaaaL96 261 VPusUfsuuuUfaAfUfuaggCfaGfacauascsa 345 UGUAUGUCUGCCUAAUUAAAAAA 429 AD-75702 usgsucugCfcUfAfAfuuaaaaaaaaL96 262 VPusUfsuuuUfuUfAfauuaGfgCfagacasusa 346 UAUGUCUGCCUAAUUAAAAAAAU 430 AD-75709 ususguauUfaUfGfCfuaccugcaaaL96 263 VPusUfsugcAfgGfUfagcaUfaAfuacaasusa 347 UAUUGUAUUAUGCUACCUGCAAA 431 AD-75710 ususaugcUfaCfCfUfgcaaaaaaaaL96 264 VPusUfsuuuUfuUfGfcaggUfaGfcauaasusa 348 UAUUAUGCUACCUGCAAAAAAAA 432

TABLE-US-00006 TABLE 4B BAPCS iRNAs-Modified Sequences Sense Sequence 5'-3' SEQ ID Duplex Name Antisense Sequence 5'-3' NO: Range in NM_011318.2 AD-197584.1 csasaugaGfcUfAfCfuaauuauaaL96 NM_011318.2_401-421_s VPusUfsauaAfaUfUfaguaGfcUfcauugsusc NM_011318.2_399-421_as AD-197510.1 asusuuuaCfaCfUfGfuguuuccgaaL96 NM_011318.2_322-342_s

VPusUfscggAfaAfCfacagUfgUfaaaaususc NM_011318.2_320-342_as AD-197583.1
ascsaaugAfgCfUfAfcuaauuuauaL96 NM_011318.2_400-420_s
VPusAfsuaaAfuUfAfguagCfuCfauuguscsu NM_011318.2_398-420_as AD-75728.3
gscsagaaUfuUfUfAfcacuguguuaL96 NM_011318.2_319-338_s
VPusAfsacaCfaGfUfguaaAfaUfucugcsasg NM_011318.2_317-338_as AD-197508.1
gsasauuuUfaCfAfCfuguguuuccaL96 NM_011318.2_320-340_G21A_s
VPusGfsgaaAfcAfCfagugUfaAfaauucsusg NM_011318.2_318-340_C1U_as AD-75737.3
csasgaauUfuUfAfCfacuguguuuaL96 NM_011318.2_320-339_C21A_s
VPusAfsaacAfcAfGfuguaAfaAfuucugscsa NM_011318.2_318-339_C21A_as AD-197550.1
csuscagaGfuCfUfUfuucuccuacaL96 NM_011318.2_364-384_s
VPusGfsuagGfaGfAfaaagAfcUfcugagsasg NM_011318.2_362-384_as AD-197582.1
gsascaauGfaGfCfUfacuaauuuaaL96 NM_011318.2_399-419_s
VPusUfsaaaUfuAfGfuagcUfcAfuugucsusc NM_011318.2_397-419_as AD-75732.3
asgsacaaUfgAfGfCfuacuaauuuaL96 NM_011318.2_400-419_s
VPusAfsaauUfaGfUfagcuCfaUfugucuscsu NM_011318.2_398-419_as AD-197726.1
asusggaaAfgCfCfUfuggguaaaaaL96 NM_011318.2_556-576_s
VPusUfsuuuAfcCfCfaaggCfuUfuccaususg NM_011318.2_554-576_as AD-197544.1
cscscgcuCfuCfAfGfagucuuuucaL96 NM_011318.2_358-378_s
VPusGfsaaaAfgAfCfucugAfgAfgcgggsasa NM_011318.2_356-378_as AD-197551.1
uscsagagUfcUfUfUfucuccuacaaL96 NM_011318.2_365-385_G21A_s
VPusUfsguaGfgAfGfaaaaGfaCfucugasgsa NM_011318.2_363-385_C1U_as AD-75730.2
gsasgacaAfuGfAfGfcuacuaauuaL96 NM_011318.2_399-418_s
VPusAfsauuAfgUfAfgcucAfuUfgucucsusg NM_011318.2_397-418_as AD-197410.1
ususcaccAfgCfCfUfucuuucagaaL96 NM_011318.2_198-218_s
VPusUfscugAfaAfGfaaggCfuGfgugaasgsa NM_011318.2_196-218_as AD-197556.1
uscsuuuuCfuCfCfUfacagugucaaL96 NM_011318.2_371-391_s
VPusUfsgacAfcUfGfuaggAfgAfaaagascsu NM_011318.2_369-391_as AD-197534.1
asgsugacCfuUfUfCfccgcucucaaL96 NM_011318.2_348-368_G21A_s
VPusUfsgagAfgCfGfggaaAfgGfucacusgsu NM_011318.2_346-368_C1U_as AD-75734.2
asgsagucUfuUfUfCfuccuacaguaL96 NM_011318.2_369-388_G21A_s
VPusAfscugUfaGfGfagaaAfaGfacucusgsa NM_011318.2_367-388_G21A_as AD-197503.1
csuscugcAfgAfAfUfuuuacacugaL96 NM_011318.2_313-333_s
VPusCfsaguGfuAfAfaauuCfuGfcagagsgsu NM_011318.2_311-333_as AD-197406.1
usgsucuuCfaCfCfAfgccuucuuuaL96 NM_011318.2_194-214_C21A_s
VPusAfsaagAfaGfGfcuggUfgAfagacasasa NM_011318.2_192-214_G1U_as AD-197416.1
asgsccuuCfuUfUfCfagaagccuuaL96 NM_011318.2_204-224_s
VPusAfsaggCfuUfCfugaaAfgAfaggcusgsg NM_011318.2_202-224_as AD-197725.1
asasuggaAfaGfCfCfuuggguaaaaL96 NM_011318.2_555-575_s
VPusUfsuuaCfcCfAfaggcUfuUfccauusgsa NM_011318.2_553-575_as AD-197546.1
csgscucuCfaGfAfGfucuuuucucaL96 NM_011318.2_360-380_C21A_s
VPusGfsagaAfaAfGfacucUfgAfgagcgsgsg NM_011318.2_358-380_G1U_as AD-197724.1
csasauggAfaAfGfCfcuuggguaaaL96 NM_011318.2_554-574_s
VPusUfsuacCfcAfAfggcuUfuCfcauugsasc NM_011318.2_552-574_as AD-197408.1
uscsuucaCfcAfGfCfcuucuuucaaL96 NM_011318.2_196-216_G21A_s
VPusUfsgaaAfgAfAfggcuGfgUfgaagascsa NM_011318.2_194-216_ClU_as AD-197548.1
csuscucaGfaGfUfCfuuuucuccuaL96 NM_011318.2_362-382_s
VPusAfsggaGfaAfAfagacUfcUfgagagscsg NM_011318.2_360-382_as AD-197565.1
csasgagaCfaAfUfGfagcuacuaaaL96 NM_011318.2_397-416_s
VPusUfsuagUfaGfCfucauUfgUfcucugscsc NM_011318.2_395-416_as AD-197581.1
asgsagacAfaUfGfAfgcuacuaauaL96 NM_011318.2_396-416_s

VPusAfsuuaGfuAfGfcucaUfuGfucucusgsc NM_011318.2_394-416_as AD-197552.1
csasgaguCfuUfUfUfcuccuacagaL96 NM_011318.2_366-386_s
VPusCfsuguAfgGfAfgaaaAfgAfcucugsasg NM_011318.2_364-386_as AD-197549.1
uscsucagAfgUfCfUfuuucuccuaaL96 NM_011318.2_363-383_C21A_s
VPusUfsaggAfgAfAfaagaCfuCfugagasgsc NM_011318.2_361-383_G1U_as AD-197529.1
cscsuacaGfuGfAfCfcuuucccgcaL96 NM_011318.2_343-363_s
VPusGfscggGfaAfAfggucAfcUfguaggsusu NM_011318.2_341-363_as AD-197417.1
gscscuucUfuUfCfAfgaagccuuuaL96 NM_011318.2_205-225_s
VPusAfsaagGfcUfUfcugaAfaGfaaggcsusg NM_011318.2_203-225_as AD-197543.1
uscsccgcUfcUfCfAfgagucuuuuaL96 NM_011318.2_357-377_C21A_s
VPusAfsaaaGfaCfUfcugaGfaGfcgggasasa NM_011318.2_355-377_G1U_as AD-197555.1
gsuscuuuUfcUfCfCfuacagugucaL96 NM_011318.2_370-390_s
VPusGfsacaCfuGfUfaggaGfaAfaagacsusc NM_011318.2_368-390_as AD-197530.1
csusacagUfgAfCfCfuuucccgcuaL96 NM_011318.2_344-364_C21A_s
VPusAfsgcgGfgAfAfagguCfaCfuguagsgsu NM_011318.2_342-364_G1U_as AD-197554.1
asgsucuuUfuCfUfCfcuacaguguaL96 NM_011318.2_369-389_C21A_s
VPusAfscacUfgUfAfggagAfaAfagacuscsu NM_011318.2_367-389_G1U_as AD-197532.1
ascsagugAfcCfUfUfucccgcucuaL96 NM_011318.2_346-366_C21A_s
VPusAfsgagCfgGfGfaaagGfuCfacugusasg NM_011318.2_344-366_G1U_as AD-197412.1
csasccagCfcUfUfCfuuucagaagaL96 NM_011318.2_200-220_C21A_s
VPusCfsuucUfgAfAfagaaGfgCfuggugsasa NM_011318.2_198-220_G1U_as AD-197713.1
gsasauuuUfgGfGfUfcaauggaaaaL96 NM_011318.2_543-563_s
VPusUfsuucCfaUfUfgaccCfaAfaauucsasa NM_011318.2_541-563_as AD-197411.1
uscsaccaGfcCfUfUfcuuucagaaaL96 NM_011318.2_199-219_G21A_s
VPusUfsucuGfaAfAfgaagGfcUfggugasasg NM_011318.2_197-219_C1U_as AD-197557.1
csusuuucUfcCfUfAfcagugucaaaL96 NM_011318.2_372-392_G21A_s
VPusUfsugaCfaCfUfguagGfaGfaaaagsasc NM_011318.2_370-392_C1U_as AD-197528.1
ascscuacAfgUfGfAfccuuucccgaL96 NM_011318.2_342-362_C21A_s
VPusCfsgggAfaAfGfgucaCfuGfuaggususc NM_011318.2_340-362_G1U_as AD-197631.1
uscsaaaaGfuCfAfCfaguccguggaL96 NM_011318.2_459-479_s
VPusCfscacGfgAfCfugugAfcUfuuugasusu NM_011318.2_457-479_as AD-197541.1
ususucccGfcUfCfUfcagagucuuaL96 NM_011318.2_355-375_s
VPusAfsagaCfuCfUfgagaGfcGfgggaaasgsg NM_011318.2_353-375_as AD-197533.1
csasgugaCfcUfUfUfcccgcucucaL96 NM_011318.2_347-367_s
VPusGfsagaGfcGfGfgaaaGfgUfcacugsusa NM_011318.2_345-367_as AD-197632.1
csasaaagUfcAfCfAfguccgugguaL96 NM_011318.2_460-480_s
VPusAfsccaCfgGfAfcuguGfaCfuuuugsasu NM_011318.2_458-480_as AD-197542.1
ususcccgCfuCfUfCfagagucuuuaL96 NM_011318.2_356-376_s
VPusAfsaagAfcUfCfugagAfgCfgggaasasg NM_011318.2_354-376_as AD-197629.1
asasucaaAfaGfUfCfacaguccguaL96 NM_011318.2_457-477_G21A_s
VPusAfscggAfcUfGfugacUfuUfugauusgsu NM_011318.2_455-477_C1U_as
TABLE-US-00007 TABLE 5A APCS Single Dose Screen in Cos7 cells 10 nM 0.1 nM Avg [% Avg [% message message Duplex ID remaining] SD remaining] SD AD-75727.1 39.3 3.8 92.0 8.5 AD-75729.1 49.7 9.1 101.7 24.5 AD-75730.1 30.7 5.9 98.1 11.8 AD-75739.1 24.1 16.8 86.0 9.9 AD-75738.1 86.2 6.8 91.1 15.9 AD-75728.1 121.2 7.5 109.0 3.9 AD-75737.1 106.3 17.9 115.5 13.0 AD-75733.1 23.2 7.5 59.0 15.4 AD-75732.1 16.1 3.3 62.7 18.2 AD-75736.1 72.7 18.7 101.5 21.4 AD-75731.1 124.0 18.6 106.2 7.8 AD-75666.1 7.2 1.0 31.4 3.3 AD-75734.1 122.9 14.4 101.0 8.3 AD-75735.1 86.5 7.3 104.2 10.6 AD-75663.1 5.3 1.7 41.4 10.8 AD-75677.1 25.9 5.3 74.7 6.8 AD-75696.1 8.8 0.5 32.4 0.9 AD-75676.1 15.3 6.9 50.0 14.8 AD-75669.1 8.3 1.0 47.4 14.6 AD-75718.1 11.0 2.9 36.5 9.9 AD-75693.1 20.8 2.5 79.2 8.5 AD-75667.1 14.2 3.2 67.8 10.7 AD-75678.1 12.7

1.9 48.4 1.8 AD-75691.1 9.5 2.0 35.0 5.0 AD-75717.1 14.0 3.1 67.7 9.4 AD-75697.1 8.4 1.3 39.5 9.9 AD-75692.1 4.8 0.4 20.9 3.3 AD-75720.1 7.3 1.4 54.0 5.8 AD-75706.1 9.0 1.8 42.7 14.2 AD-75694.1 9.7 2.8 51.2 15.5 AD-75679.1 3.0 0.7 28.1 5.5 AD-75684.1 11.3 3.0 63.5 1.4 AD-75686.1 7.0 1.5 32.6 2.7 AD-75687.1 5.7 1.3 51.4 6.0 AD-75674.1 8.2 1.7 54.8 5.3 AD-75723.1 13.0 3.0 61.3 13.6 AD-75685.1 8.6 1.6 52.1 5.6 AD-75681.1 28.9 5.1 85.1 11.4 AD-75675.1 5.9 1.8 28.2 1.1 AD-75671.1 3.8 0.5 20.4 5.5 AD-75689.1 4.5 0.6 20.8 3.8 AD-75657.1 7.5 0.2 32.3 6.4 AD-75659.1 9.6 1.6 38.1 12.2 AD-75714.1 12.1 2.5 85.6 11.8 AD-75665.1 9.9 2.7 44.2 6.6 AD-75664.2 7.8 1.8 56.7 8.2 AD-75660.1 8.8 1.7 29.4 1.0 AD-75708.1 11.3 1.8 69.9 8.9 AD-75658.1 7.8 1.0 36.0 7.5 AD-75680.1 7.7 1.7 58.9 6.5 AD-75690.1 26.5 2.5 74.7 7.4 AD-75688.1 15.2 2.1 63.2 9.4 AD-75683.1 19.5 2.7 51.8 6.2 AD-75716.1 11.1 3.3 66.0 9.9 AD-75673.1 9.4 1.2 57.9 4.1 AD-75707.1 6.1 0.5 53.4 13.1 AD-75713.1 12.4 2.1 44.4 5.5 AD-75672.1 9.3 1.8 59.3 6.5 AD-75699.1 7.9 0.6 37.0 4.2 AD-75703.1 14.8 4.2 56.3 7.7 AD-75719.1 6.0 0.9 30.6 7.0 AD-75704.1 10.8 1.6 34.9 6.4 AD-75661.1 9.6 1.2 40.5 2.5 AD-75712.1 13.6 4.6 39.0 7.3 AD-75705.1 8.8 1.3 17.8 6.0 AD-75702.1 8.0 1.1 25.7 3.4 AD-75722.1 71.6 8.6 99.6 22.2 AD-75670.1 16.6 5.3 67.0 7.6 AD-75700.1 8.7 1.7 24.8 5.0 AD-75701.1 6.0 1.8 28.5 2.4 AD-75668.1 33.5 2.3 71.7 11.0 AD-75709.1 17.2 3.5 55.6 4.1 AD-75711.1 10.4 1.6 47.5 2.9 AD-75726.1 18.3 2.4 48.3 12.8 AD-75662.1 7.0 3.2 35.2 4.6 AD-75721.1 17.5 2.7 55.4 9.9 AD-75695.1 13.1 0.8 48.5 6.4 AD-75664.1 11.6 2.7 53.8 5.5 AD-75682.1 12.4 3.2 53.8 6.8 AD-75710.1 78.0 11.2 101.4 6.3 AD-75698.1 11.4 0.8 31.3 11.0 AD-75724.1 26.6 4.4 81.6 17.6 AD-75725.1 30.4 2.4 79.0 11.5 AD-75715.1 9.3 1.5 53.2 5.1

TABLE-US-00008 TABLE 5B APCS Single Dose Screen in Cos7 cells 10 nM 1 nM 0.1 nM Avg Avg Avg [% [% [% message message message Duplex ID remaining] SD remaining] SD remaining] SD AD-197584.1 10.0 1.0 9.6 1.3 33.8 1.9 AD-197510.1 3.4 0.2 6.6 1.2 34.9 5.0 AD-197583.1 7.3 0.5 8.6 0.4 35.9 3.2 AD-75728.3 8.8 2.6 11.4 1.9 39.4 6.8 AD-197508.1 5.4 0.5 10.2 3.2 46.1 5.6 AD-75737.3 11.7 1.3 18.1 1.2 48.7 3.4 AD-197550.1 5.6 1.2 17.0 8.3 49.3 10.0 AD-197582.1 9.7 0.8 15.4 1.4 49.9 5.0 AD-75732.3 9.7 0.5 14.3 2.0 54.8 6.3 AD-197726.1 20.3 3.8 23.0 3.8 58.9 3.1 AD-197544.1 13.1 0.7 23.7 2.8 60.9 7.6 AD-197551.1 7.9 0.9 15.5 0.8 64.1 9.8 AD-75730.2 8.1 0.7 22.6 2.9 65.3 4.5 AD-197410.1 10.6 0.3 21.7 1.0 65.4 11.2 AD-197556.1 12.1 0.7 25.1 2.7 65.5 8.9 AD-197534.1 14.1 0.8 21.5 3.4 66.8 10.1 AD-75734.2 18.0 0.8 36.7 4.4 67.9 5.8 AD-197503.1 10.1 0.8 20.5 3.1 68.1 2.3 AD-197406.1 8.0 0.6 24.6 5.7 68.4 6.8 AD-197416.1 8.3 2.2 23.1 2.0 69.2 6.1 AD-197725.1 25.2 3.1 25.2 1.7 69.6 11.6 AD-197546.1 12.9 1.6 23.6 2.8 71.1 5.5 AD-197724.1 18.4 1.5 45.3 11.2 71.5 9.2 AD-197408.1 16.1 0.4 37.4 7.7 72.9 5.5 AD-197548.1 9.0 0.7 24.5 7.4 72.9 7.1 AD-197565.1 17.0 2.1 26.9 4.2 73.7 5.5 AD-197581.1 17.0 0.6 34.1 6.8 75.4 3.3 AD-197552.1 6.0 1.1 19.8 5.3 75.7 7.5 AD-197549.1 8.6 0.6 24.3 3.5 76.5 7.4 AD-197529.1 18.0 1.3 43.8 8.1 76.9 4.0 AD-197417.1 13.6 2.1 30.4 3.4 77.3 4.2 AD-197543.1 17.2 0.8 37.1 6.3 77.4 6.1 AD-197555.1 12.8 1.5 27.2 1.0 78.0 11.6 AD-197530.1 16.8 1.1 31.5 2.3 79.0 3.4 AD-197554.1 15.8 1.4 35.3 4.0 79.2 3.8 AD-197532.1 17.4 1.3 33.8 3.1 80.5 1.5 AD-197412.1 16.4 1.5 27.9 6.4 80.9 9.5 AD-197713.1 35.5 6.0 41.6 4.5 83.5 8.5 AD-197411.1 21.8 1.3 41.8 2.2 83.6 8.8 AD-197557.1 32.4 2.6 53.2 9.5 84.0 8.9 AD-197528.1 15.4 1.8 51.9 10.6 84.5 4.7 AD-197631.1 38.1 4.3 53.2 6.4 87.2 10.0 AD-197541.1 27.6 1.1 61.0 6.5 87.2 10.1 AD-197533.1 23.4 0.5 43.4 11.0 87.9 5.4 AD-197632.1 31.4 1.3 56.9 5.4 88.8 5.4 AD-197542.1 21.3 4.3 41.1 4.3 91.3 4.1 AD-197629.1 15.5 0.9 40.9 5.3 91.8 5.4

Example 2. IRNA Design, Synthesis, Selection, and In Vitro Evaluation

Bioinformatics

[0517] An additional set of siRNAs targeting the human APCS (SAP), “amyloid P component, serum” (human: NCBI refseqlD NM_001639.3; NCBI GeneID: 325) as well the toxicology-species APCS ortholog from cynomolgus monkey: XM_005541312.2) was designed using custom R and Python scripts. All the siRNA designs are a perfect match to the human APCS transcript and a subset have either perfect or near-perfect matches to the cynomolgus monkey ortholog. The human NM_001639 REFSEQ mRNA, version 3, has a length of 960 bases.

[0518] A detailed list of the unmodified SAP sense and antisense strand sequences is shown in

Table 6. A detailed list of the modified SAP sense and antisense strand sequences is shown in Table 7.

In Vitro Screen

Dual-Glo® Luciferase Assay

[0519] Cos7 cells (ATCC, Manassas, VA) were grown to near confluence at 37° C. in an atmosphere of 5% CO2 in DMEM (ATCC) supplemented with 10% FBS, before being released from the plate by trypsinization. siRNA and psiCHECK2-APCS plasmid transfection was carried out by adding 5 μl of siRNA duplexes and 5 μl of psiCHECK2-APCS plasmid per well along with 5 μl of Opti-MEM plus 0.1 μl of Lipofectamine 2000 per well (Invitrogen, Carlsbad CA. cat #13778-150) and then incubated at room temperature for 15 minutes. The mixture was then added to the cells which were re-suspended in 35 μl of fresh complete media. The transfected cells were incubated at 37° C. in an atmosphere of 5% CO2.

[0520] Forty-eight hours after the siRNAs and psiCHECK2-APCS plasmid were transfected; Firefly (transfection control) and *Renilla* (fused to ANGPTL4 target sequence) luciferase were measured. First, media was removed from cells. Then Firefly luciferase activity was measured by adding 20 μl of Dual-Glo® Luciferase Reagent equal to the culture medium volume to each well and mix. The mixture was incubated at room temperature for 30 minutes before luminescence (500 nm) was measured on a Spectramax (Molecular Devices) to detect the Firefly luciferase signal. *Renilla* luciferase activity was measured by adding 20 μl of room temperature of Dual-Glo® Stop & Glo® Reagent to each well and the plates were incubated for 10-15 minutes before luminescence was again measured to determine the *Renilla* luciferase signal. The Dual-Glo® Stop & Glo® Reagent, quenches the firefly luciferase signal and sustained luminescence for the *Renilla* luciferase reaction. siRNA activity was determined by normalizing the *Renilla* (HBV) signal to the Firefly (control) signal within each well. The magnitude of siRNA activity was then assessed relative to cells that were transfected with the same vector but were not treated with siRNA or were treated with a non-targeting siRNA. All transfections were done at n=2 or greater. Table 8 shows the results of a single dose screen. Data are expressed as percent of mRNA remaining relative to untreated Cos7 cells.

TABLE-US-00009 TABLE 6 APCS iRNAs-Unmodified Sequences

Duplex Name	transSeq	SEQ ID NO:	transSeq	SEQ ID NO:	Range in NM_001639.3
AD-77749	AAUAUCAGACGCUAGGGGA	433	UCCCCUAGCGUCUGAUAUU	501	7-25
AD-77750	GGGGGACAGCCACUGUGUU	434	AACACAGUGGCUGUCCCCC	502	22-40
AD-77751	UUGUCUGCUACCCUCAUCA	435	UGAUGAGGGUAGCAGACAA	503	39-57
AD-77752	CAUCCUGGUCACUGCUUCU	436	AGAAGCAGUGACCAGGAUG	504	53-71
AD-77753	UUCUGCUAUAACAGCCCUA	437	UAGGGCUGUUAUAGCAGAA	505	68-86
AD-77754	CCCUAGGCCAGGAAUAUGA	438	UCAUAUUCCUGGCCUAGGG	506	82-100
AD-77755	CAGGAAUAUGAACAAGCCA	439	UGGCUUGUUCAUAUUCCUG	507	90-108
AD-77756	AGCCGCUGCUUUGGAUCUA	440	UAGAUCCAAAGCAGCGGCU	508	104-122
AD-77757	AUCUCUGUCCUCACCAGCA	441	UGCUGGUGAGGACAGAGAU	509	118-136
AD-77758	CAGCCUCCUGGAAGCCUUU	442	AAAGGCUUCCAGGAGGCUG	510	132-150
AD-77759	UUUGCUCACACAGACCUCA	443	UGAGGUCUGUGUGAGCAAA	511	148-166
AD-77760	CCUCAGUGGGAAGGUGUUU	444	AAACACCUUCCCACUGAGG	512	162-180
AD-77761	GGAAGGUGUUUGUAUUUCA	445	UGAAAUACAAACACCUUCC	513	170-188
AD-77762	UUCCUAGAGAAUCUGUUAA	446	UUAACAGAUUCUCUAGGAA	514	185-203
AD-77763	UUACUGAUCAUGUAAACUU	447	AAGUUUACAUGAUCAGUAA	515	200-218
AD-77764	AACUUGAUCACACCGCUGA	448	UCAGCGGUGUGAUCAAGUU	516	214-232
AD-77765	GCUGGAGAAGCCUCUACAA	449	UUGUAGAGGCUUCUCCAGC	517	228-246
AD-77766	UACAGAACUUUACCUUGUA	450	UACAAGGUAAAGUUCUGUA	518	242-260
AD-77767	ACUUUACCUUGUGUUUUCA	451	UGAAAACACAAGGUAAAGU	519	248-266
AD-77768	UUCGAGCCUAUAGUGAUCU	452	AGAUCACUAUAGGCUCGAA	520	263-281
AD-77769					

GAUCUCUCUCGUGCCUACA 453 UGUAGGCACGAGAGAGAUC 521 277-295 AD-77770
UCUCGUGCCUACAGCCUCU 454 AGAGGCUGUAGGCACGAGA 522 283-301 AD-77771
CAGCCUCUUCUCCUACAAU 455 AUUGUAGGAGAAGAGGCUG 523 294-312 AD-77772
CAAUACCCAAGGCAGGGAU 456 AUCCCUGCCUUGGGUAUUG 524 309-327 AD-77773
GGGAUAAUGAGCUACUAGU 457 ACUAGUAGCUCAUUAUCCC 525 323-341 AD-77774
UAGUUUAUAAAGAAAGAGU 458 ACUCUUUCUUUAUAAACUA 526 338-356 AD-77775
AGAGUUGGAGAGUAUAGUA 459 UACUAUACUCUCCAACUCU 527 352-370 AD-77776
UAGUCUAUACAUUGGAAGA 460 UCUUCCAAUGUAUAGACUA 528 366-384 AD-77777
UUGGAAGACACAAAGUUAA 461 UUAACUUUGUGUCUUCCAA 529 377-395 AD-77778
CACAAAGUUACAUCCAAAG 462 CUUUGGAUGUAACUUUGUG 530 385-403 AD-77779
AAAGUUAUCGAAAAGUUCA 463 UGAACUUUUCGAUAACUUU 531 400-418 AD-77780
GUUCCCGGCUCCAGUGCAA 464 UUGCACUGGAGCCGGGAAC 532 414-432 AD-77781
UGCACAUCUGUGUGAGCUA 465 UAGCUCACACAGAUGUGCA 533 428-446 AD-77782
AGCUGGGAGUCCUCAUCAA 466 UUGAUGAGGACUCCCAGCU 534 442-460 AD-77783
GUAUUGCUGAAUUUUGGAU 467 AUCCAAAAUUCAGCAAUAC 535 461-479 AD-77784
GGAUCAAUGGGACACCUUU 468 AAAGGUGUCCCAUUGAUCC 536 476-494 AD-77785
CUUUGGUGAAAAAGGGUCU 469 AGACCCUUUUUCACCAAAG 537 491-509 AD-77786
GGUCUGCGACAGGGUUACU 470 AGUAACCCUGUCGCAGACC 538 505-523 AD-77787
UUUGUAGAAGCUCAGCCCA 471 UGGGCUGAGCUUCUACAAA 539 523-541 AD-77788
GCCCAAGAUUGUCCUGGGA 472 UCCCAGGACAAUCUUGGGC 540 537-555 AD-77789
UGGGGCAGGAACAGGAUUA 473 UAAUCCUGUUCCUGCCCCA 541 551-569 AD-77790
UCCUAUGGGGGCAAGUUUA 474 UAAACUUGCCCCCAUAGGA 542 568-586 AD-77791
UUUGAUAGGAGCCAGUCCU 475 AGGACUGGCUCCUAUCAAA 543 583-601 AD-77792
GUCCUUUGUGGGAGAGAUU 476 AAUCUCUCCCACAAAGGAC 544 597-615 AD-77793
AGAUUGGGGAUUUGUACAU 477 AUGUACAAAUCCCCAAUCU 545 611-629 AD-77794
UUUGUACAUGUGGGACUCU 478 AGAGUCCCACAUGUACAAA 546 621-639 AD-77795
AUGUGGGACUCUGUGCUGA 479 UCAGCACAGAGUCCCACAU 547 628-646 AD-77796
CUGCCCCCAGAAAAUAUCA 480 UGAUAUUUUCUGGGGGCAG 548 643-661 AD-77797
UAUCCUGUCUGCCUAUCAA 481 UUGAUAGGCAGACAGGAUA 549 657-675 AD-77798
CCUAUCAGGGUACCCCUCU 482 AGAGGGGUACCCUGAUAGG 550 668-686 AD-77799
CUCUCCCUGCCAAUAUCCU 483 AGGAUAUUGGCAGGGAGAG 551 683-701 AD-77800
UCCUGGACUGGCAGGCUCU 484 AGAGCCUGCCAGUCCAGGA 552 698-716 AD-77801
UGGCAGGCUCUGAACUAUA 485 UAUAGUUCAGAGCCUGCCA 553 706-724 AD-77802
UAUGAAAUCAGAGGAUAUA 486 UAUAUCCUCUGAUUUCAUA 554 721-739 AD-77803
UAUGUCAUCAUCAAACCCU 487 AGGGUUUGAUGAUGACAUA 555 736-754 AD-77804
ACCCUUGGUGUGGGUCUGA 488 UCAGACCCACACCAAGGGU 556 750-768 AD-77805
UGUGGGUCUGAGGUCUUGA 489 UCAAGACCUCAGACCCACA 557 758-776 AD-77806
UUGACUCAACGAGAGCACU 490 AGUGCUCUCGUUGAGUCAA 558 773-791 AD-77807
GCACUUGAAAAUGAAAUGA 491 UCAUUUCAUUUUCAAGUGC 559 787-805 AD-77808
AAUGACUGUCUAAGAGAUA 492 UAUCUCUUAGACAGUCAUU 560 801-819 AD-77809
AGAUCUGGUCAAAGCAACU 493 AGUUGCUUUGACCAGAUCU 561 815-833 AD-77810
CAAAGCAACUGGAUACUAA 494 UUAGUAUCCAGUUGCUUUG 562 824-842 AD-77811
CUAGAUCUUACAUCUGCAA 495 UUGCAGAUGUAAGAUCUAG 563 839-857 AD-77812
GCAGCUCUUUCUUCUUUGA 496 UCAAAGAAGAAAGAGCUGC 564 854-872 AD-77813
UUGAAUUUCCUAUCUGUAU 497 AUACAGAUAGGAAAUUCAA 565 869-887 AD-77814
CUAUCUGUAUGUCUGCCUA 498 UAGGCAGACAUACAGAUAG 566 878-896 AD-77815
CCUAAUUAAAAAAAUAUAU 499 AUAUAUUUUUUUAAUUAGG 567 893-911 AD-77816
UAUAUAUUGUAUUAUGCUA 500 UAGCAUAAUACAAUAUAUA 568 907-925
TABLE-US-00010 TABLE 7 APCSiRNAs-Modified Sequences senseOligo antisOligo
Duplex Name Name oligoSeq SEQ ID NO: Name oligoSeq SEQ ID NO: mRNA

target sequence SEQ ID NO: AD-77749 A-156212 AAUAUCAGACGCUAGGGGAdTdT 569 A-156213 UCCCCUAGCGUCUGAUAUUdTdT 637 AAUAUCAGACGCUAGGGGG 705 AD-77750 A-156214 GGGGGACAGCCACUGUGUUdTdT 570 A-156215 AACACAGUGGCUGUCCCCCdTdT 638 GGGGGACAGCCACUGUGUU 706 AD-77751 A-156216 UUGUCUGCUACCCUCAUCAdTdT 571 A-156217 UGAUGAGGGUAGCAGACAAdTdT 639 UUGUCUGCUACCCUCAUCC 707 AD-77752 A-156218 CAUCCUGGUCACUGCUUCUdTdT 572 A-156219 AGAAGCAGUGACCAGGAUGdTdT 640 CAUCCUGGUCACUGCUUCU 708 AD-77753 A-156220 UUCUGCUAUAACAGCCCUAdTdT 573 A-156221 UAGGGCUGUUAUAGCAGAAdTdT 641 UUCUGCUAUAACAGCCCUA 709 AD-77754 A-156222 CCCUAGGCCAGGAAUAUGAdTdT 574 A-156223 UCAUAUUCCUGGCCUAGGGdTdT 642 CCCUAGGCCAGGAAUAUGA 710 AD-77755 A-156224 CAGGAAUAUGAACAAGCCAdTdT 575 A-156225 UGGCUUGUUCAUAUUCCUGdTdT 643 CAGGAAUAUGAACAAGCCG 711 AD-77756 A-156226 AGCCGCUGCUUUGGAUCUAdTdT 576 A-156227 UAGAUCCAAAGCAGCGGCUdTdT 644 AGCCGCUGCUUUGGAUCUC 712 AD-77757 A-156228 AUCUCUGUCCUCACCAGCAdTdT 577 A-156229 UGCUGGUGAGGACAGAGAUdTdT 645 AUCUCUGUCCUCACCAGCC 713 AD-77758 A-156230 CAGCCUCCUGGAAGCCUUUdTdT 578 A-156231 AAAGGCUUCCAGGAGGCUGdTdT 646 CAGCCUCCUGGAAGCCUUU 714 AD-77759 A-156232 UUUGCUCACACAGACCUCAdTdT 579 A-156233 UGAGGUCUGUGUGAGCAAAdTdT 647 UUUGCUCACACAGACCUCA 715 AD-77760 A-156234 CCUCAGUGGGAAGGUGUUUdTdT 580 A-156235 AAACACCUUCCCACUGAGGdTdT 648 CCUCAGUGGGAAGGUGUUU 716 AD-77761 A-156236 GGAAGGUGUUUGUAUUUCAdTdT 581 A-156237 UGAAAUACAAACACCUUCCdTdT 649 GGAAGGUGUUUGUAUUUCC 717 AD-77762 A-156238 UUCCUAGAGAAUCUGUUAAdTdT 582 A-156239 UUAACAGAUUCUCUAGGAAdTdT 650 UUCCUAGAGAAUCUGUUAC 718 AD-77763 A-156240 UUACUGAUCAUGUAAACUUdTdT 583 A-156241 AAGUUUACAUGAUCAGUAAdTdT 651 UUACUGAUCAUGUAAACUU 719 AD-77764 A-156242 AACUUGAUCACACCGCUGAdTdT 584 A-156243 UCAGCGGUGUGAUCAAGUUdTdT 652 AACUUGAUCACACCGCUGG 720 AD-77765 A-156244 GCUGGAGAAGCCUCUACAAdTdT 585 A-156245 UUGUAGAGGCUUCUCCAGCdTdT 653 GCUGGAGAAGCCUCUACAG 721 AD-77766 A-156246 UACAGAACUUUACCUUGUAdTdT 586 A-156247 UACAAGGUAAAGUUCUGUAdTdT 654 UACAGAACUUUACCUUGUG 722 AD-77767 A-156248 ACUUUACCUUGUGUUUUCAdTdT 587 A-156249 UGAAAACACAAGGUAAAGUdTdT 655 ACUUUACCUUGUGUUUUCG 723 AD-77768 A-156250 UUCGAGCCUAUAGUGAUCUdTdT 588 A-156251 AGAUCACUAUAGGCUCGAAdTdT 656 UUCGAGCCUAUAGUGAUCU 724 AD-77769 A-156252 GAUCUCUCUCGUGCCUACAdTdT 589 A-156253 UGUAGGCACGAGAGAGAUCdTdT 657 GAUCUCUCUCGUGCCUACA 725 AD-77770 A-156254 UCUCGUGCCUACAGCCUCUdTdT 590 A-156255 AGAGGCUGUAGGCACGAGAdTdT 658 UCUCGUGCCUACAGCCUCU 726 AD-77771 A-156256 CAGCCUCUUCUCCUACAAUdTdT 591 A-156257 AUUGUAGGAGAAGAGGCUGdTdT 659 CAGCCUCUUCUCCUACAAU 727 AD-77772 A-156258 CAAUACCCAAGGCAGGGAUdTdT 592 A-156259 AUCCCUGCCUUGGGUAUUGdTdT 660 CAAUACCCAAGGCAGGGAU 728 AD-77773 A-156260 GGGAUAAUGAGCUACUAGUdTdT 593 A-156261 ACUAGUAGCUCAUUAUCCCdTdT 661 GGGAUAAUGAGCUACUAGU 729 AD-77774 A-

156262 UAGUUUAUAAAGAAAGAGUdTdT 594 A-156263
ACUCUUUCUUUAUAAACUAdTdT 662 UAGUUUAUAAAGAAAGAGU 730 AD-77775 A-
156264 AGAGUUGGAGAGUAUAGUAdTdT 595 A-156265
UACUAUACUCUCCAACUCUdTdT 663 AGAGUUGGAGAGUAUAGUC 731 AD-77776 A-
156266 UAGUCUAUACAUUGGAAGAdTdT 596 A-156267
UCUUCCAAUGUAUAGACUAdTdT 664 UAGUCUAUACAUUGGAAGA 732 AD-77777 A-
156268 UUGGAAGACACAAAGUUAAdTdT 597 A-156269
UUAACUUUGUGUCUUCCAAdTdT 665 UUGGAAGACACAAAGUUAC 733 AD-77778 A-
156270 CACAAAGUUACAUCCAAAGdTdT 598 A-156271
CUUUGGAUGUAACUUUGUGdTdT 666 CACAAAGUUACAUCCAAAG 734 AD-77779 A-
156272 AAAGUUAUCGAAAAGUUCAdTdT 599 A-156273
UGAACUUUUCGAUAACUUUdTdT 667 AAAGUUAUCGAAAAGUUCC 735 AD-77780 A-
156274 GUUCCCGGCUCCAGUGCAAdTdT 600 A-156275
UUGCACUGGAGCCGGGAACdTdT 668 GUUCCCGGCUCCAGUGCAC 736 AD-77781 A-
156276 UGCACAUCUGUGUGAGCUAdTdT 601 A-156277
UAGCUCACACAGAUGUGCAdTdT 669 UGCACAUCUGUGUGAGCUG 737 AD-77782 A-
156278 AGCUGGGAGUCCUCAUCAAdTdT 602 A-156279
UUGAUGAGGACUCCCAGCUdTdT 670 AGCUGGGAGUCCUCAUCAG 738 AD-77783 A-
156280 GUAUUGCUGAAUUUUGGAUdTdT 603 A-156281
AUCCAAAAUUCAGCAAUACdTdT 671 GUAUUGCUGAAUUUUGGAU 739 AD-77784 A-
156282 GGAUCAAUGGGACACCUUUdTdT 604 A-156283
AAAGGUGUCCCAUUGAUCCdTdT 672 GGAUCAAUGGGACACCUUU 740 AD-77785 A-
156284 CUUUGGUGAAAAAGGGUCUdTdT 605 A-156285
AGACCCUUUUUCACCAAAGdTdT 673 CUUUGGUGAAAAAGGGUCU 741 AD-77786 A-
156286 GGUCUGCGACAGGGUUACUdTdT 606 A-156287
AGUAACCCUGUCGCAGACCdTdT 674 GGUCUGCGACAGGGUUACU 742 AD-77787 A-
156288 UUUGUAGAAGCUCAGCCCAdTdT 607 A-156289
UGGGCUGAGCUUCUACAAAdTdT 675 UUUGUAGAAGCUCAGCCCA 743 AD-77788 A-
156290 GCCCAAGAUUGUCCUGGGAdTdT 608 A-156291
UCCCAGGACAAUCUUGGGCdTdT 676 GCCCAAGAUUGUCCUGGGG 744 AD-77789 A-
156292 UGGGGCAGGAACAGGAUUAdTdT 609 A-156293
UAAUCCUGUUCCUGCCCCAdTdT 677 UGGGGCAGGAACAGGAUUC 745 AD-77790 A-
156294 UCCUAUGGGGGCAAGUUUAdTdT 610 A-156295
UAAACUUGCCCCCAUAGGAdTdT 678 UCCUAUGGGGGCAAGUUUG 746 AD-77791 A-
156296 UUUGAUAGGAGCCAGUCCUdTdT 611 A-156297
AGGACUGGCUCCUAUCAAAdTdT 679 UUUGAUAGGAGCCAGUCCU 747 AD-77792 A-
156298 GUCCUUUGUGGGAGAGAUUdTdT 612 A-156299
AAUCUCUCCCACAAAGGACdTdT 680 GUCCUUUGUGGGAGAGAUU 748 AD-77793 A-
156300 AGAUUGGGGAUUUGUACAUdTdT 613 A-156301
AUGUACAAAUCCCCAAUCUdTdT 681 AGAUUGGGGAUUUGUACAU 749 AD-77794 A-
156302 UUUGUACAUGUGGGACUCUdTdT 614 A-156303
AGAGUCCCACAUGUACAAAdTdT 682 UUUGUACAUGUGGGACUCU 750 AD-77795 A-
156304 AUGUGGGACUCUGUGCUGAdTdT 615 A-156305
UCAGCACAGAGUCCCACAUdTdT 683 AUGUGGGACUCUGUGCUGC 751 AD-77796 A-
156306 CUGCCCCCAGAAAAUAUCAdTdT 616 A-156307
UGAUAUUUUCUGGGGGCAGdTdT 684 CUGCCCCCAGAAAAUAUCC 752 AD-77797 A-
156308 UAUCCUGUCUGCCUAUCAAdTdT 617 A-156309
UUGAUAGGCAGACAGGAUAdTdT 685 UAUCCUGUCUGCCUAUCAG 753 AD-77798 A-
156310 CCUAUCAGGGUACCCCUCUdTdT 618 A-156311
AGAGGGGUACCCUGAUAGGdTdT 686 CCUAUCAGGGUACCCCUCU 754 AD-77799 A-

156312 CUCUCCCUGCCAAUAUCCUdTdT 619 A-156313 AGGAUAUUGGCAGGGAGAGdTdT 687 CUCUCCCUGCCAAUAUCCU 755 AD-77800 A-156314 UCCUGGACUGGCAGGCUCUdTdT 620 A-156315 AGAGCCUGCCAGUCCAGGAdTdT 688 UCCUGGACUGGCAGGCUCU 756 AD-77801 A-156316 UGGCAGGCUCUGAACUAUAdTdT 621 A-156317 UAUAGUUCAGAGCCUGCCAdTdT 689 UGGCAGGCUCUGAACUAUG 757 AD-77802 A-156318 UAUGAAAUCAGAGGAUAUAdTdT 622 A-156319 UAUAUCCUCUGAUUUCAUAdTdT 690 UAUGAAAUCAGAGGAUAUG 758 AD-77803 A-156320 UAUGUCAUCAUCAAACCCUdTdT 623 A-156321 AGGGUUUGAUGAUGACAUAdTdT 691 UAUGUCAUCAUCAAACCCU 759 AD-77804 A-156322 ACCCUUGGUGUGGGUCUGAdTdT 624 A-156323 UCAGACCCACACCAAGGGUdTdT 692 ACCCUUGGUGUGGGUCUGA 760 AD-77805 A-156324 UGUGGGUCUGAGGUCUUGAdTdT 625 A-156325 UCAAGACCUCAGACCCACAdTdT 693 UGUGGGUCUGAGGUCUUGA 761 AD-77806 A-156326 UUGACUCAACGAGAGCACUdTdT 626 A-156327 AGUGCUCUCGUUGAGUCAAdTdT 694 UUGACUCAACGAGAGCACU 762 AD-77807 A-156328 GCACUUGAAAAUGAAAUGAdTdT 627 A-156329 UCAUUUCAUUUUCAAGUGCdTdT 695 GCACUUGAAAAUGAAAUGA 763 AD-77808 A-156330 AAUGACUGUCUAAGAGAUAdTdT 628 A-156331 UAUCUCUUAGACAGUCAUUdTdT 696 AAUGACUGUCUAAGAGAUC 764 AD-77809 A-156332 AGAUCUGGUCAAAGCAACUdTdT 629 A-156333 AGUUGCUUUGACCAGAUCUdTdT 697 AGAUCUGGUCAAAGCAACU 765 AD-77810 A-156334 CAAAGCAACUGGAUACUAAdTdT 630 A-156335 UUAGUAUCCAGUUGCUUUGdTdT 698 CAAAGCAACUGGAUACUAG 766 AD-77811 A-156336 CUAGAUCUUACAUCUGCAAdTdT 631 A-156337 UUGCAGAUGUAAGAUCUAGdTdT 699 CUAGAUCUUACAUCUGCAG 767 AD-77812 A-156338 GCAGCUCUUUCUUCUUUGAdTdT 632 A-156339 UCAAAGAAGAAAGAGCUGCdTdT 700 GCAGCUCUUUCUUCUUUGA 768 AD-77813 A-156340 UUGAAUUUCCUAUCUGUAUdTdT 633 A-156341 AUACAGAUAGGAAAUUCAAdTdT 701 UUGAAUUUCCUAUCUGUAU 769 AD-77814 A-156342 CUAUCUGUAUGUCUGCCUAdTdT 634 A-156343 UAGGCAGACAUACAGAUAGdTdT 702 CUAUCUGUAUGUCUGCCUA 770 AD-77815 A-156344 CCUAAUUAAAAAAAUAUAUdTdT 635 A-156345 AUAUAUUUUUUUAAUUAGGdTdT 703 CCUAAUUAAAAAAAUAUAU 771 AD-77816 A-156346 UAUAUAUUGUAUUAUGCUAdTdT 636 A-156347 UAGCAUAAUACAAUAUAUAdTdT 704 UAUAUAUUGUAUUAUGCUA 772

TABLE-US-00011 TABLE 8 APCS Single Dose Screen in Cos7 cells 10 nM avg [% message Duplex ID remaining] stdev AD-77749 55.2 5.0 AD-77750 40.8 4.4 AD-77751 78.6 6.4 AD-77752 11.2 2.5 AD-77753 13.0 3.2 AD-77754 17.9 4.4 AD-77755 20.1 3.3 AD-77756 16.1 2.4 AD-77757 13.2 1.6 AD-77758 11.4 1.5 AD-77759 9.5 1.0 AD-77760 19.8 6.8 AD-77761 4.8 1.0 AD-77762 9.3 3.1 AD-77763 24.6 6.2 AD-77764 13.2 1.8 AD-77765 11.0 1.1 AD-77766 9.4 3.4 AD-77767 10.2 1.8 AD-77768 59.3 4.8 AD-77769 13.2 1.5 AD-77770 21.9 2.1 AD-77771 21.3 5.2 AD-77772 18.7 4.6 AD-77773 11.0 2.0 AD-77774 59.4 6.2 AD-77775 12.2 1.2 AD-77776 16.1 1.1 AD-77777 10.6 1.3 AD-77778 17.3 1.5 AD-77779 30.4 2.2 AD-77780 78.7 18.9 AD-77781 38.2 5.4 AD-77782 21.0 3.2 AD-77783 12.3 7.1 AD-77784 8.2 2.0 AD-77785 14.2 2.6 AD-77786 21.9 5.5 AD-77787 39.3 3.4 AD-77788 33.6 12.3 AD-77789 35.9 6.6 AD-77790 27.3 3.4 AD-77791 77.0 8.1 AD-77792 25.5 2.7 AD-77793 29.1 3.4 AD-77794 62.8 9.0 AD-77795 19.6 3.7 AD-77796 51.6 8.2 AD-77797 29.7 2.7 AD-77798 17.7 0.9 AD-77799 33.1 4.4 AD-77800 38.6 1.7 AD-77801 20.4 2.0 AD-77802 17.1 1.7 AD-77803 78.0 4.6 AD-77804 18.1 1.9 AD-77805 10.8 1.7 AD-77806 11.3 2.5 AD-77807 10.3 1.7 AD-77808 34.2 1.4 AD-77809 25.5 11.9 AD-77810 27.1 15.7 AD-77811 31.6

14.4 AD-77812 18.9 10.7 AD-77813 13.6 6.8 AD-77814 19.7 12.2 AD-77815 27.5 7.5 AD-77816 23.0 10.3

TABLE-US-00012 Informal Sequence Listing >gi|206597534|ref|NM_001639.3| Homo sapiens amyloid P component, serum (APCS), mRNA SEQ ID NO: 1
GGGCATGAATATCAGACGCTAGGGGGACAGCCACTGTGTTGTCTGCTACC
CTCATCCTGGTCACTGCTTCTGCTATAACAGCCCTAGGCCAGGAATATGA
ACAAGCCGCTGCTTTGGATCTCTGTCCTCACCAGCCTCCTGGAAGCCTTT
GCTCACACAGACCTCAGTGGGAAGGTGTTTGTATTTCCTAGAGAATCTGT
TACTGATCATGTAAACTTGATCACACCGCTGGAGAAGCCTCTACAGAACT
TTACCTTGTGTTTTCGAGCCTATAGTGATCTCTCTCGTGCCTACAGCCTC
TTCTCCTACAATACCCAAGGCAGGGATAATGAGCTACTAGTTTATAAAGA
AAGAGTTGGAGAGTATAGTCTATACATTGGAAGACACAAAGTTACATCCA
AAGTTATCGAAAAGTTCCCGGCTCCAGTGCACATCTGTGTGAGCTGGGAG
TCCTCATCAGGTATTGCTGAATTTTGGATCAATGGGACACCTTTGGTGAA
AAAGGGTCTGCGACAGGGTTACTTTGTAGAAGCTCAGCCCAAGATTGTCC
TGGGGCAGGAACAGGATTCCTATGGGGGCAAGTTTGATAGGAGCCAGTCC
TTTGTGGGAGAGATTGGGGATTTGTACATGTGGGACTCTGTGCTGCCCCC
AGAAAATATCCTGTCTGCCTATCAGGGTACCCCTCTCCCTGCCAATATCC
TGGACTGGCAGGCTCTGAACTATGAAATCAGAGGATATGTCATCATCAAA
CCCTTGGTGTGGGTCTGAGGTCTTGACTCAACGAGAGCACTTGAAAATGA
AATGACTGTCTAAGAGATCTGGTCAAAGCAACTGGATACTAGATCTTACA
TCTGCAGCTCTTTCTTCTTTGAATTTCCTATCTGTATGTCTGCCTAATTA
AAAAAATATATATTGTATTATGCTACCTGCAAAAAAAAA
>gi|982224943|ref|XM_005541312.2| PREDICTED: Macaca fascicularis amyloid P component, serum (APCS), mRNA SEQ ID NO: 2
AATAGCCTGAAGTCTAAGGGCGTGAATATCAGACGCTAGGGGGACAGCCA
CTGTGTTATCTGCTGCCCTCATCCTGGTCACTGCTCCTGCTATAATAGCC
CTAGGCCAGGAATATGAACAAGCTGCTGCTTTGGGTCTCTGTCCTCACCA
GCCTCCTGGAAGCCTTTGCTCACACAGACCTCAGTGGGAAGGTGTTTGTA
TTTCCTAGAGAATCTGTTACTGATCATGTAAACTTGATCACACAGCTGGA
GAAGCCTCTGCAGAACTTTACCTTGTGTTTTCGAGCCTACACTGATCTCT
CCCGTCCCTACAGCCTCTTCTCCTACAATACCCAGGGAAAGGATAATGAG
CTAGTAGTTTATAAAGAAAGAGTTGGAGATTATAGTCTCTACATTGGAAG
ACAAAAAGTTACATTCAAAGTTATTGAAAAGTTCCCGGCTCCAGTGCACA
TCTGTGTTAGCTGGGAGTCCTCATCAGGTATTGCTGAATTTTGGATCAAT
GGGACACCTTTGGTGAAAAAGGGTCTGCGACAGGGTTACTCTGTGGAAGC
TCATCCCAAGATTGTCCTGGGGCAGGAGCAGGATTCCTATGGCGGCAAGT
TTGATAAGAGCCAGTCTTTTGTGGGAGAGATTGGGGATTTGTACATGTGG
AACTCTGTACTGCCCCCAGAAGAGATCCTGTCTGCCTATCAGGGTACCCC
AGTCCCTGCCAATATCCTGGACTGGCGGGCTCTGAACTATGAAATCAAAG
GATATGTCATCATCAAACCCCTGGTGTGGGTCTGAGGTCTTGACTCAACA
AGATCACTTGAAAATGAAATGACCGTCTAAGAGATCTGCTCAAAGCAACT
GGATACTAGATCTTATACCTGCAGCTCTTTCTTCTTTGAATTCCCTATCT
GTATGTCTGCCGAATTAAAAACTGTATATTGTATTATGCTACCTGCATTT
GTTTAGTGCTTATCATAGTCCCATATCTTTATCTTATATCTACTACTTAT
CTATCTGCTACTTATCTGGTAATTGGTGTTTCATTATCCTGAACAATCAA
ATTGCCAAGTATGGGGAGGAAAACCTATAAGTAACTAGAAAGGTGTATCA
CAAAACCAGGGCACTCAATGAGCAGGACAGTGGCAATACTTAAAAGCTAC
TAACAATCATTTTAATAACGATTCAAAATATGTGTGGAATTGCAATCACA
ATTCTTCTGTTTTTCCAGCTACAAATCGATCAAAAAGATGTCTGAGGTTG

CATCAGGGAGACAACTATAATGAAATAAGTAAGCAATAGTTGAGAACCTA
ACAGCAAGTAATGAGACAGGAAGAATAAAACTAGAAGTCAGAATGAAAAC
CAAAAAAGTATGAGACTGGGATAGTTTTGTATGAAGACAAACATAACTTT
ATGTCTCAAAAACCTCAGTAGGACTGTACAATGCTTGATTTTCATAACGC
TGTCCTGATGTCATCAAAAAATGTTATACTCTGAATATTGCTTACATGAA
TATGCTGTCTGAAATTATTCCATGGATTCACACTTCTTACAGGACGTGAT
TTAAACCTCTAAATATGACATTGCCCTCCATAACCTGGTCCCACTTACTC
TCTGAAATGTACACAGAGAATACACACAGAATACTCTCCGAAATGGACTT
TCCATGTACATTTCAGATACAACTGTTAGCAGAGCAAGTGACATCCTAAC
TATCACACTGCAAAATAATTGTCAGCTGCTATTACTAGTAGTTATTTTTA
TGGTTATGAAAAGCATGGAACACAGTGCCTGGCTCATCATCAACAGATAT
GAGTTGCATTCCCAGGGAAGCAAGGATCCCAGAGGGAGTTCAGGCTATAT
ATATTTGAAATAAGACAGGCTTTTCTCACTGATAATAAGTAGAGCATGAT
GTCAGTGGAATCAGAAAGATAGCATCCAGAACAGTTTCAGGATGGTAACC
AATGTGATGATTTCAAACTCATGAGGCCATTTGGCCAGAATGAACAGAGG
CAGTAGCCAGATATGAAGAAATAGGATAGAATCTAGAAACTCCATAATGG
GGCAATGGGGTAGAAGCAGAGAGGACCAGACAGCACCTGTGAAGACTTGC
TGAAGTACTTAAATGGAGCAGCAGGGGCTTCATAAGAACATTTCAACCTC
ATTTCTGGTGAGGTGGTAAGATGGTTTGGATATGGTTTGTTTGCCTCCAC
CAAACTTCATCTTGAACTTTTATTCTCAATATGGTGTTGTTGGAAGGTGA
GTCCTAGTGAAAGGCATTTGAGCTATGGGGGTGGGTCTCTCATGAATGTT
TTGGTGCAGTTCTCACGGTAATGAGTGAGTTCTCACTCTTATGAAAGTAC
ATTAACTCTTGTGGGATCAGATTGGTTCTTGCCAGATTTAGCTGTTATAA
AACCAGGGTGTCCTTTAGGTTTTGTTTTCTTTGCACATGTTCACTTCCCT
TTTGACTTTCTCCACCATGTTTTGATACAACACAAAAACCCTCACCAGAA
GCCAGAGGCGTGATCTTGAACTTCTCAGCCTGCAGAGCAATGAGCTAAAT
AAACCTCTTTT >gi|226958496|ref|NM_011318.2| Mus musculus serum amyloid P-component (Apcs), mRNA SEQ ID NO: 3
GCAGAACGAAGGAGGATCTGGGAGTACCTCACATGGTATTACTTCTCTCC
ACCCTTCATTGTCATCCAAGGTACATACAAAACCTGAAATCTGAAAAGCA
TAGGCAGACACCACACTTTTGTTCCACACCCAAGTAACAGCTGCTGCTGT
CATACCCTGGACCAAGCATGGACAAGCTACTGCTTTGGATGTTTGTCTTC
ACCAGCCTTCTTTCAGAAGCCTTTTGTCAGACAGACCTCAAGAGGAAAGT
ATTTGTGTTCCCCAGAGAATCTGAAACTGATCATGTGAAGCTGATCCCAC
ATCTAGAGAAACCTCTGCAGAATTTTACACTGTGTTTCCGAACCTACAGT
GACCTTTCCCGCTCTCAGAGTCTTTTCTCCTACAGTGTCAAGGGCAGAGA
CAATGAGCTACTAATTTATAAAGAAAAAGTTGGAGAATACAGCCTATACA
TCGGACAATCAAAAGTCACAGTCCGTGGTATGGAAGAATACCTTTCTCCA
GTACACCTATGTACCACTTGGGAGTCCTCCTCTGGCATTGTTGAATTTTG
GGTCAATGGAAAGCCTTGGGTAAAAAAGTCTCTGCAGAGGGAATACACTG
TGAAAGCCCCACCCAGTATAGTCCTGGGACAGGAGCAGGATAACTACGGA
GGAGGGTTTCAAAGGTCACAGTCCTTTGTAGGAGAGTTTTCAGATTTATA
CATGTGGGACTATGTGCTGACCCCACAAGACATTCTATTTGTGTACAGAG
ATTCCCCTGTCAATCCTAATATTTTGAATTGGCAGGCTCTTAACTATGAA
ATAAATGGCTACGTAGTCATCAGGCCCCGTGTCTGGGATTGAGATCTTAC
AACAAAACCTCATGGACATCAGATGGCCGATGTGTAAGAGGTCAAGGCAG
CAGAATTCACTCTATCTGGAGCTTTTTCTTCTTTGTGAACATCTTGTATA
CATATCTGCCAAATAAAAATCCTCTCCAATTCCAAAAAAAAAAAAAAAAA
AAAAAAAAAAAA >gi|148747487|ref|NM_017170.2| Rattus norvegicus amyloid P component, serum (Apcs), mRNA SEQ ID NO: 4

ATGGACAAGCTGCTGCTTTGGATGTCTGTCTTCACCAGCCTTCTTTCAGA
AGCCTTTGCTCAGACAGACCTCAATCAGAAGGTATTTGTGTTCCCCAGAG
AATCTGAAACTGATTATGTGAAGCTGATCCCATGGCTAGAAAAACCGCTG
CAGAATTTTACACTGTGTTTCCGAGCCTACAGTGACCTTTCCCGCTCTCA
GAGTCTTTTCTCCTACAGTGTCAACAGCAGAGACAATGAGCTACTAATTT
ATAAAGACAAAGTTGGACAATATAGCCTATACATTGGAAATTCAAAAGTC
ACAGTCCGTGGTTTAGAAGAATTCCCTTCTCCAATACATTTCTGTACCAG
CTGGGAGTCCTCCTCTGGTATTGCTGAATTTTGGGTCAATGGAAAGCCTT
GGGTAAAAAAGGGTTTGCAGAAGGGATACACTGTGAAATCCTCACCCAGT
ATTGTCCTGGGACAGGAGCAGGATACGTATGGAGGAGGGTTTGATAAGAC
ACAGTCCTTTGTGGGAGAGATTGCAGATTTGTACATGTGGGACAGTGTGC
TGACCCCAGAGAACATTCATTCTGTGGACAGAGGTTTCCCACCCAATCCT
AATATTTTGGATTGGCGGGCCCTGAATTATGAAATAAATGGTTATGTAGT
CATCAAGCCCCGTATGTGGGACAACAAAAGCTCATGA Reverse complement of SEQ ID NO: 1 SEQ ID NO: 5
TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTGCAGGTAGCATAATACAATA
TATATTTTTTTAATTAGGCAGACATACAGATAGGAAATTCAAAGAAGAAA
GAGCTGCAGATGTAAGATCTAGTATCCAGTTGCTTTGACCAGATCTCTTA
GACAGTCATTTCATTTTCAAGTGCTCTCGTTGAGTCAAGACCTCAGACCC
ACACCAAGGGTTTGATGATGACATATCCTCTGATTTCATAGTTCAGAGCC
TGCCAGTCCAGGATATTGGCAGGGAGAGGGGTACCCTGATAGGCAGACAG
GATATTTTCTGGGGGCAGCACAGAGTCCCACATGTACAAATCCCCAATCT
CTCCCACAAAGGACTGGCTCCTATCAAACTTGCCCCCATAGGAATCCTGT
TCCTGCCCCAGGACAATCTTGGGCTGAGCTTCTACAAAGTAACCCTGTCG
CAGACCCTTTTTCACCAAAGGTGTCCCATTGATCCAAAATTCAGCAATAC
CTGATGAGGACTCCCAGCTCACACAGATGTGCACTGGAGCCGGGAACTTT
TCGATAACTTTGGATGTAACTTTGTGTCTTCCAATGTATAGACTATACTC
TCCAACTCTTTCTTTATAAACTAGTAGCTCATTATCCCTGCCTTGGGTAT
TGTAGGAGAAGAGGCTGTAGGCACGAGAGAGATCACTATAGGCTCGAAAA
CACAAGGTAAAGTTCTGTAGAGGCTTCTCCAGCGGTGTGATCAAGTTTAC
ATGATCAGTAACAGATTCTCTAGGAAATACAAACACCTTCCCACTGAGGT
CTGTGTGAGCAAAGGCTTCCAGGAGGCTGGTGAGGACAGAGATCCAAAGC
AGCGGCTTGTTCATATTCCTGGCCTAGGGCTGTTATAGCAGAAGCAGTGA
CCAGGATGAGGGTAGCAGACAACACAGTGGCTGTCCCCCTAGCGTCTGAT
ATTCATGCCC Reverse complement of SEQ ID NO: 2 SEQ ID NO: 6
AATAGCCTGAAGTCTAAGGGCGTGAATATCAGACGCTAGGGGGACAGCCA
CTGTGTTATCTGCTGCCCTCATCCTGGTCACTGCTCCTGCTATAATAGCC
CTAGGCCAGGAATATGAACAAGCTGCTGCTTTGGGTCTCTGTCCTCACCA
GCCTCCTGGAAGCCTTTGCTCACACAGACCTCAGTGGGAAGGTGTTTGTA
TTTCCTAGAGAATCTGTTACTGATCATGTAAACTTGATCACACAGCTGGA
GAAGCCTCTGCAGAACTTTACCTTGTGTTTTCGAGCCTACACTGATCTCT
CCCGTCCCTACAGCCTCTTCTCCTACAATACCCAGGGAAAGGATAATGAG
CTAGTAGTTTATAAAGAAAGAGTTGGAGATTATAGTCTCTACATTGGAAG
ACAAAAAGTTACATTCAAAGTTATTGAAAAGTTCCCGGCTCCAGTGCACA
TCTGTGTTAGCTGGGAGTCCTCATCAGGTATTGCTGAATTTTGGATCAAT
GGGACACCTTTGGTGAAAAAGGGTCTGCGACAGGGTTACTCTGTGGAAGC
TCATCCCAAGATTGTCCTGGGGCAGGAGCAGGATTCCTATGGCGGCAAGT
TTGATAAGAGCCAGTCTTTTGTGGGAGAGATTGGGGATTTGTACATGTGG
AACTCTGTACTGCCCCCAGAAGAGATCCTGTCTGCCTATCAGGGTACCCC
AGTCCCTGCCAATATCCTGGACTGGCGGGCTCTGAACTATGAAATCAAAG

GATATGTCATCATCAAACCCCTGGTGTGGGTCTGAGGTCTTGACTCAACA
AGATCACTTGAAAATGAAATGACCGTCTAAGAGATCTGCTCAAAGCAACT
GGATACTAGATCTTATACCTGCAGCTCTTTCTTCTTTGAATTCCCTATCT
GTATGTCTGCCGAATTAAAAACTGTATATTGTATTATGCTACCTGCATTT
GTTTAGTGCTTATCATAGTCCCATATCTTTATCTTATATCTACTACTTAT
CTATCTGCTACTTATCTGGTAATTGGTGTTTCATTATCCTGAACAATCAA
ATTGCCAAGTATGGGGAGGAAAACCTATAAGTAACTAGAAAGGTGTATCA
CAAAACCAGGGCACTCAATGAGCAGGACAGTGGCAATACTTAAAAGCTAC
TAACAATCATTTTAATAACGATTCAAAATATGTGTGGAATTGCAATCACA
ATTCTTCTGTTTTTCCAGCTACAAATCGATCAAAAAGATGTCTGAGGTTG
CATCAGGGAGACAACTATAATGAAATAAGTAAGCAATAGTTGAGAACCTA
ACAGCAAGTAATGAGACAGGAAGAATAAAACTAGAAGTCAGAATGAAAAC
CAAAAAAGTATGAGACTGGGATAGTTTTGTATGAAGACAAACATAACTTT
ATGTCTCAAAAACCTCAGTAGGACTGTACAATGCTTGATTTTCATAACGC
TGTCCTGATGTCATCAAAAAAATGTTATACTCTGAATATTGCTTACATGAA
TATGCTGTCTGAAATTATTCCATGGATTCACACTTCTTACAGGACGTGAT
TTAAACCTCTAAATATGACATTGCCCTCCATAACCTGGTCCCACTTACTC
TCTGAAATGTACACAGAGAATACACACAGAATACTCTCCGAAATGGACTT
TCCATGTACATTTCAGATACAACTGTTAGCAGAGCAAGTGACATCCTAAC
TATCACACTGCAAAATAATTGTCAGCTGCTATTACTAGTAGTTATTTTTA
TGGTTATGAAAAGCATGGAACACAGTGCCTGGCTCATCATCAACAGATAT
GAGTTGCATTCCCAGGGAAGCAAGGATCCCAGAGGGAGTTCAGGCTATAT
ATATTTGAAATAAGACAGGCTTTTCTCACTGATAATAAGTAGAGCATGAT
GTCAGTGGAATCAGAAAGATAGCATCCAGAACAGTTTCAGGATGGTAACC
AATGTGATGATTTCAAACTCATGAGGCCATTTGGCCAGAATGAACAGAGG
CAGTAGCCAGATATGAAGAAATAGGATAGAATCTAGAAACTCCATAATGG
GGCAATGGGGTAGAAGCAGAGAGGACCAGACAGCACCTGTGAAGACTTGC
TGAAGTACTTAAATGGAGCAGCAGGGGCTTCATAAGAACATTTCAACCTC
ATTTCTGGTGAGGTGGTAAGATGGTTTGGATATGGTTTGTTTGCCTCCAC
CAAACTTCATCTTGAACTTTTATTCTCAATATGGTGTTGTTGGAAGGTGA
GTCCTAGTGAAAGGCATTTGAGCTATGGGGGTGGGTCTCTCATGAATGTT
TTGGTGCAGTTCTCACGGTAATGAGTGAGTTCTCACTCTTATGAAAGTAC
ATTAACTCTTGTGGGATCAGATTGGTTCTTGCCAGATTTAGCTGTTATAA
AACCAGGGTGTCCTTTAGGTTTTGTTTTCTTTGCACATGTTCACTTCCCT
TTTGACTTTCTCCACCATGTTTTGATACAACACAAAAACCCTCACCAGAA
GCCAGAGGCGTGATCTTGAACTTCTCAGCCTGCAGAGCAATGAGCTAAAT
AAACCTCTTTT Reverse complement of SEQ ID NO: 3 SEQ ID NO: 7
TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTGGAATTGGAGAGGATTTTTAT
TTGGCAGATATGTATACAAGATGTTCACAAAGAAGAAAAAGCTCCAGATA
GAGTGAATTCTGCTGCCTTGACCTCTTACACATCGGCCATCTGATGTCCA
TGAGGTTTTGTTGTAAGATCTCAATCCCAGACACGGGGCCTGATGACTAC
GTAGCCATTTATTTCATAGTTAAGAGCCTGCCAATTCAAAATATTAGGAT
TGACAGGGGAATCTCTGTACACAAATAGAATGTCTTGTGGGGTCAGCACA
TAGTCCCACATGTATAAATCTGAAAACTCTCCTACAAAGGACTGTGACCT
TTGAAACCCTCCTCCGTAGTTATCCTGCTCCTGTCCCAGGACTATACTGG
GTGGGGCTTTCACAGTGTATTCCCTCTGCAGAGACTTTTTTACCCAAGGC
TTTCCATTGACCCAAAATTCAACAATGCCAGAGGAGGACTCCCAAGTGGT
ACATAGGTGTACTGGAGAAAGGTATTCTTCCATACCACGGACTGTGACTT
TTGATTGTCCGATGTATAGGCTGTATTCTCCAACTTTTTCTTTATAAATT
AGTAGCTCATTGTCTCTGCCCTTGACACTGTAGGAGAAAAGACTCTGAGA

GCGGGAAAGGTCACTGTAGGTTCGGAAACACAGTGTAAAATTCTGCAGAG
GTTTCTCTAGATGTGGGATCAGCTTCACATGATCAGTTTCAGATTCTCTG
GGGAACACAAATACTTTCCTCTTGAGGTCTGTCTGACAAAAGGCTTCTGA
AAGAAGGCTGGTGAAGACAAACATCCAAAGCAGTAGCTTGTCCATGCTTG
GTCCAGGGTATGACAGCAGCAGCTGTTACTTGGGTGTGGAACAAAAGTGT
GGTGTCTGCCTATGCTTTTCAGATTTCAGGTTTTGTATGTACCTTGGATG
ACAATGAAGGGTGGAGAGAAGTAATACCATGTGAGGTACTCCCAGATCCT
CCTTCGTTCTGC Reverse complement of SEQ ID NO: 4 SEQ ID NO: 8
TCATGAGCTTTTGTTGTCCCACATACGGGGCTTGATGACTACATAACCAT
TTATTTCATAATTCAGGGCCCGCCAATCCAAAATATTAGGATTGGGTGGG
AAACCTCTGTCCACAGAATGAATGTTCTCTGGGGTCAGCACACTGTCCCA
CATGTACAAATCTGCAATCTCTCCCACAAAGGACTGTGTCTTATCAAACC
CTCCTCCATACGTATCCTGCTCCTGTCCCAGGACAATACTGGGTGAGGAT
TTCACAGTGTATCCCTTCTGCAAACCCTTTTTTACCCAAGGCTTTCCATT
GACCCAAAATTCAGCAATACCAGAGGAGGACTCCCAGCTGGTACAGAAAT
GTATTGGAGAAGGGAATTCTTCTAAACCACGGACTGTGACTTTTGAATTT
CCAATGTATAGGCTATATTGTCCAACTTTGTCTTTATAAATTAGTAGCTC
ATTGTCTCTGCTGTTGACACTGTAGGAGAAAAGACTCTGAGAGCGGGAAA
GGTCACTGTAGGCTCGGAAACACAGTGTAAAATTCTGCAGCGGTTTTTCT
AGCCATGGGATCAGCTTCACATAATCAGTTTCAGATTCTCTGGGGAACAC
AAATACCTTCTGATTGAGGTCTGTCTGAGCAAAGGCTTCTGAAAGAAGGC
TGGTGAAGACAGACATCCAAAGCAGCAGCTTGTCCAT

Claims

1. (canceled)
2. A double stranded ribonucleic acid (RNAi) agent for inhibiting expression of a serum amyloid P component (APCS) gene, comprising a sense strand and an antisense strand forming a double stranded region, wherein the antisense strand comprises at least contiguous nucleotides differing by no more than 3 nucleotides from any one of the antisense sequences listed in any one of Tables 3A, 3B, 4A, 4B, 6, and 7.
3.-**12**. (canceled)
13. The double stranded RNAi agent of claim 2, wherein at least one nucleotide of the said RNAi agent comprises a nucleotide modification.
14. (canceled)
15. The double stranded RNAi agent of claim 2, wherein all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand comprise a nucleotide modification.
16. (canceled)
17. (canceled)
18. The double stranded RNAi agent of claim 13, further comprising at least one phosphorothioate internucleotide linkage.
19. The double stranded RNAi agent of claim **1**, wherein the region of complementarity between said sense strand and said antisense strand is at least 17 nucleotides in length; 19 to 30 nucleotides in length: 21 nucleotides in length; 21 to 23 nucleotides in length; or 19 nucleotides in length.
20.-**23**. (canceled)
24. The double stranded RNAi agent of claim **1**, wherein each strand is no more than 30 nucleotides in length; each strand is independently 19-30 nucleotides in length; or each strand is independently 19-25 nucleotides in length.
25. (canceled)
26. (canceled)

27. The double stranded RNAi agent of claim 24, wherein the sense strand is 21 nucleotides in length and the antisense strand is 23 nucleotides in length.
28. The double stranded RNAi agent of claim **1**, wherein at least one strand comprises a 3′ overhang of at least 1 nucleotide; or a 3′ overhang of at least 2 nucleotides.
29. (canceled)
30. The double stranded RNAi agent of claim **1**, further comprising a ligand.
31. (canceled)
32. The double stranded RNAi agent of claim 30, wherein the ligand is an N-acetylgalactosamine (GalNAc) derivative.
33. The double stranded RNAi agent of claim 32, wherein the ligand is ##STR00016##
34. The double stranded RNAi agent of claim 33, wherein the double stranded RNAi agent is conjugated to the ligand as shown in the following schematic ##STR00017## and wherein X is O or S.
35. The double stranded RNAi agent of claim 34, wherein the X is O.
36. (canceled)
37. A cell containing the double stranded RNAi agent of claim **1**.
38. (canceled)
39. A pharmaceutical composition for inhibiting expression of a serum amyloid P component (APCS) gene comprising the double stranded RNAi agent of claim **1**.
40.-**44**. (canceled)
45. A method of inhibiting expression of a serum amyloid P component (APCS) gene in a cell, the method comprising contacting the cell with the double stranded RNAi agent of claim **1**, thereby inhibiting expression of the APCS gene in the cell.
46.-**50**. (canceled)
51. A method of treating a subject having a disease that would benefit from reduction in serum amyloid P component (APCS) gene expression, comprising administering to the subject a therapeutically effective amount of the double stranded RNAi agent of claim **1**, thereby treating said subject.
52.-**62**. (canceled)
63. A method of inhibiting the expression of an APCS protein (SAP) in a subject, the method comprising administering to said subject a therapeutically effective amount of the double stranded RNAi agent of claim **1**, thereby inhibiting the expression of APCS in said subject.